

Power Cube Family



Integrated Controller for Managing
Predefined and Programmable Heating Cycles

Master Controller V3+

Installation, Use and Maintenance Manual

Document	Date	Software
FI040K0012v1100hUK	07/07/2020	MCPC35xx



Read this manual carefully before installing, operating or carrying out maintenance on the device. Keep the manual in a safe place for future reference, and in perfect condition. This manual must accompany the device described therein in the case of change of ownership, and until the device is broken up.

SYMBOLS



The equipment is marked with this symbol wherever the user should refer to this manual in order to avoid possible damage. The same symbol appears in the manual at points where warnings or particularly important instructions, essential for safe, correct operation of the device, are given.



The equipment is marked with this symbol in the areas where there is dangerous voltage. Only trained maintenance personnel should carry out work in these areas. The same symbol appears in the manual at points where warnings essential for safe, are given.



This symbol appears in the manual at points where suggestions, additional information or other relevant notes are given.

REVISION TABLE

Rev.	Date	Author	Reference	Description
1000	2014/03/10	TP2 - SP	-	First Emission
1010	2014/07/31	TP2 - SP	-	Small corrections
1020	2014/09/09	TP2 - SP	-	Added Profinet - Small corrections
1030	2014/12/01	TP2 - SP	-	Software upgrade to MCPC3120
1040	2016/02/08	TP2 - SP	-	Small corrections - Software upgrade to MCPC3126
1050	2016/11/09	TP2 - SP	6	Updated CE Certificates
			6.5	Added Manufacturer data
1060	2016/12/19	TP2 - SP	6	Updated CE Certificates
1070	2017/03/07	TP2 - SP	-	Software upgrade
1080	2017/07/26	TP2 - SP	-	Small corrections
1100	2020/07/07	DTP - SP	-	Software upgrade to MCPC35xx

Warranty terms

The warranty on all CEIA products, extended to the period agreed with the Sales Department, is applicable to goods supplied from our factory, and for every constituent part thereof, with the exception of the batteries. Any form of tampering with the device, and in particular opening its container, is strictly forbidden and will invalidate the warranty.

Customer Satisfaction Report

C.E.I.A. objective is the maximum quality of products and services offered to the customer. Any comments and suggestions useful for achieving this objective will be highly appreciated and can be sent to us by compiling and returning the form available at:

<http://www.ceia-power.com/satisfaction>

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Read this manual carefully before installing or operating the device and before carrying out maintenance operations.

1 SAFETY INSTRUCTIONS – WARNINGS

1.1 Warnings

1.1.1 General warnings

- Follow the instructions contained in this manual for all operations relating to installation, use and maintenance of the device.

CEIA cannot be held responsible for any damage resulting from procedures which are not expressly indicated in this manual, or from any lack of attention, either partial or total, of the procedures described therein.

- Whenever there is any suggestion that the level of protection has been reduced, the device should be taken out of service and secured against any possibility of unintentional use, and authorised service technicians should be called.

The level of protection is considered to have been reduced when:

- *the device shows visible signs of wear and tear, especially on parts that ensures the device protections (boxes, gaskets, glands, fixing screws, etc.);*
- *the ground connections are not compliant;*
- *the device has suffered mechanical or electrical stress (shocks, bumps, etc.);*
- *the device does not operate correctly, respect on how described in this manual;*
- *the device has been stored for long periods in sub-optimal conditions;*
- *the device has suffered severe stress during transport;*
- *the inside of the device has come into contact with liquids.*
- Handle the device with care and without excessive force during installation, use and maintenance.

SAFETY INSTRUCTIONS - WARNINGS**1.1.2 Installation warnings**

Observe current regulations regarding electrical and personal safety for both the operator and the installer when installing the device.

- Choose the installation site carefully. Avoid placing the device in locations where it may be directly exposed to sunlight or in places that are close to sources of heat. In addition, avoid places that are subject to vibrations, dust, humidity, rain, excessively high or low temperatures.
- This device contains electrical and electronic components, and may therefore be susceptible to fire. Do not install in explosive atmosphere or in contact with inflammable material, Do not use water or foam in the case of fire when the device is powered up.
- After installation the device should be stable, and not subject to vibration or accidental movement. All connecting cables should be properly fastened down, in order to avoid knocks and accidental damage and to obtain optimum performance.
- The operator must be able to access the controls of the device easily, without running personal risks.
- Before connecting the device to the power supply, ensure that the power supply voltage corresponds to that indicated on the plate affixed to the device. Verification that the power supply conforms to the values specified plate and to the regulations in force is the total responsibility of the customer.
- The device must be connected to a power supply circuit fitted with a switch or other device which allows the power to be cut off.
- If the device is to be powered via an external autotransformer to regulate the voltage, ensure that the common terminal of the autotransformer is connected to the neutral of the power-supply circuit.
- All connecting cables should be properly fastened down, in order to avoid knocks and accidental damage and to obtain optimum performance.
- If the cable length is excessive, do not cut it, but twist the exceeding part.

1.1.3 Use warnings

- To avoid damage due to lightning, disconnect the power-supply cable during storms.
- Never cut off the safety features of the device.

1.1.4 Maintenance warnings

- Before any maintenance, cleaning or movement, the device must first be disconnected from all sources of power.
- Always remove the plug by hand when disconnecting the power supply cable, never by pulling on the cable.
- Do not wash the device with liquid detergents or chemical substances. Any cleaning should be done using a slightly-damp, non-abrasive cloth.
- Read the chapter on "Maintenance" carefully before calling the service centre. Whatever the problem, only specialised service personnel authorised to work with CEIA equipment should be called.
- Any damaged parts of the device should be replaced with original components only.
- Any maintenance or repair of the device while open and energised should be avoided, and in any case should only be carried out by trained personnel who are fully aware of the risks which the operation entails, following the instructions given in the Maintenance section.



1.2 Residual risks

1.2.1 Installation

- Before providing the device with electrical power, check that the voltage from the power mains corresponds to the voltage shown on the device's electrical specifications plaque.

1.2.2 Maintenance

- The capacitors within the device may remain charged even after the device has been disconnected from all sources of power. Wait 60 seconds before beginning repair work on the device.

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2 DESCRIPTION

2.1 General description

2.1.1 Description

The **Master Controller V3+** is a computerized industrial control unit housed in a robust stainless steel container, designed for automatic and semi-automatic management of heating cycles carried out using CEIA Power Cube Induction Generators.

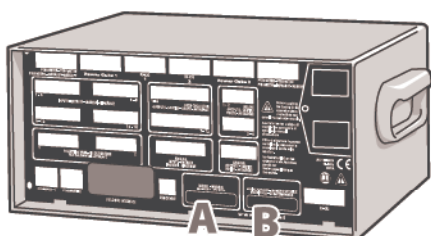
The Controller comes with a set of predefined programs. All operating parameters for each phase of the heating cycle can be programmed within a wide range of values. This allows maximum flexibility in setting the heating and cooling times, the power supplied by the induction generators, the quantity of solder wire and the operating temperatures for the piece being worked on.

Together with the optical temperature sensors (SH/SLE series), and the soldering wire feeder (PW3-WF series), the **CEIA Master Controller V3+** allows the creation of efficient induction brazing systems and achievement of completely repeatable heating processes and finished products.

An RS-232 serial interface, or an optional Field Bus interface, allows connection to a PLC or other external logic for remote loading and running of operating programs.

The photo-coupler-type static inputs have programmable P/N logic, and together with the static outputs are subject to a programmable auxiliary voltage of 12/24 V.

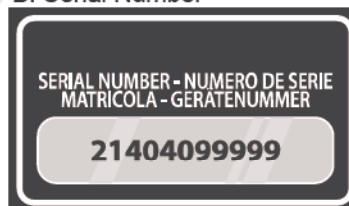
2.1.2 Device identification



A: Model



B: Serial Number



2.2 Correct and Forbidden use of the device

2.2.1 Correct use of the device

The Master Controller V3+ is an electronic device for the automatic and semi-automatic management of heating cycles executed by induction generators CEIA PowerCube.

2.2.2 Forbidden use of the device

The electrical plug must only be inserted into a socket containing a grounding contact. Any interruption in the protective conductor either inside or outside the device, or the detaching of the protective grounding terminal, renders the device dangerous. Intentional interruption is forbidden.

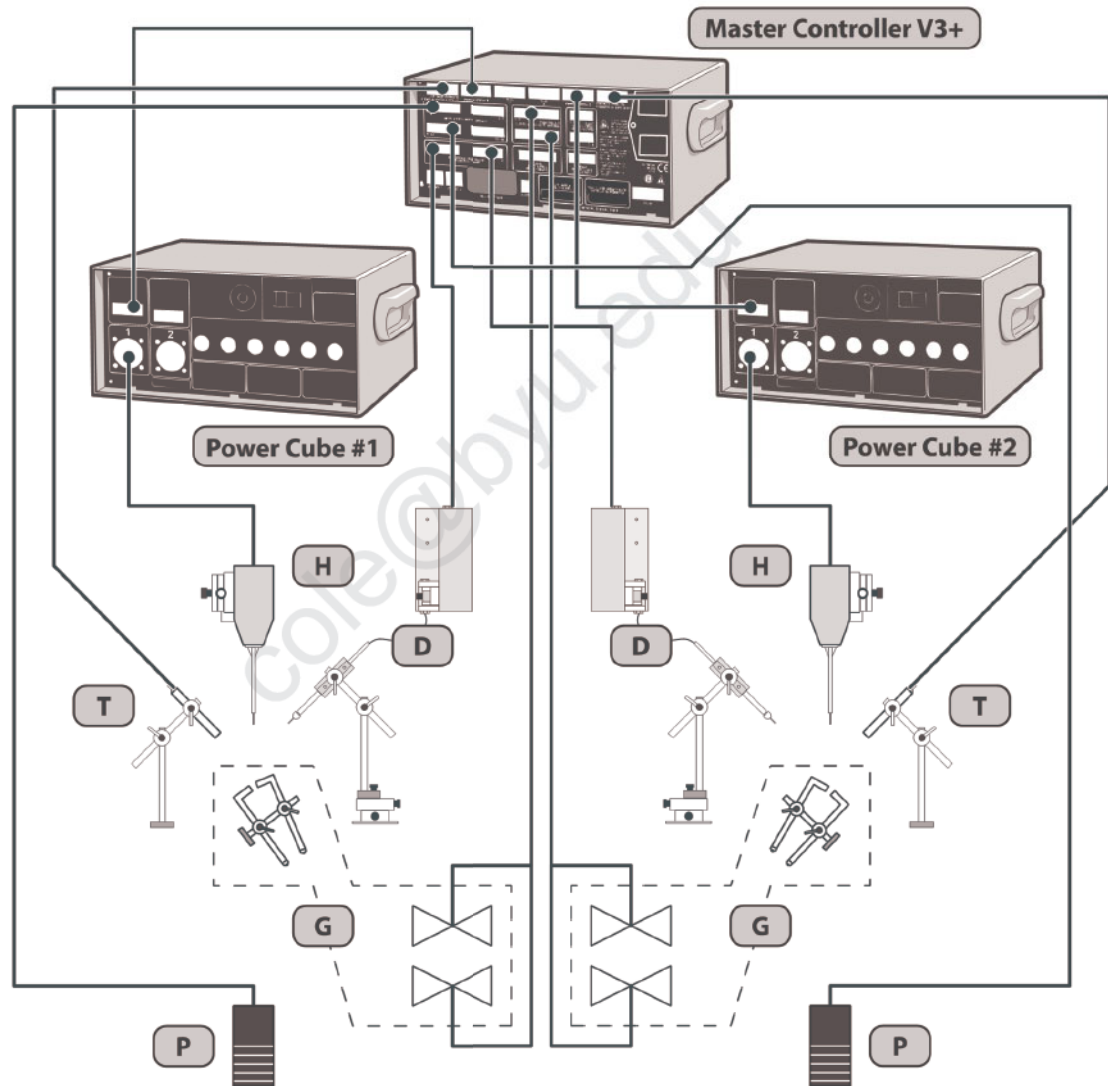
2.3 Functional diagram

The apparatus is designed to be connected to:

- a single **Power Cube** generator for the action of a single heating head;
- a single **Power Cube** generator for the alternating action of two heating heads;
- two **Power Cube** generators for the simultaneous action of two heating heads.

The following is the functional diagram of an application example with the Master Controller V3+ at its maximum expansion:

2.3.1 Application example for two PW3-xx/900 generators and two simultaneous heating heads



*H: Heating head on precision support base; T: Temperature optical sensor;
G: Cooling and anti-oxidizing gas diffusion system; P: Activation switch.
D: Soldering Wire Feeder.*

DESCRIPTION

2.4 Components

2.4.1 Control unit

The device includes a power supply stage and a programmable command section. The apparatus is guided by a microcomputer, which, thanks to its special programming, has the capacity to simultaneously control the entire soldering process of two soldering/heating points (ignition; regulation of the temperature, power provided, period of activity, synchronisation of all the phases, etc.).

The Master Controller monitors the correct functioning of the machine by means of a self-diagnosis device and signals, by display, any necessary interventions on the part of the operator.

On the front panel are:

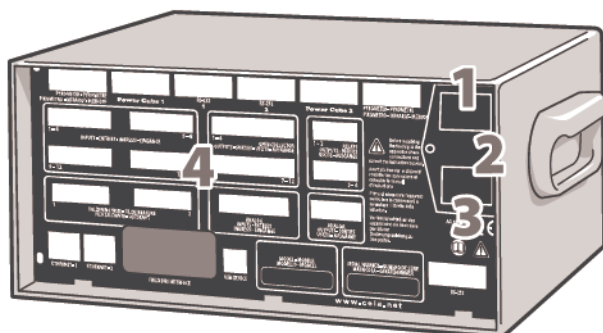


A	Graphic display, indicating the messages referred to the functioning, programming and self-diagnosis of the system.
B	Control keypad composed by the following function keys
C	Flash Card reader/writer to load and save different working parameters sets.
D	USB port (not used).

Key	Function
P	To enter into and exit from the programming phase To move from the submenus to the main menu
	To scroll through the sequence of instructions To choose the parameters to be changed To change the parameter values
STATUS	To check the status of the working parameters of the generator, inputs/outputs, optical pyrometers and of the external sensors.

Key	Function
E	To move from the main menu to the selected submenus To confirm the entered data
	To scroll through the sequence of instructions To choose the parameters to be changed To change the parameter values
QUICK ACCESS	To access the quick menu programmable by the user.

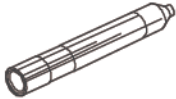
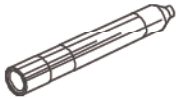
On the back panel of the generator are the following:

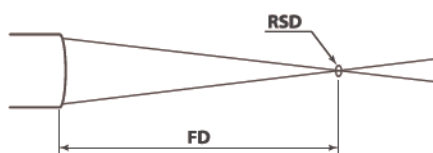
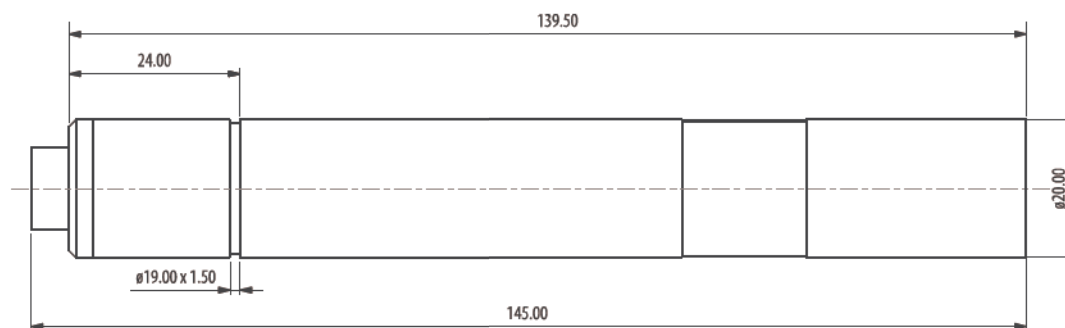


1	Power supply plug
2	Power supply fuse (1A, 250V ; time lag)
3	ON/OFF switch
4	Connections

2.4.2 Optical pyrometers (optional)

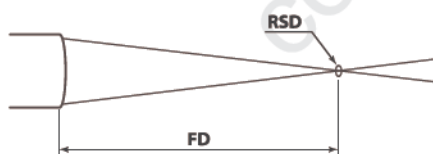
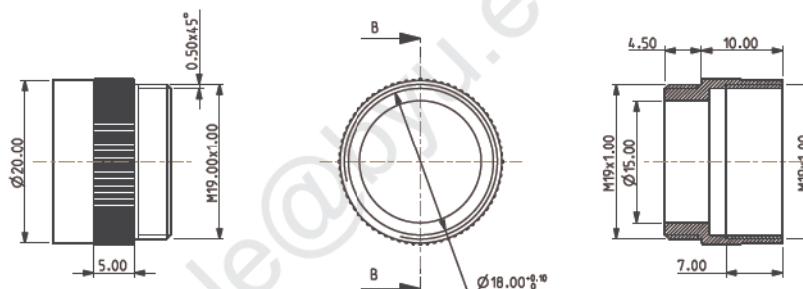
The **Controller** allows connection up to two pyrometers with led pointing, one for each heating point, in order to control the temperature of the piece being worked upon. The sensors are available in different models:

Single spectrum models		
 SH15/SLE	Range of reading from 80° to 700°C	SH15/SLE-550-D1
	Range of reading from 120° to 900°C	SH15/SLE-550-D2
	Range of reading from 200° to 1600°C	SH15/SLE-550-D3
	Range of reading from 500° to 2000°C	SH15/SLE-550-D4
Double spectrum models		
 SH2C/SLE	Range of reading from 300° to 2200°C (from 300° to 600° in single spectrum mode)	SH2C/SLE
		SH2C/SLE-240

DESCRIPTION**2.4.2.1 SH15/SLE**

Model	Reading Spot Diameter RSD	Focus Distance FD
SH15/SLE-D1	12.5 mm	550 mm
SH15/SLE-D2	4.5 mm	
SH15/SLE-D3	2 mm	
SH15/SLE-D4	2 mm	

The SH15/SLE pyrometer can be fitted with one of the CEIA Close-up lens, in order to change the focus distance, as shown in the following table:

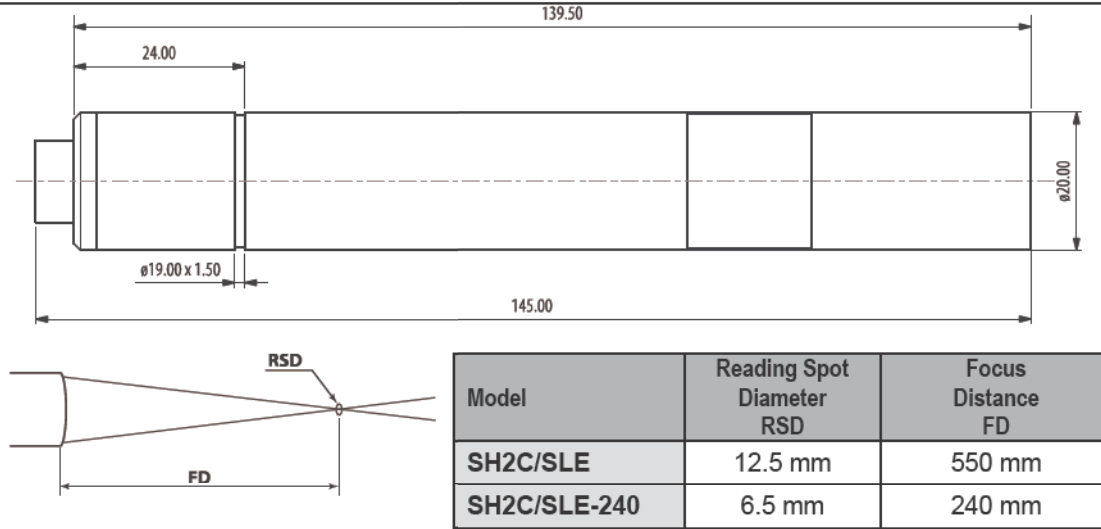


Model	Close-up lens	Reading Spot Diameter RSD	Focus Distance FD
SH15/SLE-D1	CL240/SH15	4.5 mm	240 mm
	CL120/SH15	2.5 mm	120 mm
	CL60/SH15	0.5 mm	60 mm
SH15/SLE-D2	CL240/SH15	1.5 mm	240 mm
	CL120/SH15	1 mm	120 mm
	CL60/SH15	< 0.4 mm	60 mm
SH15/SLE-D3	CL240/SH15	0.6 mm	240 mm
	CL120/SH15	< 0.4 mm	120 mm
SH15/SLE-D4	CL240/SH15	0.6 mm	240 mm
	CL120/SH15	< 0.4 mm	120 mm



When the additional close-up lens is mounted on the pyrometer, the correction factor shall be enabled through the **HEATING > CLOSE-UP LENS PYR.** Command.

2.4.2.1 SH2C/SLE



The electric connection cable, between the pyrometer and the Controller V3+, is available with a length of 1,5, 4 and 6 m (see the Accessory list for order codes).
The minimum bend radius for the cable is 60 mm.

2.5 Safety features

The **Master Controller V3+** is fitted with a list of safety features described below, in order to warrant the proper protection level, during the correct use.

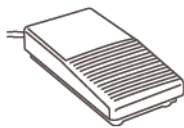
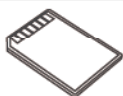
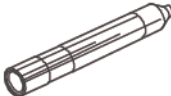
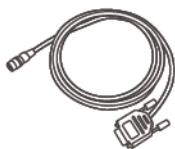
IP Protection degree	IP20
Self-diagnosis system	An internal self-diagnosis system checks constantly the functionality of the system. In case of fault a message is shown on the display of the control panel and of the generator.
Password	Access to parameters and programs of the Controller is protected by passwords. USER and SUPER USER access level are available.

DESCRIPTION

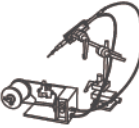
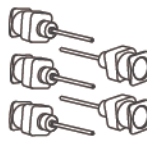
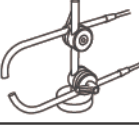
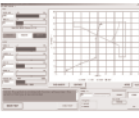
2.6 Options

Option		Order code	Symbol
	Master Controller V3+ Standard version	MASTER-C-V3+	-
	Wire Feeder option Management up to two independent wire feeders (one for each soldering point) <i>Wire feeder not included. See the Accessories table.</i>	V3+/WF	
	Thermal Profile option Thermal Profile management	V3+/TP	
	IXC option Web Server management	V3+/IXC	
	Field Bus interface	V3+/FB-ETHERCAT V3+/FB-ETHERNET/IP V3+/FB-MODBUS-1 V3+/FB-PROFIBUS-1 V3+/FB-PROFINET	
	Tin Soldering option Version optimized for Tin Soldering. <i>Requires the –WF option</i>	V3+/TS	
	DIAS option Management of DIAS thermocamera. <i>Requires the –IXC option</i>	V3+/DIAS	
	Stand-Alone option Management up to two optical sensors and two wire feeders, without the connection with a CEIA generator. <i>Includes –WF and –TS options</i>	V3+/STANDALONE	
	Sensors Only option Management up to two CEIA SLE pyrometers, without the connection with a CEIA generator.	V3+/SO	
	Analog Sensors option Temperature reading by external analog sensors (0-10V, 4-20mA, 0-20mA).	V3+/ANALOG2S	
	Synchronized Sections option Allows the simultaneous activation of two CEIA generators on the same work piece. <i>Requires the –TP option</i>	V3+/SYNSEC	

2.7 Accessories

Accessory		Order code
	On/Off Pedal for the activation of working cycles	2,5 m 3457
		4,0 m 46059
	RS-232 cable (M-F) for the connection between the Controller and the Generator (1 m length)	19623
	RS-232 cable (M-F) for the connection between the Controller and the Generator (3 m length)	23243
	RS232 cable (F-F) for the connection between the Controller and a Computer (3 m length)	50086
	Program Card with 100 memory cells	55265
	Single spectrum optical pyrometer	Range: 80° - 700°C SH15/SLE-550-D1
		Range: 120° - 900°C SH15/SLE-550-D2
		Range: 200° - 1600°C SH15/SLE-550-D3
		Range: 500° - 2000°C SH15/SLE-550-D4
	Double spectrum optical pyrometer	Range: 300° - 2200°C SH2C/SLE
		SH2C/SLE-240
	Cable for the connection between the Controller and the SH15 Optical Pyrometer	1,5m length 49438
		4m length 49439
		6m length 66685
	Cable for the connection between the Controller and the SH2C Optical Pyrometer	1,5m length 63272
		4m length 63273
		6m length 72366
	SH15/SLE Close-up additional lens	Focus 60 mm CL60/SH15
		Focus 120 mm CL120/SH15
		Focus 240 mm CL240/SH15
	90° angle reading accessory	SLE-90D-BD
	Lens cleaning accessory by air purge	SLE-PURGE
	Accessory for the pyrometer cool-down and integrated lens cleaning system	SLE-PURGE-COOL

DESCRIPTION

Accessory		Order code
	SH23 optical sensor base	21871
	ES3M micrometric optical sensor base, complete with swivel	23497
	Additional swivel	21788
--	Insulated thermocouple, type K, for CEIA Controllers	Cable 3m length 56840
		Cable 6m length 58172
	Wire feeder (micrometric support non included)	0,3 to 0,65 mm silver PW3-WF/0.5
		0,7 to 1,1 mm silver PW3-WF/1
		0,8 to 1,2 mm tin PW3-WF/ST1
		1,3 to 1,7 mm tin PW3-WF/S1,5
		Wire presence check PW3-WF/XX-WS
		Additional check of wire feeding length PW3-WF/XX-WLC
		Wire presence and feeding length check PW3-WF/XX-WS-WLC
-	2,5 x 0,2 mm silver strip feeder	PW3-WF/2.5x0.2
	Kit of 5 stainless steel needles (truncated tip)	Inner diameter $\varnothing 0,45$ mm 74917
		Inner diameter $\varnothing 0,6$ mm 74916
		Inner diameter $\varnothing 0,7$ mm 74666
		Inner diameter $\varnothing 0,8$ mm 74915
		Inner diameter $\varnothing 1,0$ mm 74665
		Inner diameter $\varnothing 1,2$ mm 74664
		Inner diameter $\varnothing 1,5$ mm 74920
		Inner diameter $\varnothing 1,8$ mm 74698
	Anti-oxidizing gas diffusion system	SG100
	Cooling and Anti-oxidizing gas diffusion system	SG101
	Kit for the Temperature inputs calibration verification	66736
	PowerCube Assistant Free Software providing useful advice for coil optimisation with series 50, 200 and 900 CEIA generators.	Downloadable from the reserved area at www.ceia.net upon registration

2.8 Technical characteristics

Heating System Set-up	Alternating between the two heads (if connected to a single Power Cube generator)	
	Simultaneous on two heads (if connected to two Power Cube generators)	
	Capable of driving Power Cube generators in continuous operation	
Management and Control	Activation of cycle: via external contact, RS-232 or Field Bus interface (with -FB option)	
	Control and Time Programming of 2 antioxidant gas diffusers, 2 heating heads and 2 wire dispensers	
	Available settings: heating power; soldering temperature (resolution 1/10°C until 1000°C, then 1°C); wire quantity and speed; antioxidant gas supply times; maximum solder force; wire feeder; piston advance delay	
	Control of the temperature of the article being processed: via optical temperature sensors, thermocouple or external sensors	
Predefined programs	SH15/SLE sample rate <i>0.5 milliseconds</i>	
	Manual	
	A TIMER A	Fully time-programmable cycle phases with controlled-temperature holding time
	B TIMER B	Fully time-programmable cycle phases with programmable heating time and optional temperat.control
	C TIMER C	Fully time-programmable cycle phases, heating programmable with two times and two power levels
	S SYNCHRONISED	As TIMER A with holding time synchronised by an external event
	AUT FULL AUTO	as TIMER A, with management of the automatic solder wire feeder
	A2W FULL AUTO 2 WIRES	as FULL AUTO, with the possibility of applying two independent solder wire feeders at the same soldering point (available only with -TS option):
	TP THERMAL PROFILE	functioning with thermal profile (available only with -TP option)
	T2W THERMAL PROFILE 2 WIRES	as THERMAL PROFILE, with the possibility of applying two independent solder wire feeders at the same soldering point (available only with -TS and -TP option):
Control inputs	2 inputs for the connection of CEIA optical temperature sensors	
	2 insulated digital inputs for connecting the cycle activation pedals	
	2 insulated digital inputs for halting the operating cycle	
	4 0-10V (or 0-20 mA, or 4-20mA) inputs (2 used for power control and 2 for temperature control)	
	2 insulated inputs (RS232) for connection to external programmable units	
	1 plug for a Field Bus module (accessory)	
	2 insulated digital inputs for external flow meter	
	2 insulated digital synchronization inputs for temperature holding timer	
	2 solder wire dispenser encoder inputs	
	2 "wire present" inputs	
Outputs for activation of external slave devices	6 insulated auxiliary digital inputs	
	2 serial outputs for connecting and managing two independent Power Cube generators	
	2 outputs for activation of antioxidant gas diffusers	
	2 outputs for "Generator ON" signals	open collector,
	2 outputs for "Temperature reached" signals	current carrying
	2 outputs for activation of rapid cooling gas diffusers	capacity 12/24Vcc;
	2 outputs for "End of cycle" signals	400mA
	2 output for system ready signal (relay output)	
	2 0-10V (or 0-20 mA, or 4-20mA) outputs for temperature reading	
	2 0-10V (or 0-20 mA, or 4-20mA) outputs for power reading	
Internal memory	2 solder wire dispenser driver outputs	
	2 auxiliaries open collector outputs	
Removable data flash card	100 working parameter sets can be stored	
	100 different storable thermal processes made up of up to 20 segments per process (TP versions)	
Removable data flash card	100 sets of working parameters storable on each card	
	100 different storable thermal processes made up of up to 20 segments per process (TP versions)	

DESCRIPTION

Self-Diagnosis	Check of temperature and of the cooling water presence	Correct inductor dimensioning check
	Internal malfunction	Power Cube missed connection
	Supply voltage too high	Supply voltage too low
	Generators parameter screenshot	Phase missing (if connected to 90, 180, 360 and 720 PowerCube generators)
	Working cycle malfunction	
	Reading/writing malfunction of internal memory cells	Programming access control through a password
	Reading/writing malfunction of Program Card memory cells	
Power supply	Inductor short circuit	Soldering wire presence
	Supply voltage	230Vac $\pm 10\%$ monophas - 50/60 Hz
	Max Power Absorption	60 W
Operating conditions	Working temperature	+5° ÷ +55°C
	Storage temperature	-25° ÷ +70°C
	Relative humidity	0-95% (without condensation)
Container	Stainless steel construction	
	Dimensions	275 x 265 x 140 (WxDxH) mm
	Weight	6 kg
Safety features	Insulation from the mains voltage;	
	Low operating voltage: no risk for the operator	
	Complies with international standards currently applicable for Electrical Safety (EN 60204-1) and Electromagnetic Compatibility (EN 61000-6-2, EN 61000-6-4)	

3 INSTALLATION



Before starting any installation operation, read carefully the **Safety instructions - Warnings** section of this manual.

3.1 Unpacking

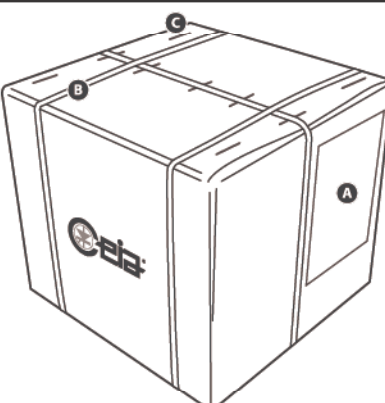
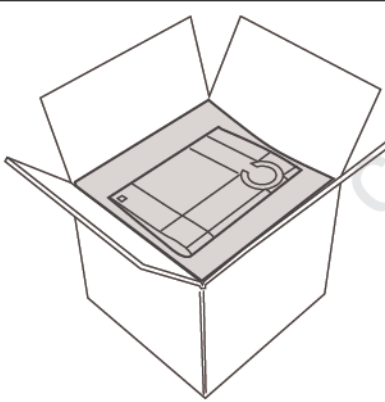
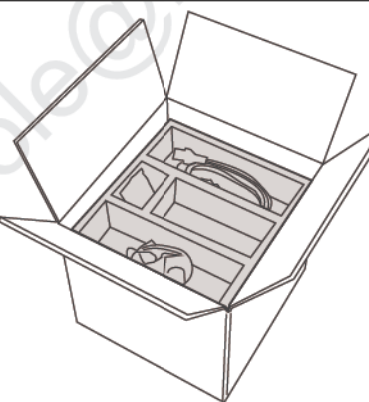
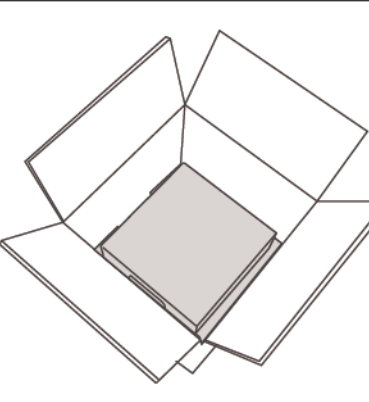
3.1.1 Packing transportation

The Master Controller V3+ is a small package and its weight is 8.5 kg.

3.1.2 Unpacking procedure



Conserve the package for possible future uses.

	Read the Packing List (A) to check the packing content.	
		Using pliers, extract all metallic clips (B) from the upper surface of the packing. Throw them away.
 Open the box and extract the Instruction Manual and its underlying support	 Extract the support with the supply cable, the serial cable and other contents from the box	 Extract the Power Controller from the box, holding it from its handlings

3.1.3 Content List

- The packing consist of:
- Master Controller V3+
 - Power supply cable
 - Serial connection cables
 - Instruction Manual

3.2 Mechanical Installation

3.2.1 Minimum requirements

After installation the device should be stable, and not subject to vibration or accidental movement.

3.2.2 Assembly

Any mechanical assembly procedure is request,

3.2.3 Movement

The device is lightweight and two comfortable handlings are provided for its movement and transportation.

3.2.4 Mechanical installation

Position and fasten the soldering workstation components according to the processing required in order to allow installation, bearing in mind:

- the length of the connecting cables;
- the electrical and hydraulic requirements;
- the modes of fastening.
- the accessibility of the commands

3.2.5 Positioning of the optical pyrometer

3.2.5.1 SH15/SLE sensors



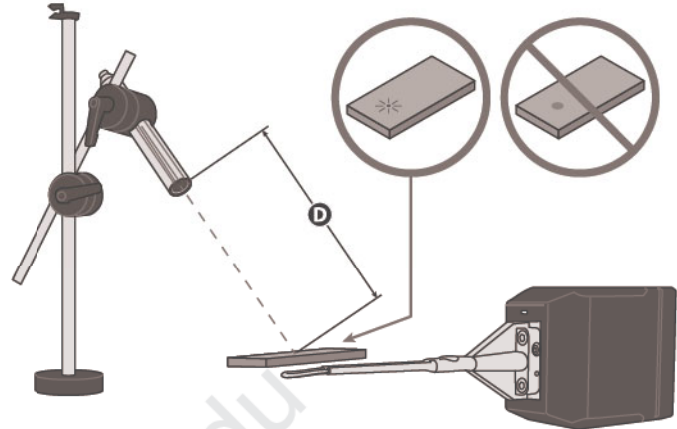
During normal use, it is advisable to deactivate the led beam by means of an appropriate command (see Programming chapter).

For the best temperature reading with pieces of substantial size the pyrometer should be pointed at the zone which, during the heating, reaches the maximum temperature.

Adjust the distance (**D**) between the pyrometer and the piece to be heated so that the luminous aiming beam given out by the pyrometer is focused on the piece.

This distance is indicated on paragraph 2.4.2.

The optimum aiming is obtained when the illuminated area shows its smallest size.



3.2.5.2 UPP® protocol sensors

To install and position this models, please refers to the their own instruction booklet.



The power supply of UPP®Protocol sensors shall be connected to terminals J7, J8 or J12, +V and GND contacts, after setting +V to 24V.



WARNING! The maximum total power on +24V output is 30W.

3.2.5.1 DIAS thermal imaging camera

On models fitted with the IXC card, it is possible to connect and use a DIAS thermal imaging camera, connected via Ethernet.

To install and position this models, please refers to the their own instruction booklet.



During the camera settings, performed with an external PC connected to the same network , ensure that parameters TEMP SENSOR TYPE 1 and 2 have a value other than THERMAL CAMERA.

Define the ROI points (Region Of Interest) on the camera, following the camera itself instructions. ROI 1 will be associated to the temperature sensor 1 and ROI2 will be associated to the eventual temperature sensor 2.

Once ROIs are set, modify the TEMP SENSOR TYPE 1 and 2 values to THERMAL CAMERA.

The DIAS thermal cameras have an automatic **shutter** function, that periodically closes the camera lens, in order to calibrate the camera without interferences.

To avoid this function to interfere with the heating cycle, especially on short ones, it is necessary to disable it directly on the thermal camera (see related documents) and to set **DIAS SHUTTER T1** and **DIAS SHUTTER T2** parameters on values higher than 0, so as the Controller will manually manage the shutter to start outside the heating cycles.

INSTALLATION

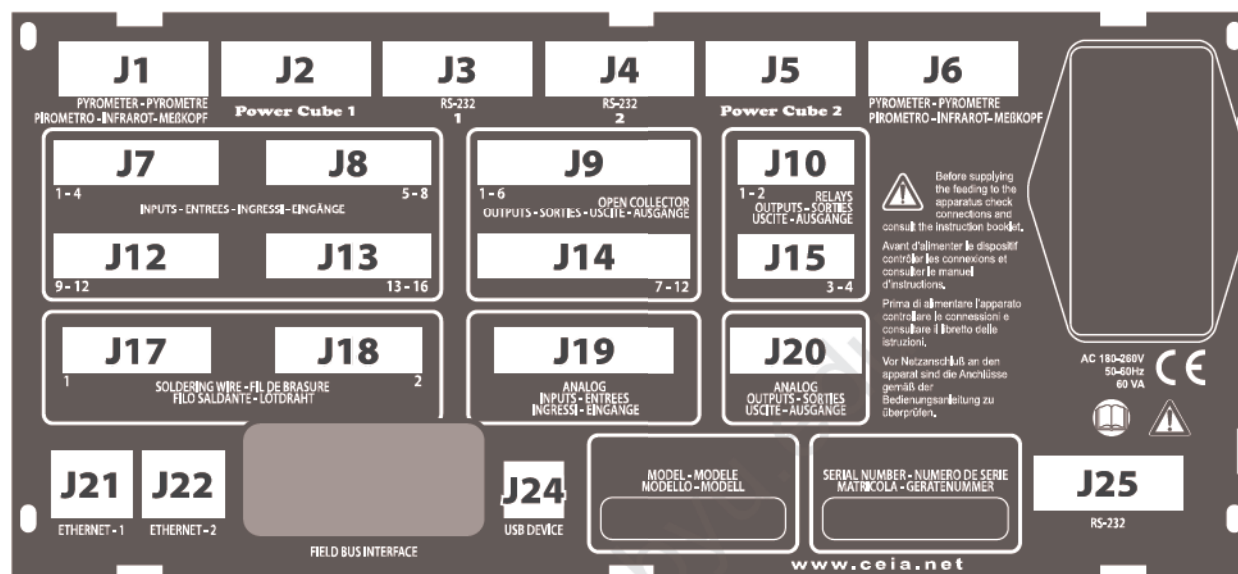
3.3 Electrical installation

3.3.1 Minimum requirements

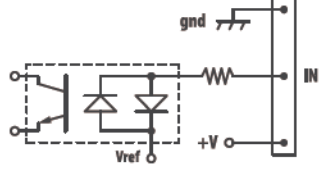
Verify that the voltage from the power mains corresponds to the voltage shown on the device's electrical specifications plaque.

3.3.2 Connection between all system components

On the figure below, the back panel of the Master Controller is shown, with the connections available, needed for the complete installation of the soldering workstation.

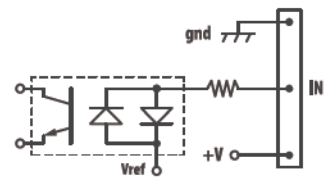
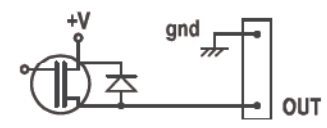


3.3.3 Input Output connections

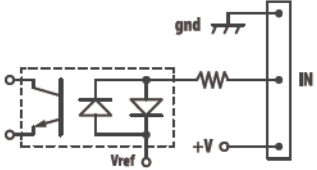
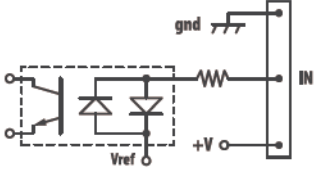
Conn.	#	Label	In/Out	Type	Value range	Function
J1	1	Gnd	Input	Sub-D9	-	Pyrometer 1 Connection with a CEIA pyrometer /SLE through its own cable or to a thermocouple type K, S or C through the contacts 8 (IN1+) and 5 (IN1-).
	2	+15V				
	3	-15V				
	4	Data				
	5	IN1-				
	6	IN2-				
	7	IN2+				
	8	IN1+				
	9	Laser ON				
J2	2	RS232 IN	In/Out	Sub-D9	-	PowerCube 1 Connection with a CEIA PowerCube generator
	3	RS232 OUT				
	5	Gnd				
J3	2	RS232 IN	In/Out	Sub-D9	-	Connection to Serial Interface 1 (auxiliary) or to an UPP® protocol optical sensor for the Soldering Point 1
	3	RS232 OUT				
	5	Gnd				
J4	2	RS232 IN	In/Out	Sub-D9	-	Connection to Serial Interface 2 (for external controls) or to an UPP® protocol optical sensor for the Soldering Point 2
	3	RS232 OUT				
	5	Gnd				
J5	2	RS232 IN	In/Out	Sub-D9	-	PowerCube 2 Connection with a second CEIA PowerCube generator
	3	RS232 OUT				
	5	Gnd				
J6	1	Gnd	Input	Sub-D9	-	Pyrometer 2 Connection with a second CEIA pyrometer /SLE through its own cable or to a thermocouple type K, S or C through the contacts 8 (IN1+) and 5 (IN1-).
	2	+15V				
	3	-15V				
	4	Data				
	5	IN1-				
	6	IN2-				
	7	IN2+				
	8	IN1+				
	9	Laser ON				
J7	1	Gnd	-	-	Debounce time: 2ms	
	2	Activation 1	Input	-		
	3	+V	-	-	12 or 24V (programmable)	
	4	Max.Temp.Acq.1	Input	-		
	5	Gnd	-	-		
	6	Reset 1	Input	-		
	7	+V	-	-	12 or 24V (programmable)	
	8	Thermoreg. 1	Input	-	Debounce time: 2ms (Only on TS)	
	9	Gnd	-	-		

Reference voltage Vref is programmable as **P+N** (default value), **P** or **N**.
See following table for the external possible connections.

INSTALLATION

Conn.	#	Label	In/Out	Type	Value range	Function
J8	1	Gnd	-	-		 Reference voltage Vref is programmable as P+N (default value), P or N. See following table for the external possible connections.
	2	Synchr 1	Input	-		
	3	+V	-	-	12 or 24V (programmable)	
	4	Synchr 2	Input	-		
	5	Gnd	-	-		
	6	Ext. Fluxometer 1	Input	-		
	7	+V	-	-	12 or 24V (programmable)	
	8	Ext. Fluxometer 2	Input	-		
	9	Gnd	-	-		
J9	1	Gnd	Output	Open Collector	I _{max} 0,4 A for each output Maximum power on all outputs: 30W	 +V: 12 or 24V (programmable)
	2	Anti-Oxidizing Gas 1				
	3	Gnd				
	4	Cooling Gas 1				
	5	Gnd				
	6	Anti-Oxidizing Injection 1 ¹				
		Wire movem. 1 ¹				
		Temp. below min. 1 ²				
	7	Gnd				
	8	End Cycle 1				
	9	Gnd				
	10	Generator ON 1				
	11	Gnd				
	12	Piece in temperature 1				
		Temperat. over max. 1 ³				

1. If **WIRE > Out J9-6 J14-6** is set to **INJECTION**, the output drives the injection of anti-oxidizing liquid, if set to **WIRE MOV. SIGNAL**, the output indicates the movement of the soldering wire feeder.
2. If **MODE > OPERATING MODE** is set to **TIMER (A)**, **TIMER (B)** or **SYNCHRONISED**, and **TEMPERATURE CONTROL** is set to one of the "in-window" values, the output meaning is "Temperature below the minimum value".
3. If **MODE > OPERATING MODE** is set to **TIMER (A)**, **TIMER (B)** or **SYNCHRONISED**, and **TEMPERATURE CONTROL** is set to one of the "in-window" values, the output meaning is "Temperature over the maximum value".

Conn.	#	Label	In/Out	Type	Value range	Function
J10	1	N.C.	Output	N.C.	Vmax 48 Vdc Imax 0,5 A	Relay 1 : Section 1 Ready
	2	Common		Common		
	3	N.O.		N.O.		
	4	N.C.	Output	N.C.	Vmax 48 Vdc Imax 0,5 A	Relay 2 : Section 2 Ready
	5	Common		Common		
	6	N.O.		N.O.		
J12	1	Gnd	-	-	Debounce time: 2ms	
	2	Activation 2	Input	-		
	3	+V	-	-	12 or 24V (programmable)	
	4	Max.Temp.Acq.2	Input	-		Reference voltage Vref is programmable as P+N (default value), P or N. See following table for the external possible connections.
	5	Gnd	-	-		
	6	Reset 2	Input	-		
	7	+V	-	-	12 or 24V (programmable)	
	8	Thermoreg. 2	Input	-	Debounce time: 2ms (Only on TS)	
	9	Gnd	-	-		
J13	1	Gnd	-	-		
	2	Memory Cell Load Strobe	Input	-		
	3	+V	-	-	12 or 24V (programmable)	
	4	Memory Cell Select Bit 1	Input	-		Reference voltage Vref is programmable as P+N (default value), P or N. See following table for the external possible connections.
	5	Gnd	-	-		
	6	Memory Cell Select Bit 2	Input	-		
	7	+V	-	-	12 or 24V (programmable)	
	8	Memory Cell Select Bit 3	Input	-		
	9	Gnd	-	-		

INSTALLATION

Conn.	#	Label	In/Out	Type	Value range	Function
J14	1	Gnd	Output	Open Collector	I _{max} 0,4 A for each output Maximum power on all outputs: 30W	 +V: 12 or 24V (programmable)
	2	Anti-Oxidizing Gas 2				
	3	Gnd				
	4	Cooling Gas 2				
	5	Gnd				
	6	Anti-Oxidizing Injection 2 ¹				
		Wire movem. 2 ¹				
		Temp.below min. 2 ²				
	7	Gnd				
	8	End Cycle 2				
	9	Gnd				
	10	Generator ON 2				
12		Piece in temperature 2				
		Temperat.over max. 2 ³				

- If **WIRE > Out J9-6 J14-6** is set to **INJECTION**, the output drives the injection of anti-oxidizing liquid, if set to **WIRE MOV.SIGNAL**, the output indicates the movement of the soldering wire feeder.
- If **MODE > OPERATING MODE** is set to **TIMER (A)**, **TIMER (B)** or **SYNCHRONISED**, and **TEMPERATURE CONTROL** is set to one of the "in-window" values, the output meaning is "Temperature below the minimum value".
- If **MODE > OPERATING MODE** is set to **TIMER (A)**, **TIMER (B)** or **SYNCHRONISED**, and **TEMPERATURE CONTROL** is set to one of the "in-window" values, the output meaning is "Temperature over the maximum value".

J15	1	N.C.	Output	N.C.	V _{max} 48 Vdc I _{max} 0,5 A	RELAY 3 AUXILIARY or TEMPERATURE OUT OF TOLERANCE 1*
	2	COMMON		COMMON		
	3	N.O.		N.O.		
	4	N.C.	Output	N.C.	V _{max} 48 Vdc I _{max} 0,5 A	RELAY 4 AUXILIARY or TEMPERATURE OUT OF TOLERANCE 2*
	5	COMMON		COMMON		
	6	N.O.		N.O.		
J17	1	+5V	Output	+5V		Soldering wire feeder 1
	2	Encoder IN	Input	Encoder		
	3	Gnd		Gnd		
	4	Motor -	Output	Motor -		
	5	Motor +		Motor +		
	6	Piston -		Piston -		
	7	Piston +		Piston +		
	8	Wire	Input	Wire		

* Only on Master Controller V3+-TP models

Conn.	#	Label	In/Out	Type	Value range	Function
J18	1	+5V	Output	+5V		Soldering wire feeder 2
	2	Encoder IN	Input	Encoder		
	3	Gnd		Gnd		
	4	Motor -	Output	Motor -		
	5	Motor +		Motor +		
	6	Piston -		Piston -		
	7	Piston +		Piston +		
	8	Wire	Input	Wire		
J19	1	Gnd	Input	-	When 0-10 V mode is selected: Zin: 100 kohm Vmax: ± 50 Vdc When 0-20 mA or 4-20 mA mode are selected: Vmax: 14 Vdc	Analogic inputs : 0 ÷ 10 V OR Analogic inputs : 0 ÷ 20 mA OR Analogic inputs : 4 ÷ 20 mA
	2	Gnd				
	3	Power Adj. 1 Pyrometer 1 IN ¹				
	4	Gnd				
	5	Power Adj. 2 Pyrometer 2 IN ¹				
	6	Gnd				
	7	Temp. reg. 1				
	8	Gnd				
	9	Temp .reg. 2				
	10	Gnd				
J20	1	Temperature indication 1	Output	-	When 0-10 V mode is selected: Zout: 500 ohm Vout: 0 ÷ 10 Vdc When 0-20 mA or 4-20 mA modes are selected: Vout max: 14 Vdc	Analogic outputs : 0 ÷ 10 V OR Analogic outputs : 0 ÷ 20 mA OR Analogic outputs : 4 ÷ 20 mA
	2	Gnd				
	3	Temperature indication 2				
	4	Gnd				
	5	Power indication 1				
	6	Gnd				
	7	Power indication 2				
J21	-	-	In/Out	Ethernet	-	Data logger (only with -X option)
J22	-	-	In/Out	Ethernet	-	DIAS Thermo camera (only with -X option)
J24-J25		Reserved for internal use.				

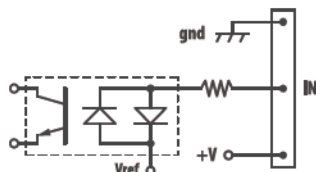
¹ With -PYAN option

3.3.3.1 Possible external connections on J7, J8, J12 and J13

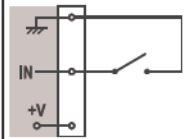
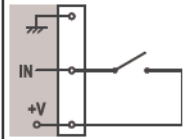
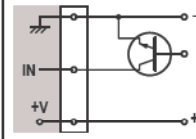
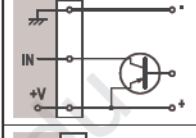
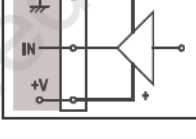
The voltage **+V** of the 16 digital inputs of **Master Controller V3+** can be programmed to 12 or to 24 Vdc according to different sensor types.

More than this, the reference voltage **Vref** can be programmed to three different values **P+N** (default value), **P** and **N**.

According to this parameters, the possible external connections are described here beside:



The programming of the voltage **+V** affects not only the 16 digital inputs but also the 12 open collector outputs of **Master Controller V3+**.

External connection	Vref		
	P+N	N	P
	YES	YES	NO
	YES	NO	YES
	YES	YES	NO
	YES	NO	YES
	NO	YES	YES

3.3.4 Inputs description

Activation 1 and 2

Input for the activation of the work cycle on the corresponding workstation. The debounce time for this input is programmable from 2 to 20 ms (default value is 2 ms).

Manual Activation 1 and 2

When one of this input is activated, in the corresponding workstation the programmed timing for the heating are excluded (TIMER A functioning with $t_a = 0$). The corresponding heating last for the time the activation switch comes held pressed.

Reset 1 and 2

Immediate interruption of the work cycle on the corresponding workstation. If the workstation is in the heating phase, the cooling phase is also executed, using the programmed time.

The activation of Reset input is indicated by the display, as shown in the example beside.

1> * RESET *

2> T= 260C S=====L

TIME=2.3s POW.=75%

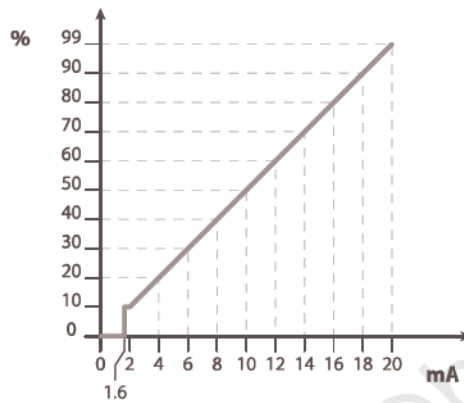
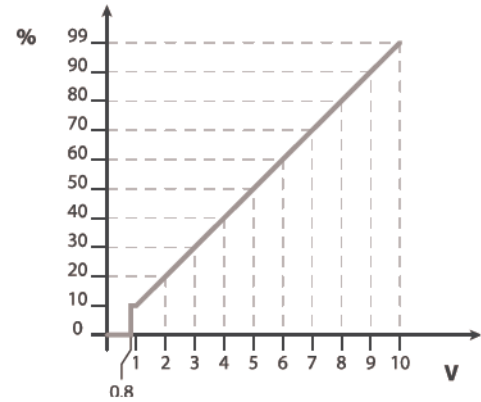
Power adjustment 1 and 2 (analogic inputs)

The power is adjustable referring to analogic inputs, programmable as voltage (0-10V) or current (0-20mA or 4-20mA).

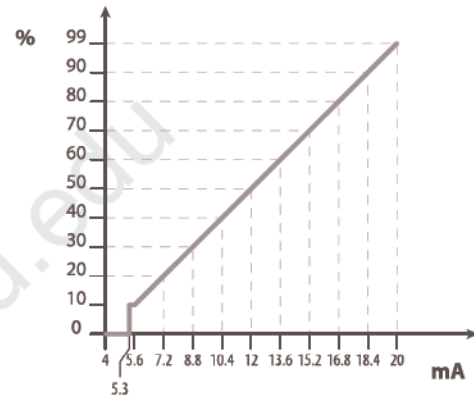
The graph shown beside represent how the power will be set depending on the voltage of this input.

The two graphs below represent how the power will be set depending on the current of this input.

Analog input = 0-10V



Analog input = 0-20mA



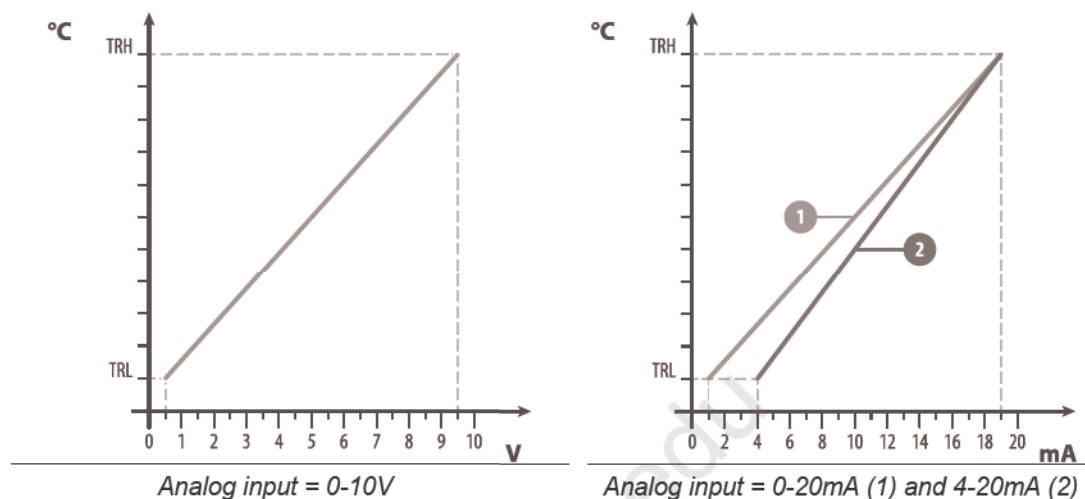
Analog input = 4-20mA

INSTALLATION

Temperature regulation 1 and 2

Like power 1, the temperatures 1 and 2 can be regulated with reference to the analog inputs (Temp. reg. 1 and Temp. reg. 2 on connector J19); these can be programmed in terms of voltage (0V-10V) or in terms of current (0-20mA or 4-20mA).

The graphics and the table shown here below, shows the Temperature settings according to this input.



Temperature sensor	TRL	TRH
SH15/SLE-D1	80	700
SH15/SLE-D2	120	900
SH15/SLE-D3	200	1600
SH15/SLE-D4	500	2000
SH2C/SLE	300	2200
Thermocouple K	50	1300
Thermocouple S	50	1700
Thermocouple C	50	2320

In order to select Temperature control from the analogic input, it is necessary to set the **HEATING > TEMP SENSOR TYPE 1/2** parameter on the **ANALOG** value.

In order to define the Temperature reading range, set the **HEATING > AN.TEMP.MIN 1/2** and **HEATING > AN.TEMP.MAX 1/2** values, corresponding to the voltage or current range limits.



WARNING! When the Temperature reading is set from the analog input, check that the **MODE > AN.INP.NOISE REJ.** parameter is set to **NO**.

External thermoregulator

Input provided for the connection of an external ON/OFF thermoregulator.

Once the temperature selected has been reached, the sensor has to activate the input, allowing the apparatus to complete the cycle. The functioning with external temperature sensor is indicated by the display with the message **"*EX*"**, during the phase of temperature maintenance of the head.

The debounce time for this input is 0.2 seconds.

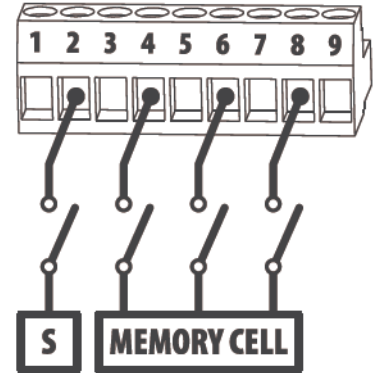
Memory cell selection through binary inputs



This function is available only if BINARY INP.CELL SEL = ENABLED.

This function allows to use the 4 binary inputs of connector J13 to load one of the first 8 memory cells. When pin 2 is closed, the Controller reads the status of pins 4, 6 and 8, and loads data from the relative memory cell.

Pin status			Memory cell to load
4	6	8	
OPEN	OPEN	OPEN	1
CLOSED	OPEN	OPEN	2
OPEN	CLOSED	OPEN	3
CLOSED	CLOSED	OPEN	4
OPEN	OPEN	CLOSED	5
CLOSED	OPEN	CLOSED	6
OPEN	CLOSED	CLOSED	7
CLOSED	CLOSED	CLOSED	8



This operation will be performed only if both work cycles are idle and the indicated memory cell contains valid data.

If the operation is carried out correctly, the Controller sends a pulsed signal to the **End Cycle 1** output (J9 pin 8). The signal consists of a number of pulses equal to the memory cell number just loaded. Each pulse last for 20ms and there are 20ms between each pulse.

3.3.5 Outputs description

Anti-Oxidizing Gas 1 and 2

Output for the activation of gas diffusers, through electric valves (12/24Vcc - 300mA max.).



WARNING! Use only electric valves at 12V or 24V, 3W max.

Cooling Gas 1 and 2

At the end of the soldering cycle there is a period of time in which the piece cools. During this period, the Master Controller closes a relay that can activate an additional gas diffusion circuit; this serves to speed up the cooling time of the piece.



WARNING! Use only electric valves at 12V or 24V, 3W max.

Anti Oxidizing Liquid 1 and 2

Port provided for the connection of a system of liquid injection into the nozzle of the wire distributor. Open collector output, closed and earthed during liquid injection.

Movement of the soldering wire feeder 1 and 2

Port provided for the signalization of the soldering wire feeder movement. Open collector output, closed and earthed during motor movement.

Temperature below the minimum value 1 and 2

Port provided for the signalization of a temperature below the window minimum value. Open collector output, closed and earthed during the detected temperature below the minimum.



WARNING! Anti Oxidizing Liquid, Movement of Soldering Wire Feeder and Temperature below the minimum value, uses the same terminals.

To determine which of the first two is in use, is the parameter *OUT J9-6 J14-6* described on paragraph 4.6.7 of this document. The third one is used only in case the OPERATING MODE is set to *TIMER (A)*, *TIMER (B)* or *SYNCHRONISED*, and *TEMPERATURE CONTROL* is set to one of the "in-window" values.

Temperature out of tolerance

This is an output used when the Operating Mode "THERMAL PROFILE" is selected. In case of a good heating cycle the output remains idle; in case of a non-conforming heating cycle, the output is activated immediately or with a pulse of 200ms at the end of the cycle. See parameters **T.OUT TOLER:SIGNAL** and **T.OUT TOLER:TIME**.

End cycle 1 and 2

The Master Controller send, at the end of each soldering cycle (cycle ended without interruptions), a signal (0,2 sec.). In case the controller is receiving a start cycle signal during this 0.2 seconds, the end cycle signal is interrupted.

Generator ON 1 and 2

The Master Controller gives a signal for the whole time that the PowerCube generator is ON, through the activation of the open collector output.

Piece in temperature 1 and 2

The Master Controller gives a signal for the whole time that the piece being worked has the programmed temperature, through the activation of the open collector output. This signal isn't used in THERMAL PROFILE mode.

Temperature over the maximum value 1 and 2

Port provided for the signalization of a temperature over the window maximum value.
Open collector output, closed and earthed during the detected temperature over the maximum.



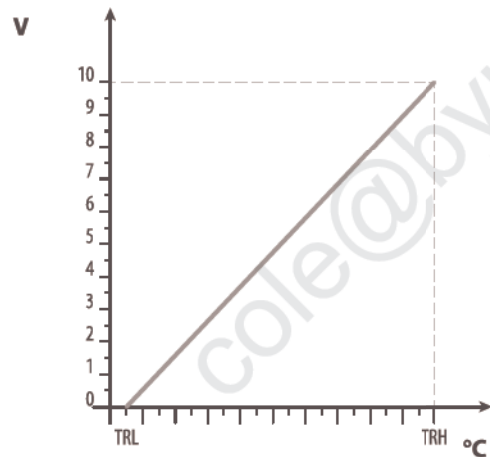
WARNING! Piece in temperature and **Temperature over the maximum value**, uses the same terminals.
The second one is used only in case the OPERATING MODE is set to TIMER (A), TIMER (B) or SYNCHRONISED, and TEMPERATURE CONTROL is set to one of the "in-window" values.

Ready Relay 1 and 2

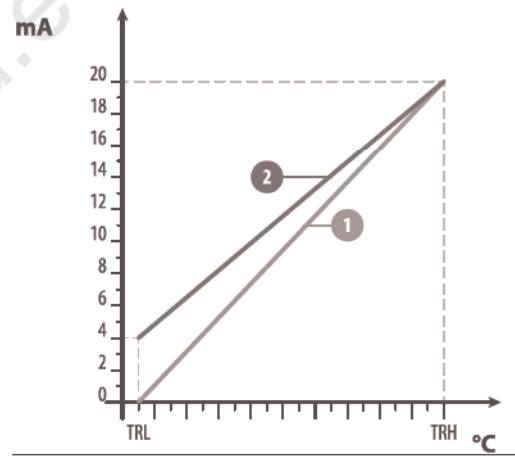
The Master Controller gives a signal if ready to start a new soldering, through the activation of a relay (30Vca/cc; 750mA max.). (This signal isn't used with PowerCube 180 and 360 generators with double head).

Temperature indication 1 and 2 (analogic outputs)

The analogic output can be programmed to indicate the temperature with voltage (0-10V) or with current (0-20mA or 4-20mA).



Analog output = 0-10V



Analog output = 0-20mA (1) and 4-20mA (2)

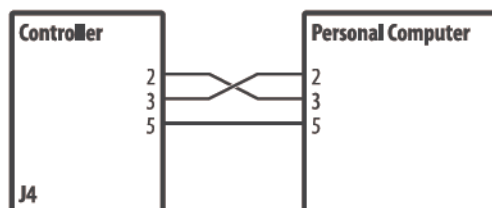
Temperature sensor	TRL	TRH
SH15/SLE-D1	80	700
SH15/SLE-D2	120	900
SH15/SLE-D3	200	1600
SH15/SLE-D4	500	2000
SH2C/SLE	300	2200
Thermocouple K	50	1300
Thermocouple S	50	1700
Thermocouple C	50	2320
UPP sensor*	TRL	TRH

* Depending on sensor characteristics

3.3.6 Serial line connection

Connect to the RS232 port on the Controller through the serial cable (included), with a terminal configured to the following line parameters

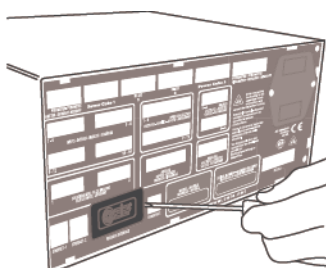
- Baud rate: 9600 (or 57600, according to BR parameter)
- 8 Data bit
- No parity
- Stop bit: 1
- Flow Control: none



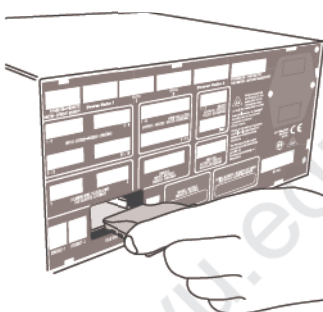
3.3.7 Installation of a Field Bus module



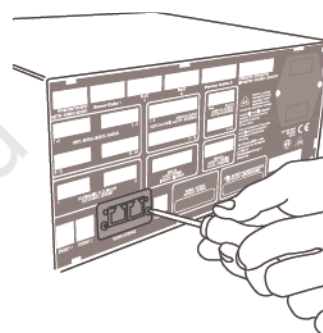
Before installing the module, ensure that the Controller has been switched off.



With a 2mm Allen wrench, remove the plastic cover from the Field Bus slot.



Insert the Field Bus module on the internal guides and push it until the end.



Lock the module in place, using a T9 Torx.

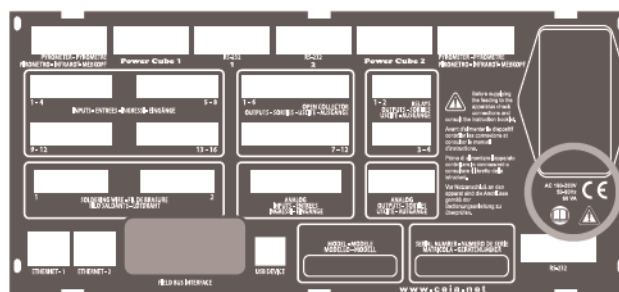
3.3.8 Power supply connection

Supply voltage indication

The supply voltage is shown on the back panel of the Controller, as described in the figure beside.

Connection

Connect the Controller to the ground and to the voltage specified in the back panel, through its own power supply cable (included).



The external line shall have conductors with a section of minimum 1 mm².
Power all the devices connected to the Controller following their own requirements.



WARNING! The protective and neutral conductors must be common to those of the CEIA generator.

3.4 Timing

3.4.1 Operating modes

The Master Controller allows different kinds of functioning selectable with an appropriate programming of the system.

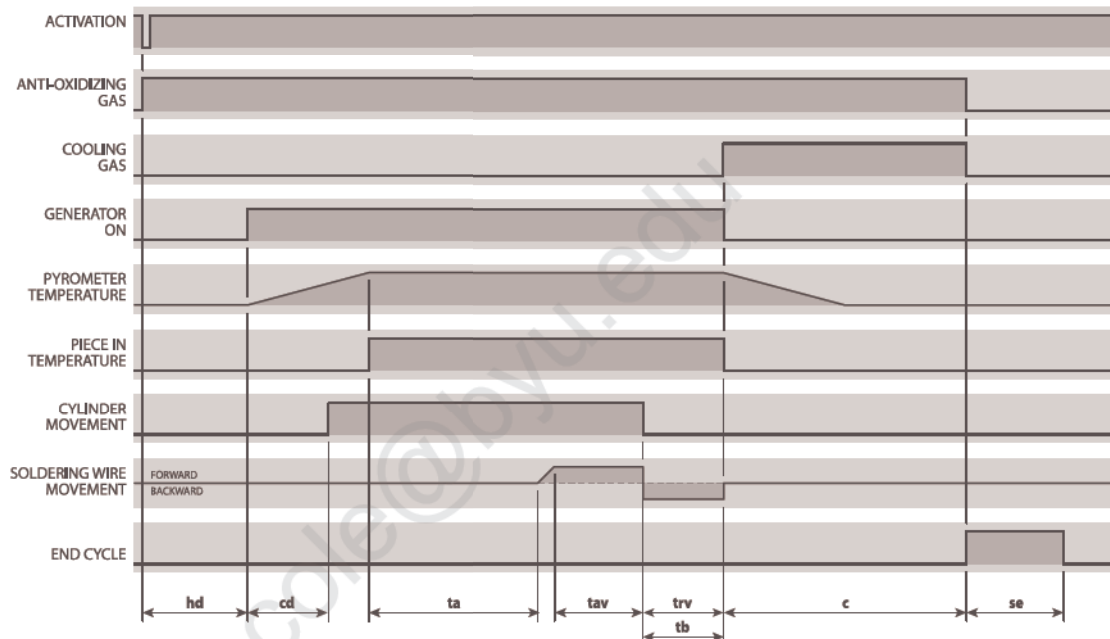


The work cycles described below are simplified and are determined by an ensemble of programming parameters that may be defined by the user depending on the type of processing desired.

3.4.2 Fully automatic **AUTO**

With **WF** option

A complete work cycle of the soldering station managed by the Master Controller V3+, with FULL AUTO functioning, consists of the following phases:



- activation of the on/off pedal;
- immediate activation of the circulation of anti-oxidizing inert gas;
- activation of the heating of the piece under processing (after a time period **hd** determined by the programming parameter "Heating delay 1[2]")
- nearing of the soldering wire to the piece under processing (after **cd** time given by the programming parameter "Cylinder delay 1[2]")
- once the desired temperature of the piece being under processing has been reached, maintenance of that temperature is performed. During this time the output "Piece in temperature" is activated (see paragraph 3.3.4).
- waiting for a time period **ta** ("Holding time A 1[2]" parameter), before feeding the soldering wire
- advancing of the soldering wire and placing on the piece under processing (wire length and speed depending on parameters "Feed 1[2]" and "Feed rate 1[2]" and generating the time **tav**);
- rewind of soldering wire for a time **trv** depending from parameters "REWIND LENGTH 1[2]" e "REWIND RATE 1[2]", together with the maintenance of the temperature for a time period **tb** ("Holding time B 1[2]" parameter), necessary to an optimum soldering;



The generator is turned off only after time **tb** (maintenance of the temperature) and time **trv** (rewind of soldering wire), are both elapsed.

INSTALLATION

- cooling of the piece being worked upon with maintenance of the anti-oxidizing gas and activation of the additional gas circuit, for a time period *c* ("Cooling time 1[2]");
- sending a signal to the external control unit to indicate that the cycle has properly ended (*se*) through the "End cycle" output (see chapter "Installation").

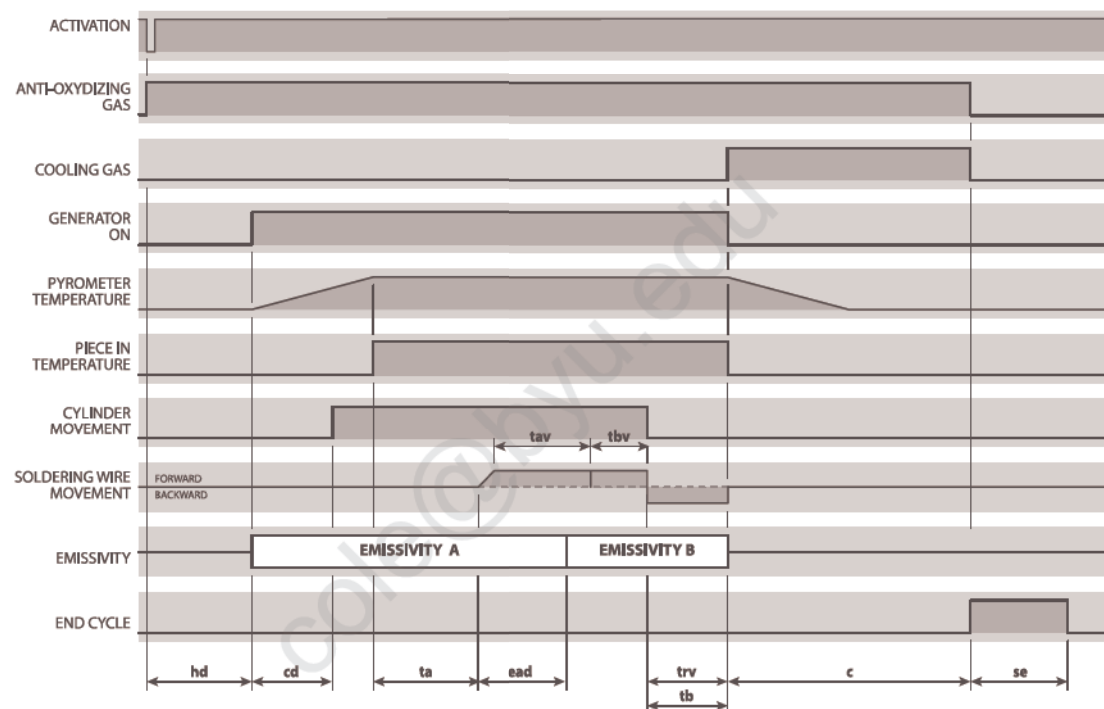


If time *cd* is set too much long, the soldering wire will be neared to the processed piece anyway, when the temperature set is reached, and this is valid for both the wire feeders, independently.

The feeding of the soldering wire will start after a time *t* as programmed in the parameter CYL STROKE TIME (see paragraph 4.6.7 – *Additional programming parameters*).

With (TS) option

A complete work cycle of the soldering station managed by the **Master Controller V3+**, with **FULL AUTO** functioning, consists of the following phases:



- activation of the on/off foot pedal;
- immediate activation of the circulation of anti-oxidizing inert gas;
- activation of the heating of the piece being worked upon (after a time period *hd* determined by the programming parameter "Heating delay 1[2]")
- nearing of the wire applicator to the piece under processing (after *cd* time given by the programming parameter "Cylinder delay 1[2]")
- once the desired temperature of the piece being under processing has been reached, maintenance of that temperature is performed. During this time the output "*Piece in temperature*" is activated;
- waiting for a time period *ta* ("*Holding time A 1[2]*" parameter), before feeding the soldering wire
- advancing of the soldering wire and placing on the piece under processing (wire length and speed depending on parameters "*Feed A 1[2]*" and "*Feed rate A 1[2]*" originating a time *tav*). In the double emissivity functioning (EMISSIVITY ADJUSTMENT=YES), the emissivity value changes: from the initial value ("*EMISSIVITY A 1 [2]*" parameter) to a new value which considers the emissivity variation due to the melting of the soldering wire ("*EMISSIVITY B 1 [2]*" parameter). This change of emissivity occurs after a time period *ead* (EMISSIVITY ADJUSTMENT DELAY 1 [2]" parameter).

- after the time **tav**, the quantity of soldering wire and its speed changes according to the values set on parameters "*Feed B 1[2]*" and "*Feed rate B 1[2]*" for a time **tbv**.
- rewind of soldering wire for a time **trv** depending from parameters "*REWIND LENGTH 1[2]*" e "*REWIND RATE 1[2]*", together with the maintenance of the temperature for a time period **tb** ("*Holding time B 1[2]*" parameter), necessary to an optimum soldering;



The generator is turned off only after time **tb** (maintenance of the temperature) and time **trv** (rewind of soldering wire), are both elapsed.

- cooling of the piece being worked upon with maintenance of the anti-oxidizing gas and activation of the additional gas circuit, for a time period **c** ("*Cooling time 1[2]*").
- sending a signal to the external control unit to indicate that the cycle has properly ended (time **se**).



If the temperature control must be disabled (parameter *Temperature control* = *NO*), the workcycle does not correspond to the one described above; in this case, see paragraph "*Functioning without the temperature control*"

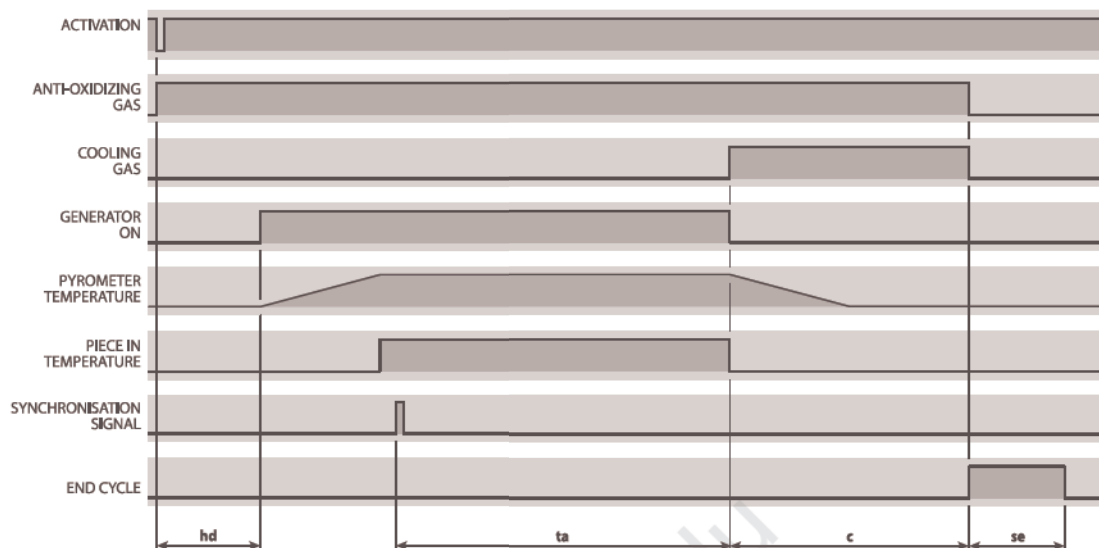


If time **cd** is set too much long, the soldering wire will be neared to the processed piece anyway, when the temperature set is reached, and this is valid for both the wire feeders, independently.
The feeding of the soldering wire will start after a time **t** as programmed in the parameter *CYL STROKE TIME* (see paragraph 4.6.7 – *Additional programming parameters*).

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3.4.3 Synchronised functioning

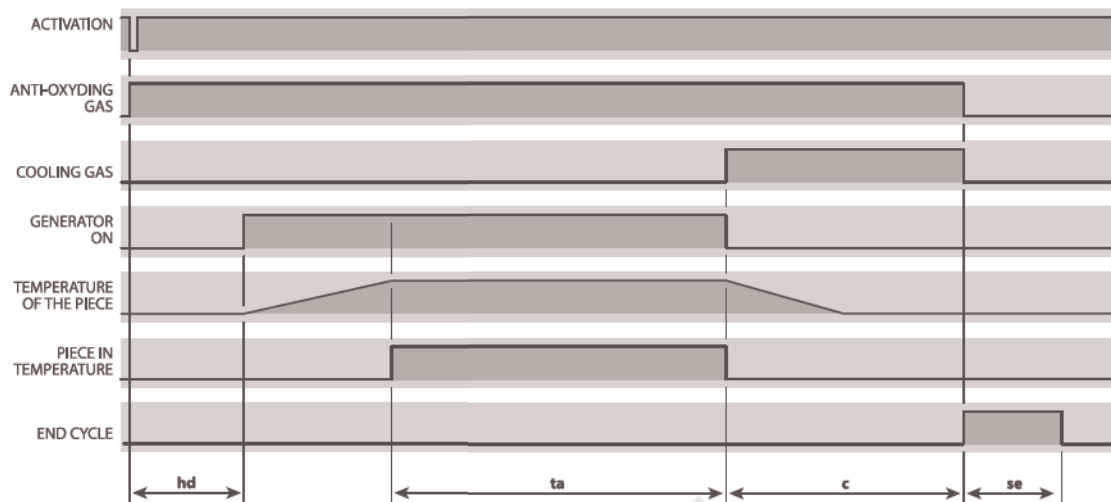
A complete work cycle of the soldering station managed by the **Master Controller V3+**, with "SYNCHRONISED" functioning consists of the following phases:



- activation of the on/off foot pedal;
- immediate activation of the circulation of anti-oxidizing inert gas;
- activation of the heating of the piece under processing (after a time period **hd** determined by the programming parameter "Heating delay 1[2]") and reaching of the selected temperature ("Temperature 1[2]" parameter)
- the Controller awaits the activation of the synchronisation signal, and then maintains the temperature for a time period **ta** ("Holding time A 1[2]" parameter);
- cooling of the piece being worked upon with maintenance of the anti-oxidizing gas and activation of the additional gas circuit, for a time period **c** ("Cooling time 1[2]").
- sending a signal to the external control unit to indicate that the cycle has properly ended (**se**) through the "End cycle" output (see chapter "Installation")

3.4.4 Functioning type A - timer-controlled with temperature control **A**

A complete work cycle of the soldering station managed by the **Master Controller V3+**, with **TIMER A** functioning, consists of the following phases:



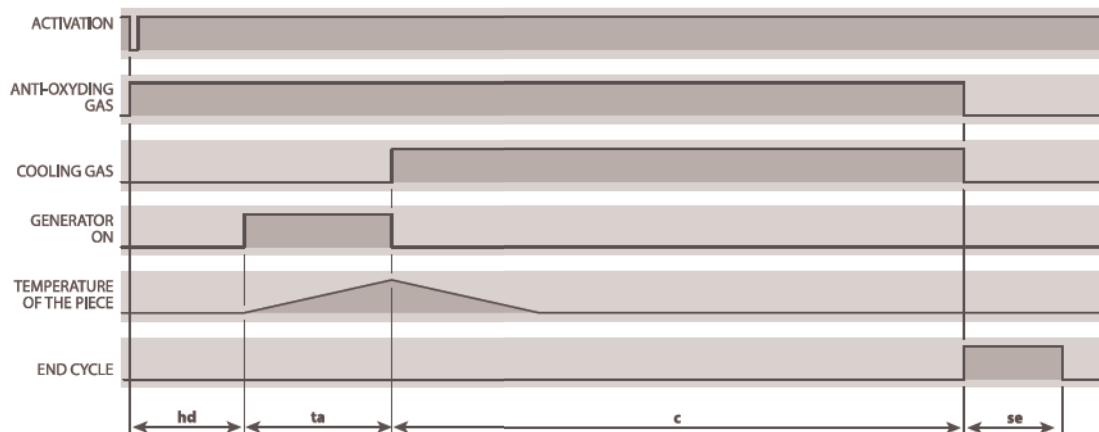
- activation of the on/off foot pedal;
- immediate activation of the circulation of anti-oxidizing inert gas;
- activation of the heating of the piece being worked upon (after a time period **hd** determined by the programming parameter "Heating delay 1[2]") and activation of the "generator ON" output;
- after the piece has reached the programmed temperature, this latter is maintained for a time **ta** ("Holding time A 1[2]" parameter), during this time the "piece in temperature" output is activated (see chapter "Installation");
- cooling of the piece being worked upon with maintenance of the anti-oxidizing gas and activation of the additional gas circuit, for a time period **c** ("Cooling time 1[2]");
- sending a signal to the external control unit to indicate that the cycle has properly ended (**se**) through the "End cycle" output (see chapter "Installation").



When the parameter **ta** is set to **0**, the heating time of the head corresponds to the time the on/off foot pedal is pressed, so allowing to prolong the working cycle indefinitely.

3.4.5 Functioning type B - timer-controlled with one heating phase **B**

A complete work cycle of the soldering station managed by the **Master Controller V3+**, with "TIMER B" functioning consists of the following phases:



- activation of the on/off foot pedal;
- immediate activation of the circulation of anti-oxidizing inert gas;
- activation of the heating of the piece being worked upon (after a time period **hd** determined by the programming parameter "Heating delay 1[2]") and activation of "generator ON" output;
- heating of the piece for a time period **ta** ("Holding time A 1[2]" parameter);
- cooling of the piece being worked upon with maintenance of the anti-oxidizing gas and activation of the additional gas circuit, for a time period **c** ("Cooling time 1[2]").
- sending a signal to the external control unit to indicate that the cycle has properly ended (**se**) through the "End cycle" output (see chapter "Installation").



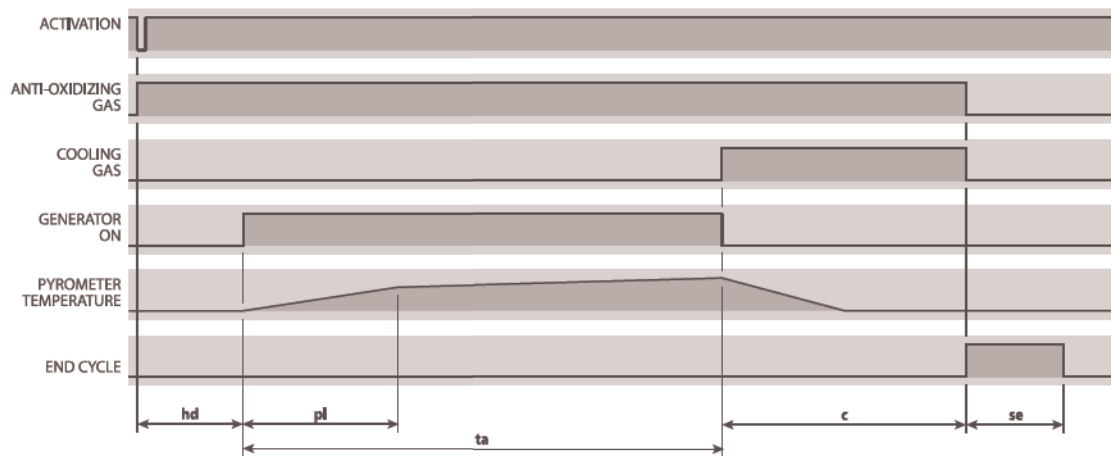
When the parameter **ta** is set to 0, the heating time of the head corresponds to the time the on/off foot pedal is pressed, so allowing to prolong the working cycle indefinitely.



If in the soldering station a pyrometer is present, the piece will be maintained at the programmed temperature, if this is reached within time **ta**. When time **ta** elapses, the cycle continues to the following phase even if the programmed temperature has not been reached.

3.4.6 Functioning type C – timer-controlled with two heating phases

A complete work cycle of the soldering station managed by the **Master Controller V3+**, with "TIMER C" functioning consists of the following phases:



- activation of the on/off foot pedal;
- immediate activation of the circulation of anti-oxidizing inert gas;
- activation of the heating of the piece (after a set time period **hd** determined by the programming parameter "Heating delay 1[2]" and of the "generator on" output. During this phase, the piece is heated up on the basis of the power value selected by the "Power A 1[2]" parameter; the duration of this phase (**pl** in the illustration) is set by the "Power A time 1[2]" parameter.
- Second phase in heating the piece, which corresponds to the power value selected by the "Power B 1[2]" parameter. The total length of time that the piece is heated (**ta** in the illustration) is given by the "Holding time 1[2]" parameter
- cooling of the piece being worked upon with maintenance of the anti-oxidizing gas and activation of the additional gas circuit, for a time period **c** ("Cooling time 1[2]").
- sending a signal to the external control unit to indicate that the cycle has properly ended (**se**) through the "End cycle" output (see chapter "Installation").



When the parameter **ta** is set to 0, the heating time of the head corresponds to the time the on/off foot pedal is pressed, so allowing to prolong the working cycle indefinitely.

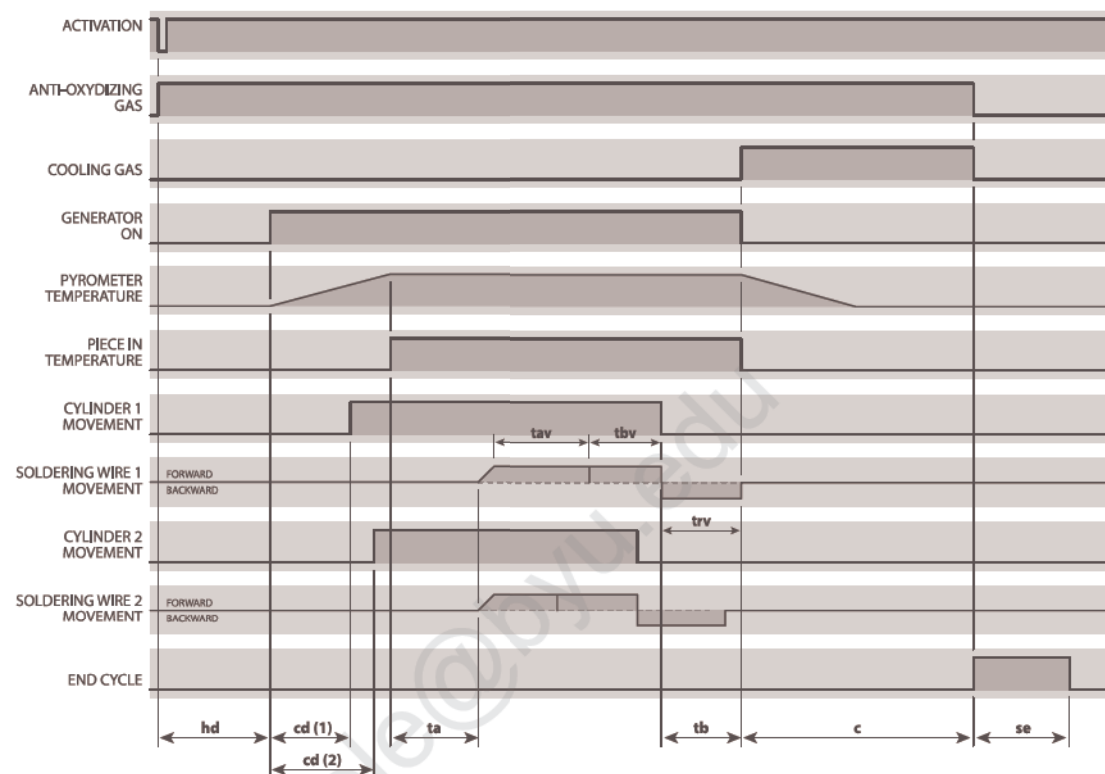
3.4.7 Functioning with two soldering wire feeders **A2W**



This operating mode is available ONLY with **(TS)** option.

In this operating mode, the soldering point 2 is not used. The parameters related to it are not relevant, except for **Feed 2, Feed rate 2, Cylinder delay 2, Injection cycle 2** and **Injection time 2** that are necessary for the management of the second soldering wire.

A complete work cycle consists of the following phases:



- activation of the on/off foot pedal;
- immediate activation of the circulation of anti-oxidizing inert gas;
- activation of the heating of the piece under processing (after a time period **hd** determined by the programming parameter "Heating delay 1")
- nearing of the wire applicator 1 to the piece under processing (after **cd** time given by the programming parameters "Cylinder delay 1")
- nearing of the wire applicator 2 to the piece under processing (after **cd** time given by the programming parameters "Cylinder delay 1")
- once the desired temperature of the piece being under processing has been reached, maintenance of that temperature is performed. During this time the output "*Piece in temperature*" is activated;
- Waiting for a time **ta** (parameter "Holding time A 1"), before feeding the soldering;
- advancing of the soldering wires from cylinder 1 and placing on the piece under processing (time depending on parameters "Feed A 1" and "Feed rate A 1" generating a time **tav**).
- advancing of the soldering wires from cylinder 2 and placing on the piece under processing (time depending on parameters "Feed A 2" and "Feed rate A 2" generating a time **tav**).
- after the time **tav**, the quantity of soldering wire and its speed changes according to the values set on parameters "Feed B 1[2]" and "Feed rate B 1[2]" for a time **tbv**.

- rewind of soldering wire for a time **trv** depending from parameters "*REWIND LENGTH 1[2]*" e "*REWIND RATE 1[2]*", together with the maintenance of the temperature for a time period **tb** ("*Holding time B 1[2]*" parameter), necessary to an optimum soldering;



The generator is turned off only after time **tb** (maintenance of the temperature) and time **trv** (rewind of soldering wire), are both elapsed.

- cooling of the piece being worked upon with maintenance of the anti-oxidizing gas and activation of the additional gas circuit, for a time period **c** ("*Cooling time 1*").
- sending a signal to the external control unit to indicate that the cycle has properly ended (**se**) through the "End cycle" output (see chapter "Installation").



When the parameter **ta** is set to **0**, the heating time of the head corresponds to the time the on/off foot pedal is pressed, so allowing to prolong the working cycle indefinitely.



If time **cd** is set too much long, the soldering wire will be neared to the processed piece anyway, when the temperature set is reached, and this is valid for both the wire feeders, independently.
The feeding of the soldering wire will start after a time **t** as programmed in the parameter CYL STROKE TIME (see paragraph 4.6.7 – *Additional programming parameters*).

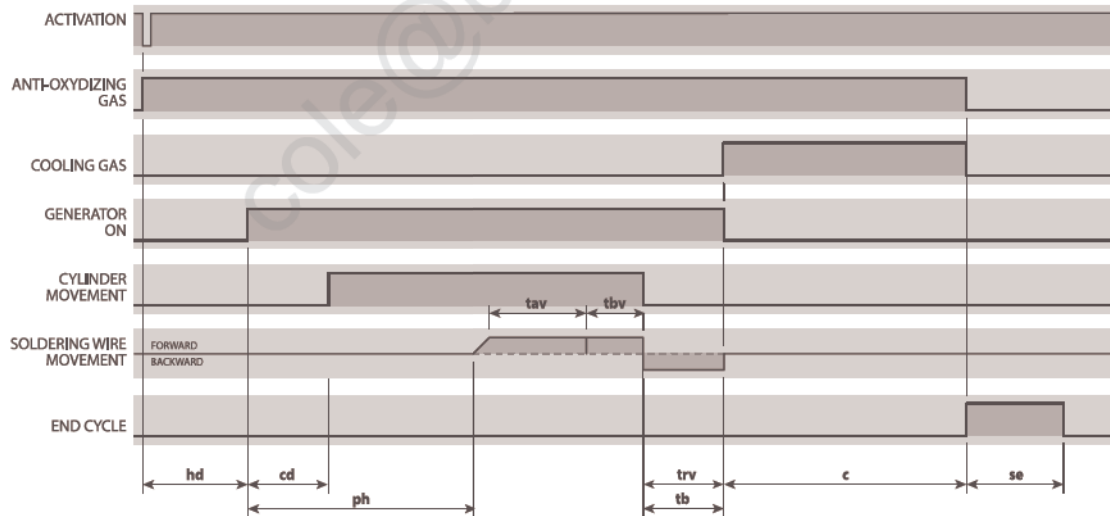
3.4.8 Functioning without temperature control (Temperature Control = NO)



This operating mode is available ONLY with **TS** option.

Should temperature control be disconnected (*Temperature control = OFF*) phases of soldering cycle are not regulated according to the piece temperature, but by parameters of special programming.
The advancing of the soldering wire feeder will be time-activated.

A complete work cycle consists of the following phases:



- activation of the on/off foot pedal;
- immediate activation of the circulation of anti-oxidizing inert gas;

INSTALLATION

- activation of the heating of the piece under processing (after a time period **hd** determined by the programming parameter "*Heating delay 1[2]*"). Piece heating is divided into two phases:
 1st phase: the piece is being heated at a first power level, fixed by "*Power A 1 [2]*" parameter; the duration of this phase is due to "*Power A Time 1 [2]*" parameter;
 2nd phase: the piece is being heated at a second power level, due to "*Power B 1 [2]*" parameter up to the end of the heating (time **tb** on the graph);
- nearing of the wire applicator to the piece under processing (after "**cd**" time given by the programming parameters "*Cylinder delay 1[2]*")
- waiting for a time **ph** (parameter "*Pre-heating time 1[2]*"), before feeding the soldering wire
- advancing of the soldering wire, and placing on the piece under processing (time depending on parameters "*Feed 1[2]*" and "*Feed rate 1[2]*" generating a time **tav**);
- after the time **tav**, the quantity of soldering wire and its speed changes according to the values set on parameters "*Feed B 1[2]*" and "*Feed rate B 1[2]*" for a time **tbv**.
- rewind of soldering wire for a time **trv** depending from parameters "*REWIND LENGTH 1[2]*" e "*REWIND RATE 1[2]*";
- cooling of the piece being worked upon with maintenance of the anti-oxidizing gas and activation of the additional gas circuit, for a time period **c** ("*Cooling time 1[2]*");
- sending a signal to the external control unit to indicate that the cycle has properly ended (**se**) through the "End cycle" output (see chapter "Installation").

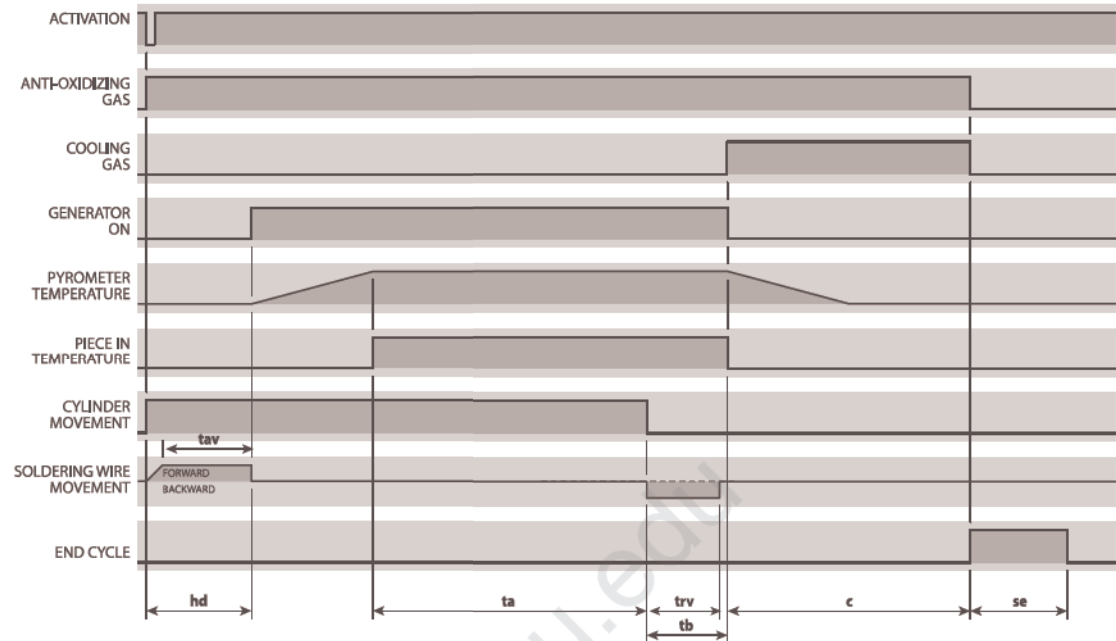
This kind of working is possible only with **FULL AUTO** and **FULL AUTO 2 WIRES**.



If time **cd** is set too much long, the soldering wire will be neared to the processed piece anyway, when the temperature set is reached, and this is valid for both the wire feeders, independently.
 The feeding of the soldering wire will start after a time **t** as programmed in the parameter CYL STROKE TIME (see paragraph 4.6.7 – *Additional programming parameters*).

3.4.9 Functioning with pre-distribution of soldering wire/strip

When is necessary to distribute the soldering wire before turning on the generator, in order to position it on the piece being worked, the additional parameter *forward feed* = *PRE-HEAT* can be set.



- activation of the on/off foot pedal;
- immediate activation of the circulation of anti-oxidizing inert gas;
- immediate nearing of the wire applicator to the piece under processing;
- advancing of the soldering wire, and placing on the piece under processing (time depending on parameters "Feed 1[2]" and "Feed rate 1[2]" generating a time *tav*);



In this OPERATING MODE, the parameters *Feed B 1 [2]* and *Feed rate B 1 [2]* are not used.

- heating of the piece (after a time *hd* determined by the parameter "Heating delay 1" or after time *tav*);
- heating of the piece for a time period *ta* ("Holding time A 1[2]" parameter);
- rewind of soldering wire for a time *trv* depending from parameters "REWIND LENGTH 1[2]" e "REWIND RATE 1[2]", together with the maintenance of the temperature for a time period *tb* ("Holding time B 1[2]" parameter), necessary to an optimum soldering;



The generator is turned off only after time *tb* (maintenance of the temperature) and time *trv* (rewind of soldering wire), are both elapsed.

- cooling of the piece being worked upon with maintenance of the anti-oxidizing gas and activation of the additional gas circuit, for a time period *c* ("Cooling time 1").
- sending a signal to the external control unit to indicate that the cycle has properly ended (*se*) through the "End cycle" output (see chapter "Installation").

This kind of working is possible only with **FULL AUTO, FULL AUTO 2 WIRES, THERMAL PROFILE** and **THERMAL PROFILE 2 WIRES**.

3.4.10 Functioning with thermal profile



This kind of working is possible ONLY with the  option.

A complete work cycle with “Thermal Profile” functioning consist of the following phases:

- activation of the on/off pedal;
- immediate activation of the circulation of the anti-oxidizing inert gas;
- activation of the heating of the piece (after a time period **hd** determined by the parameter “Heating Delay 1[2]”) and of the “generator ON” output;
- execution of the programmed steps of the thermal profile;
- cooling of the piece with activation of the additional gas circuit for a time period **c** (“Cooling Time 1[2]”);
- activation of the “End Cycle” output to indicate that the cycle has ended; at the same time activation of the “Temperature out of tolerance” output if during the execution of the thermal profile the difference between the measured temperature and the setpoint was higher than the programmed tolerance.

A Thermal Profile consist of up to 20 steps (as selected with the parameter “Number of Steps 1[2]”) which are executed in sequence.

A step can be of “HEATING”, of “TEMP.CONTROL.” or “COOLING” type.

An “Heating” **step type** is used to heat a piece to the set temperature using the chosen power. It is useful to heat a piece as quickly as possible or, when the piece is cold, to bring the temperature inside the sensor range. It is characterized by the following parameters:

- **Temperature max**: when the measured temperature is equal to or higher than this parameter, the step ends;
- **Time max**: this is the maximum allowed duration of the step. If this time expires, the step ends (note that this should be considered a timeout, the normal way to switch to the next step should be thanks to the condition on temperature max).
- **Power**: the generator power is set at this value during the step.

A “Temperature Controlled” **step type** is used to take the temperature of the piece to the desired value in a well-defined amount of time. It is characterized by the following parameters:

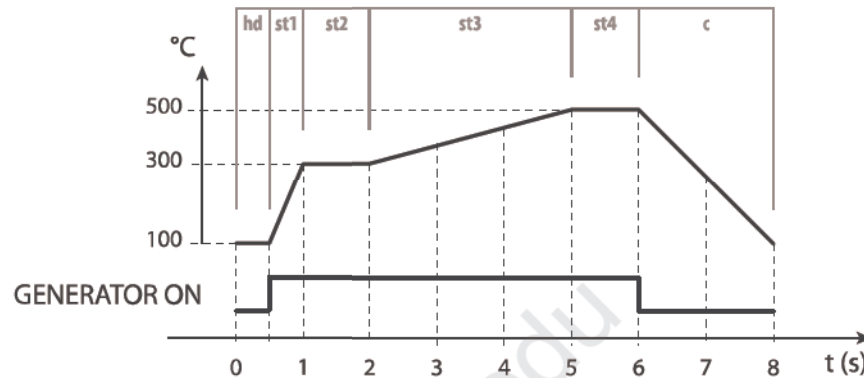
- **Temperature final**: this is the temperature desired at the end of the step;
- **Time**: this is the time requested to reach the final temperature set for the step starting from the temperature set at the previous step. The step ends when this time expires;
- **Power max**: the generator power is set at this value during the step, so this will be the maximum power, while the average power will be lower because of the temperature control;
- **Tolerance**: this parameter defines how big can become the difference between the desired and the real temperature during the step. If the tolerance is exceeded then a signal is given at the end of the cycle. The tolerance is intended for both positive and negative differences, for example if the desired temperature is 400°C and the tolerance is 30°C, then the accepted range is 370°C-430°C. If this parameter is set to 0, the tolerance check is not performed on the step.

A “**COOLING**” step type is used to wait the temperature of the piece to reach the desired value, in a well-defined amount of time. In this step, the Generator does not supply any power.

It is characterized by the following parameters:

- **TEMP. MIN:** when the measured temperature is equal to or lower than this parameter, the step ends;
- **TIME MAX:** this is the maximum allowed duration of the step. If this time expires, the step ends (note that this should be considered a timeout, the normal way to switch to the next step should be thanks to the condition on temperature min.).

Example:



To obtain this thermal profile set the following parameters:

- Heating Delay 1 = 0,5 s
- Number of steps 1 = 4
- H1 Step01:Type = HEATING
- H1 Step01:Temper.Max = 300
- H1 Step01:Power = 50% (or as needed by the application)
- H1 Step01:Time Max = 5s (this is intended as a timeout, choose a value that is high enough for the piece to reach the temperature)
- H1 Step01:Toleran = 5°C (or as needed by the application)
- H1 Step02:Type = TEMP.CONTROL.
- H1 Step02:Temper. = 300
- H1 Step02:Power Max = 50% (or as needed by the application)
- H1 Step02:Time = 1s
- H1 Step02:Toleran = 10°C (or as needed by the application)
- H1 Step03:Type = TEMP.CONTROL.
- H1 Step03:Temper. = 500
- H1 Step03:Power Max = 50% (or as needed by the application)
- H1 Step03:Time = 3s
- H1 Step03:Toleran = 10°C (or as needed by the application)
- H1 Step04:Type = TEMP.CONTROL.
- H1 Step04:Temper. = 500
- H1 Step04:Power Max = 50% (or as needed by the application)
- H1 Step04:Time = 1s
- H1 Step04:Toleran = 5°C (or as needed by the application)
- Cooling time 1 = 2s



In case that during one of the programmed steps, the temperature of the piece results out of the temperature range, set by the tolerance values, the output “Temperature out of range” is activated during the “End of Cycle” signal.

3.4.10.1 Wire feed during thermal profile

To activate the feeding of soldering wire, set the parameter "Hx STEP: WIRE" to a value higher than 0. When the step indicated by the parameter is reached, and the temperature set is reached, the soldering wire is fed, with quantity and rate as set in "WIRE" menu.



The step ends when its associated duration expires and wire feed ends.

If the temperature of the pieces is below the temperature set in the step, the wire feed doesn't start.

3.4.10.2 Thermal profile without temperature control

On Master Controller V3+, with –TS option, it is possible to use the thermal profile mode, even without any temperature control. The parameter TEMPERATURE CONTROL, under MODES menu, must be set to OFF and only TEMP.CONTROL step types must be used.

3.4.11 Functioning with thermal profile and two soldering wire feeders



This operating mode is available ONLY with  option.

In this operating mode, the soldering point 2 is not used. The parameters related to it are not relevant, except for **Feed 2**, **Feed rate 2**, **Cylinder delay 2**, **Injection cycle 2** and **Injection time 2** that are necessary for the management of the second soldering wire.

This operating mode acts exactly as per the THERMAL PROFILE mode, with the possibility to operate two soldering wire feeders on the same soldering point.

3.5 Functioning with two pyrometers on the same heating head

In order to use two pyrometers on the same piece, the following parameters have to be set on the right value:

- MODE > HEATING SYSTEM = SINGLE GENERATOR
- MODE > SINGLE POWERCUBE = ONE HEAD
- MODE > HEAD 1 MODE = 2-pyro

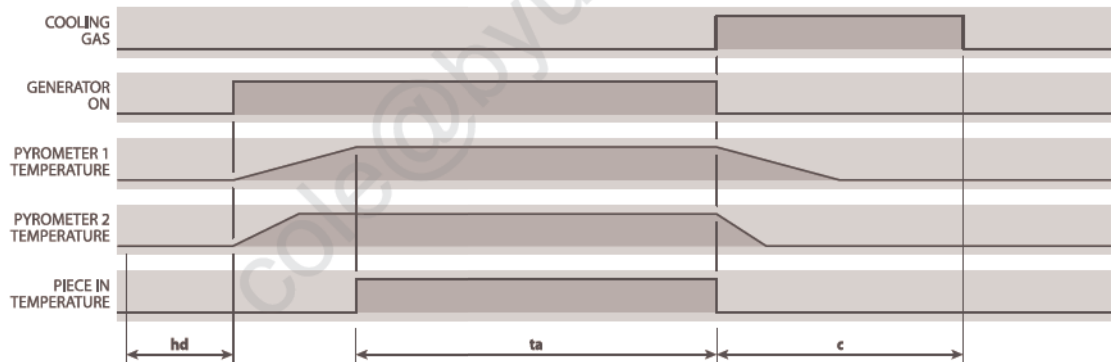
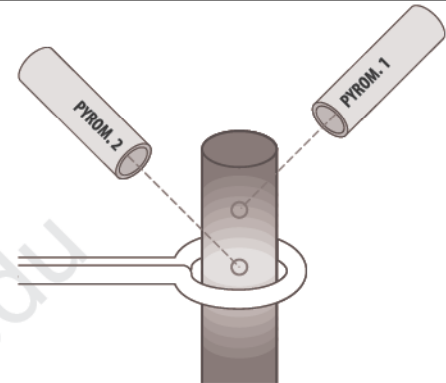


When this functionality is activated, the section 2 is locked and no cycles can be started there. In the main screen, all data related to section 2, except the temperature, will not be shown.

3.5.1 Example of a possible application

- Place the two pyrometers as shown in the figure, with the pyrometer 2 aiming at the hottest point.
- Set the desired temperature for the pyrometer 2 (parameter TEMPERATURE 2).
- Set the desired temperature for the pyrometer 1 (parameter TEMPERATURE 1).

When the temperature set for the spot 2 is reached, the generator start to operate in order to maintain that temperature on spot 2, until the temperature on spot 1 is also reached, then starts with the programmed holding time.



The work cycles that needs a temperature to be reached, in order to proceed (i.e. TIMER A with Holding time > 0) are based only on TEMPERATURE 1.

PID type adjustment cannot be used on both pyrometers.

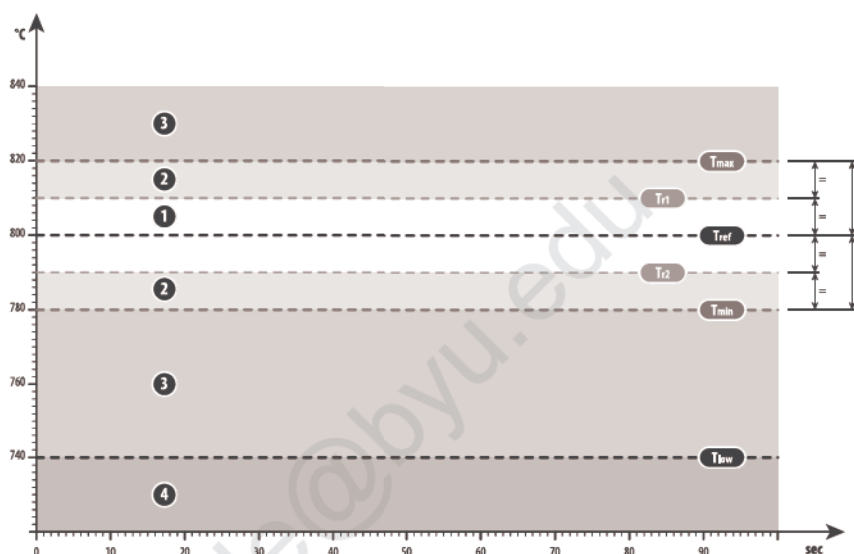
3.6 Functioning with temperature “in window”

This functioning type allows to control the heating temperature inside a programmable tolerance window.

The “in window” control is available only with the functioning modes **TIMER A (A)**, **TIMER B (B)** and **SYNCHRONIZED (S)**.

The control on live values (**TEMPERATURE CONTROL 1/2 = WINDOW LIVE VAL. (WR)**) can be used during the heating of a single piece on continuous movement. The control on maximum values (**TEMPERATURE CONTROL 1/2 = WINDOW MAX VAL. (WM)**) can be used when heating consecutive pieces with or without an empty space between each other, or a single piece in discontinuous movement.

During the movement of the piece, inside the heating coil, the controller read the temperature values, calculates the average and automatically adjust the generator power.



If the average temperature lies in the area of the graph labeled with **1**, the controller does not change the power of the generator: the value is in the range of acceptance.

If the average temperature lies in the area **2**, the controller increases / decreases the power of the generator by the set percentage.

If the value lies in the area **3**, to speed up the stabilization of the temperature, the controller operates an increase / decrease of the power, equal to twice the value set.

If the temperature drops in the area **4**, the controller realizes that there is no part to be measured.

Programmable parameters are:

- **Min. Temperature (T_{min})**: the minimum acceptable temperature for the desired temperature window;
- **Max Temperature (T_{max})**: the maximum acceptable temperature for the desired temperature window. The ideal temperature (T_{ref}) is automatically calculated as the center of the window ($T_{ref} = (T_{min} + T_{max}) / 2$);
- **Control Time**: indicates the amount of time needed to perform the temperature average. In presence of objects with irregular emissivity, it is recommended to increase the Control Time. Bigger is the control time, more stable the temperature measurement.
- **Temperature Delay**: indicates the amount of time the controller will ignore the temperature measurement, after any modification of the power. This is to give the time, to a power correction, to propagate completely on the heated pieces, to influence the temperature sensor (which could be measured away from the coil), so that to avoid power over-corrections.
- **Power Increase**: Indicates, for each control cycle, the percentage value to automatically increase or decrease the generator power. If measured temperature is higher than T_{max} or lower than T_{min} , this value will be doubled to speed up the temperature stabilization. If the

value is between T_{r1} and T_{r2} , no power change will be applied.

If this parameter is set to 0, the power change is performed manually by the operator. The controller will only signal if the read temperature is inside the requested limits.

- **Low Temperature (T_{low}):** below this limit the controller states that there is no piece to be measured by the temperature sensor. Used to synchronize the reading of the maximum temperature value of each heated piece.

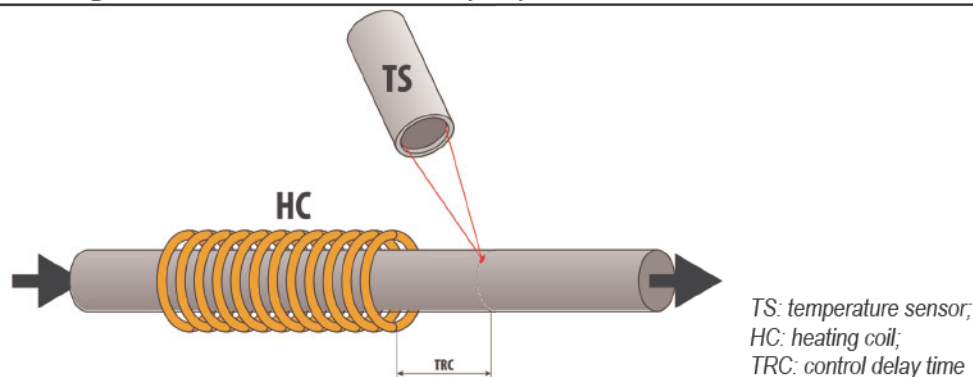
The Open Collector output “**Temp.over max.**” is activated when the piece has a temperature higher than the maximum set, and the output “**Temp.below min.**” is activated to indicate that the temperature is lower than the minimum set.

Digital inputs **Max.Temp.Acq.1** and **Max.Temp.Acq.2** (respectively for section 1 and 2) can be used in alternative or together with the parameter “**Low Temperature**” to synchronize the acquisition of the maximum temperature value of each new piece in transit.

With the “in window” functioning type, after the cycle start, in order to avoid power overcorrections at the beginning, the temperature control will be activated only after the measured temperature is inside the area ❶ ($>T_{r2}$) for the first time. For this reason, it is recommended to set an adequate starting power in order to allow the heating pieces to reach this temperature area.

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3.6.1 Functioning in window with live values (WR)

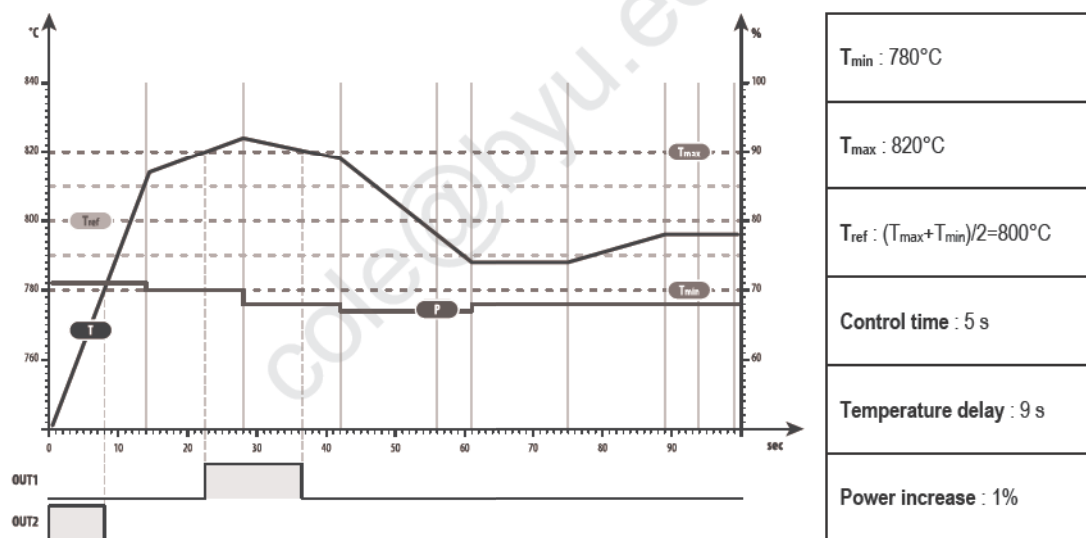


For this application type, set the temperature control on live values (**TEMPERATURE CONTROL 1/2 = WINDOW LIVE VAL.**).

The controller, according to the other parameters listed below, will adjust the generator power in order to heat the object at the desired temperature.

At the cycle start, the power set by the operator on **POWER A 1/2** will be used and, since the temperature does not reach the center of the acceptance range (T_{ref}), any power change will be performed.

Then the controller will operate increasing or decreasing the power in order to maintain the temperature inside the window set.

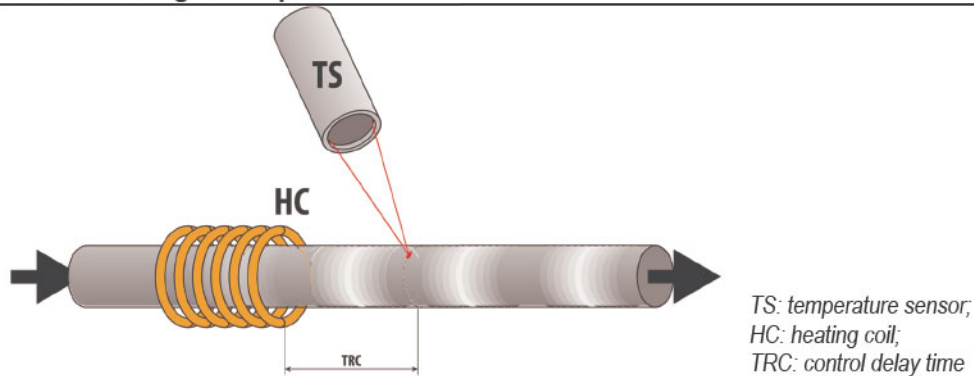


T: Measured temperature; P: Generator power; OUT1: "Temp. over max." output; OUT2: "Temp. below min." output

In the example here above, the temperature is measured and the average is calculated on time periods of 5 seconds, with a delay of 9 seconds, so, at the cycle start, and when necessary, the power is changed after $9 + 5 = 14$ seconds. If no changes are made, the next control is performed after 5 seconds.

3.6.2 Functioning in window with maximum values (WM)

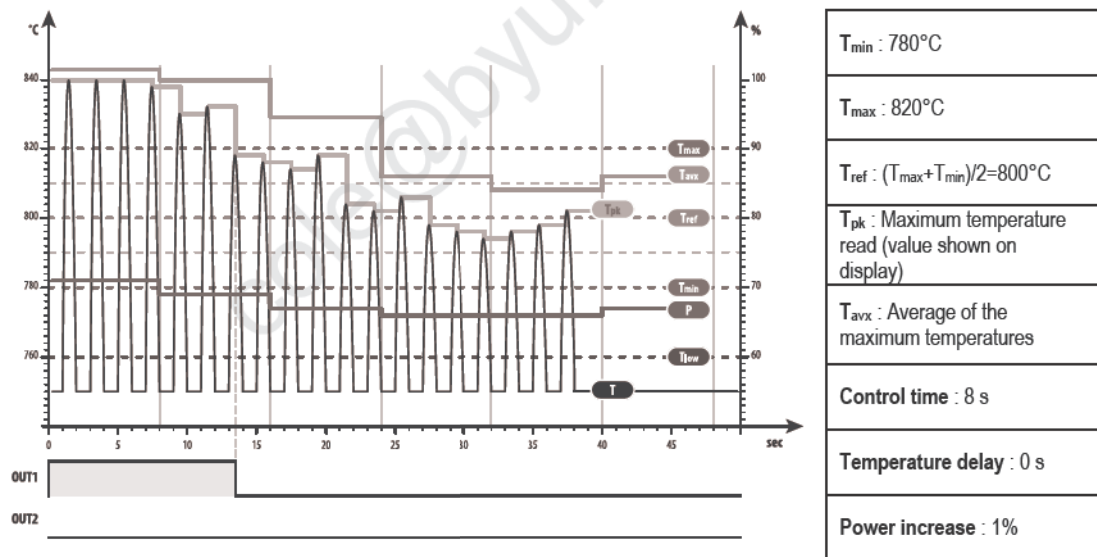
3.6.2.1 Timing on temperature



Set the temperature control on maximum values (**TEMPERATURE CONTROL 1/2 = WINDOW MAX VAL.**), the display will show the maximum temperature reached and not the actual one, as per the open collector outputs (**Temp.over max.** and **Temp.below min.**) will be driven according to the maximum measured temperature.

The search for a new temperature peak can be repeated according to two method, both working together:

- Measured temperature passes from a value lower than “Low temperature” parameter, to a higher value.
- Activation, from an external logic, of the synchronization input (J7-4).

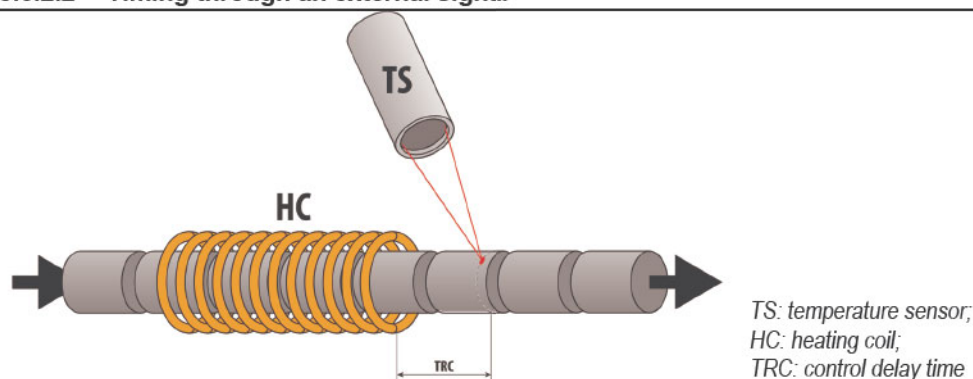


T: Measured temperature; P: Generator power; OUT1: “Temp.over max.” output; OUT2: “Temp.below min.” output

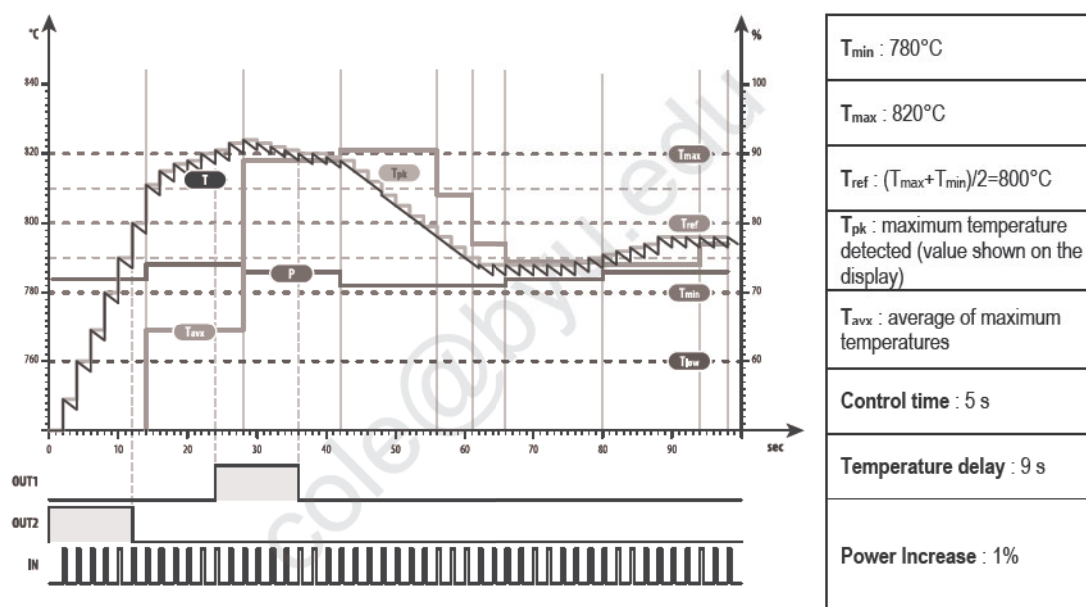
The T_{pk} track represents the maximum temperature detected during the material transit and it is the value shown on the display and used to drive the Open Collector outputs to signal if the temperature is within the acceptable range. Note that the maximum temperature tracking is reset each time the live temperature goes from lower to higher values than T_{low} .

The T_{avg} track represents the average between the maximum values, detected during a specific control time (8 seconds in this example). The decision to change the power is taken according to this value.

3.6.2.2 Timing through an external signal



In this case, the research of a new temperature peak, is repeated according to the activation, by an external logic, of the synchronization input (J7-4) or by means of the Field Bus (see Field Bus Start command).



T : Measured temperature; P : Generator power;

$OUT1$: "Temp. over max." output; $OUT2$: "Temp. below min." output; IN : Digital input "Max. Temp. Acq."

The T_{pk} track represents the maximum temperature detected during the material transit and it is the value shown on the display and used to drive the Open Collector outputs to signal if the temperature is within the acceptable range. Note that the maximum temperature tracking is reset each time the IN digital input is activated.

The T_{avx} track represents the average between the maximum values, detected during a specific control time (14 seconds in this example). The decision to change the power is taken according to this value.

3.6.3 Functioning in Stand-Alone mode (-STDLN)



This option allows the using of the Master Controller v3+ by itself, without the connection with a CEIA Generator.

Only the following Operating modes are available:

- **AUT** FULL AUTO
- **A2W** FULL AUTO 2-wires

The management of the Induction Heating generator is ignored.

A start signal, provided by a digital input, an analog input or by remote (Field Bus), allows the wire feeding process to start, following the parameters set on the Controller, as HEATING DELAY, WIRE FEED and FEED RATE.

All digital and analog outputs are available (i.e. workpiece in temperature, temperature over max, temperature below min, wire movement ...).

3.6.4 Functioning in Sensor Only mode (-SO)



This option allows the using of the Master Controller v3+ by itself, without the connection with a CEIA Generator.

The controller can manage up to two temperature sensors and allow recording in the data logger (with V3+/IXC option enabled).

The only allowed operating mode is TIMER B and the temperature control in the Window Live Values (WR).

The CEIA pyrometers (or other sensors that may be used) monitor the temperature in real time and the controller activates the digital outputs accordingly:

- Temperature above max 1/2;
- Temperature below min 1/2;

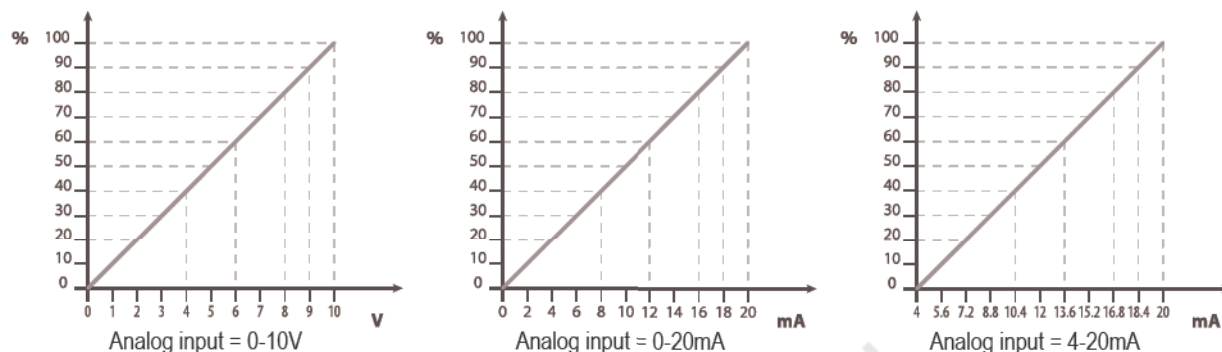
3.6.5 Functioning in Analog Sensor mode (-PYAN)



This option allows the using of external analog pyrometers.

The controller accepts inputs of type 0-10V, 0-20mA, 4-20mA, as described in the J19 connector description.

The following graphs show the percentage reading of temperature as a function of voltage or current:



In order to define the temperature range, set **HEATING > AN.TEMP.MIN 1/2** and **HEATING > AN.TEMP.MAX 1/2** corresponding to the reading range of the pyrometer used.



WARNING! When the temperature reading is set from the analog input, check that the **MODE > AN.INP.NOISE REJ.** parameter is set to **NO**.

3.6.6 Functioning in Synchronized Sections mode (-SYNS)



This option allows the simultaneous activation of two CEIA generators on the same work piece.

If the parameter **MODE > SECTIONS SYNCHRON.** is set to **NO**, the controller operates as a standard controller.

If the parameter **MODE > SECTIONS SYNCHRON.** is set to **YES**, the only possible configuration is **DUAL POWER CUBE** with **THERMAL PROFILE** as operating mode.

- Section 1 operates as master, section 2 as slave.
- The activation signal on section 1 (via serial cable, RS232, Field Bus) starts the heating in both sections. Any activation signal on section 2 is ignored.
- If a section is not ready, neither section is activated.
- If during the heating cycle one section is interrupted due to a diagnosis, the other section is also stopped.
- Each section follows its own heating cycle:
 - If section 1 ends before section 2, section 2 is also stopped at the end of the cycle of section 1;
 - If section 2 ends before section 1, section 2 maintains the temperature set in the last heating step until section 1 has also completed the cycle.
- If activated, the data logger records a unique file for both sections.

3.7 Power ON

Activate the main switch located on the back of the Controller.

3.8 Verification procedure for a correct installation

The Master Controller shows on the display every message coming from the internal self-diagnosis system (see chapter "Maintenance") related both to the Controller itself, and to the connected generators. In this case it inhibit also the working cycle start.



The internal self-diagnosis is comprehensive of the Controller and of the generator but it can not see installation errors like a wrong alignment between the pyrometer and the piece being worked.

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4 USE



Before starting any operation on the unit, read carefully the **Safety instructions - Warnings** section of this manual.

4.1 Indicators

4.1.1 Optical indicators

All optical indications given by the Master Controller V3+ are shown on the display.
Beside a work screenshot example.

```
1> T= 700C   S=====L
    TIME=----s   POW.=99%

2> T= 700C   S=====L
    TIME=----s   POW.=99%
```

4.1.2 Acoustical indicators

The Master Controller V3+ utters only a short acoustical signal when a self-diagnosis message is displayed and two short acoustical signals when the fault is solved and is ready to operate again.

4.2 Controls

Programming and Management are executed through the built-in keypad described below:

KEYS	FUNCTIONS
	Access to and exit from Programming phase Return to the previous menu
	Scrolling of the instructions sequence Selection of the parameter to be modified Parameter value modification
	Selection of the submenu Value confirmation
	To check the status of the working parameters of the generator, inputs/outputs, optical pyrometers and of the external sensors.
	To access the quick menu programmable by the user.

4.3 Power on

4.3.1 Signals at power on

Powering on the Master Controller, the display shows, for a few seconds, a screenshot indicating the Controller model, the serial number and the software version, as shown beside.

```
Master
Controller v3+
S/N xxxxxxxxxxxx
MSCT30xx
```

4.4 Use of the device

4.4.1 Start-up

Check that all the devices needed for the heating operation are placed correctly, so as to let the workstation operative.
Supply all system devices with the right voltage and activate all workstation components.
Activate the main switch located on the back of the Controller.

4.4.2 Set-up

The display of the Master Controller is divided, in this case, in two sections: the upper row shows the status of the soldering point 1, the lower row the status of the point 2 (see picture beside). Even in the case of a self-diagnosis message, this will be displayed in the row related to the soldering point that generated the message.

```

1> T= 700C  PID S====#==L
WIRE= 50,0mm POW.=99%
                2.6s

2> T= 800C  2C S=====L
WIRE= 50,0mm POW.=99%
        6+MEM006          2.6s
    
```

For each soldering point, a bar-graph indication is given to represent the inductor dimensioning: if the cursor is between the two characters "S" and "L" the dimensions are correct.
If the cursor moves to the left until the "S" character (small) starts to blink, the inductor is too small.
If the cursor moves to the right with the "L" character (large) blinking, the inductor is too big.

Beside the temperature indication, a small symbol indicates if a thermocouple (PID) or a two-colours pyrometer (2C) is connected. This latter indication appears only if the pyrometer is operating in two-colours mode, over 600°C, because for temperatures lower than that, the pyrometer uses only one sensor.

In the example above, the section 1 is connected to a thermocouple and in the section 2, an SH2C/SLE two-colours pyrometer is operative.

4.4.3 Normal use

Program the operating parameters according to the type of processing to be carried out or load the parameters by using one of the internal memory.



Each time the Master Controller is turned on, it automatically sets itself using the last parameters in use.

Place the pieces to be heated near the heating inductor, taking care not to be in contact with it.

Activate the cycle, and carry out the required operations.

4.4.4 Status screenshots

From the work screenshot, pressing the STATUS key, a list of parameters screenshots are displayed, following a cyclic scheme:

- Generator 1 parameters
- Generator 2 parameters (if connected)
- CEIA /SL pyrometers parameters (only if at least one connected sensor is SLE type)
- UPP pyrometers parameters (only if at least one connected sensor is UPP type)
- Thermal imaging camera parameters (only with –DIAS option)
- Diagnosis status
- Logger parameters (only with –IXC option)
- Field Bus parameters (only with –FB option)
- Inputs and Outputs status

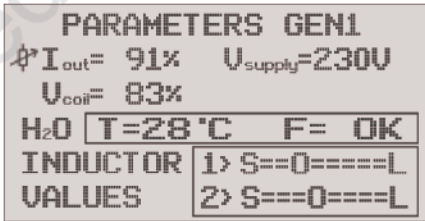
After the last screenshot, press again STATUS and the work screenshot will be displayed again.

4.4.4.1 Generators parameters screenshot

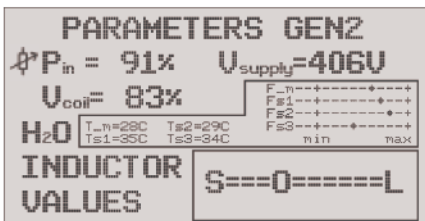


If a second generator is not connected, after the generator 1 screenshot the Controller will proceed with the next screenshot.

Generator data, displayed in the relative screenshot, vary depending on the type of generator connected. Here below an example for a Power Cube 32/900.

I_{out}	Value (%) of the current supplied by the generator to the head	 PARAMETERS GEN1 Φ I_{out} = 91% U_{supply} = 230V U_{coil} = 83% H₂O T = 28 °C F = OK INDUCTOR 1> S = 0 ===== L VALUES 2> S = 0 ===== L
U_{coil}	Value (%) of the voltage supplied by the generator to the inductor	
U_{supply}	Generator supply voltage	
H₂O	T is the temperature of the water running into the heads, and F is the flux status.	
	This symbols appears beside I_{out} (as in the picture) when the generator is controlled in current and beside U_{coil} when it is controlled in voltage.	

If a fluxometer is installed to the generator, the indication of the water flux is available as a bar-graph. Here below an example for a Power Cube 720/50.

P_{in}	Value (%) of the power absorbed by the generator	 PARAMETERS GEN2 Φ P_{in} = 91% U_{supply} = 406V U_{coil} = 83% H₂O T_m = 28C T_{s1} = 29C T_{s2} = 29C T_{s3} = 34C INDUCTOR S = 0 ===== L VALUES
H₂O	T_m	
	T_{s1}	
	T_{s2}	
	T_{s3}	
F_m	F_{s1}	
	F_{s2}	
	F_{s3}	
	This symbols appears beside P_{in} (as in the picture) when the generator is controlled in power and beside U_{coil} when it is controlled in voltage.	



If only one generator is connected, controlling two heads in alternate functioning, the values shown are referred to the last head being activated.

4.4.4.2 CEIA /SLE pyrometers parameters screenshot



In case none of the connected temperature sensor are CEIA /SL type, this screenshot will not be shown and the Controller will pass to the next screenshot.

1>	Model name for the temperature sensor connected to point 1.	CEIA /SL 1>SH15SLE553 SN:21301800001 200-1600C HW:1 RV:1 T:----- 2>SH15SLE554 SN:21301800002 500-2000C HW:1 RV:1 T:-----
2>	Model name for the temperature sensor connected to point 2.	
SN:	Serial number	
HW:	Hardware version	
RV:	Revision number	
T:	Real time temperature value. If "-----" the sensor is not reading, if "HIGH" the sensor is reading a value higher than the maximum allowed.	

4.4.4.3 UPP pyrometers parameters screenshot



In case none of the connected temperature sensor are UPP type, this screenshot will not be shown and the Controller will pass to the next screenshot.

1>	Model name for the UPP sensor connected to point 1.	UPP sensor 1>Sensor name ref:3 sw:XXXXXX sn:XXXXXX T:500 Ti:36 Tim:46 Fs:0 2>Sensor name ref:4 sw:XXXXXX sn:XXXXXX T:500 Ti:36 Tim:46 Fs:0
2>	Model name for the UPP sensor connected to point 2.	
ref:	Reference number	
sw:	Software version	
sn:	Serial number	
T:	Temperature read by the sensor	
Ti:	Actual inner temperature	
Tim:	Maximum inner temperature read by the pyrometer	
Fs:	Faults status (see Pyrometer manual for the relative messages)	

4.4.4.4 Thermal imaging camera parameters screenshot



In case that a thermal camera is not connected or enabled, this screenshot will not be shown and the Controller will pass to the next screenshot.

	Thermal camera model	THERMAL CAMERA PIROVIEW 160L ID:123ABCD 10.0C - 500.0C70.0Hz IP:192.192.192.192 ROI1:173.0C X:14 Y:32 E:100% ROI2:162.0C X:118 Y:109 E:95%	
ID:	Thermal camera serial number		
	Temperature range and frame rate		
IP:	IP address		
ROI1:	Region of Interest 1: Temperature read on the image point indicated by coordinates X,Y and set emissivity (E).		
ROI2:	Region of Interest 2: Temperature read on the image point indicated by coordinates X,Y and set emissivity (E)		

4.4.4.5 Diagnosis status

1>	Current diagnosis status on point 1	STATUS 1> OK PREVIOUS: A3 2> OK PREVIOUS: A3 MSTS3.028.0231-- S/N: 213040039999 OPT: WF
2>	Current diagnosis status on point 2	
PREVIOUS:	Diagnosis message previously shown on corresponding point	
	Software version currently installed	
S/N	Serial number	
OPT:	Installed options	

4.4.4.6 Logger parameters check



In case that the IXC card is not present, this screenshot will not be shown and the Controller will pass to the next screenshot.

F ILENAME	Name of the file where all data is actually saved	LOGGER F ILENAME: 2013-07-10_14_04_48_217(P0000-S2).CSV HOSTNAME: CEIA-000A290212DE.LOCAL TCP/IP ADDRESS: 10.0.200.196 STATUS: RUNNING PERIOD: 10 MS INFO DETAIL: N/A
HOSTNAME	Name of the host	
TCP/IP ADDRESS	IXC card network address	
STATUS	IXC card actual status	
PERIOD	Time period for data sampling	

4.4.4.7 Field Bus Status

Status	Status of the Field Bus module	F.B.STATUS Status: OK F.B. ID: MC3ST001 Net: EtherNet/IP SUP bit: NO DHCP: OFF STAT1: 255 STAT2: 255 IP: 192.168.3.11 NM: 255.255.255.0
F.B. ID	ID number of the Field Bus module	
Net:	Network type	
SUP bit	Activation of the PLC supervision for the connection between the Controller and the PLC itself	
DHCP	Status of the DHCP	
STAT1	Section 1 status	
STAT2	Section 2 status	
IP:	Module IP Address	
NM:	Network Mask	

4.4.4.8 Inputs and Outputs status

J7, J8 J12, J13	Digital inputs actual status	I/O J7 J8 J9 J10 J19-3=0.00V 0000 0000 000000 00 J19-5=0.0 mA 2468 2468 246811 2 5 J19-7=0.00V 02 J19-9=0.00V J12 J13 J14 J15 J20-1=0.00V 0000 0000 000000 00 J20-3=4.0 mA 2468 2468 246811 2 5 J20-5=0.00V 02 J20-7=0.00V
J9, J14	Digital outputs actual status	
J10, J15	Relays actual status	
J19-x	Real-time values for the analog inputs	
J20-x	Real-time values for the analog outputs	

4.4.5 Manual advancing of the soldering wire

Whenever necessary, it is possible to activate the feeding motor of the soldering wire forwards and backwards, in order to set up the work station or the unblock a possible jamming of the wire. This function is accessible through a programming menu (see related chapter).

4.4.6 Stop

When work is finished disconnect the supplies.

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4.5 Program Cards Management

The Master Controller is provided with a Program Card reader, where it is possible to load and save until 100 different sets of working parameters.

4.5.1 Loading a set of parameters from the Program card



Loading a new set of parameters from the Program Card, the actual set of parameters will be overwritten, without any possibility to restore them.

To insert the Program Card, hold it with the golden contacts down and towards the Controller, and insert it until the display shows the following message:.

LOADS NEW PARAMETERS
FROM CARD

c.1

E=Continue



Pressing **E** all parameters stored in the card, at the selected memory cell, will be restored in the Controller.

4.5.2 Saving a set of parameters on the Program Card

To save the parameters actually in use, on a Program Card, it is necessary to enter in

Programming mode, by pressing **P** and to select the menu voice MEMORY.

heating	mode
wire	> MEMORY
gas	manual

Select the command STORES PARAMETERS IN USE IN CARD.

The Controller will ask for the insertion of a Program Card into the reader.

STORES PARAMETERS
IN USE IN CARD

c.1

Insert program card

Immediately after the insertion of the card, the message will change indicating the possibility to proceed.

STORES PARAMETERS
IN USE IN CARD

c.3

Press E to start

Pressing **E**, the number beside the prefix c. (memory cell) start to blink and it is possible to change it using **↑** and **↓**.

Pressing again **E** will show a screen to enter a six characters name for the memory cell, then the parameters in use will be stored on the Program Card, at the selected memory cell.

4.6 Programming



The programming of the Master Controller parameters can only be carried out by specialized technical personnel who are authorised to access the programming section.

The access to programming parameters can be protected with two password levels: USER and SUPER USER.

When none of them is set, as per default, pressing **P** logs in as SUPER USER without any password request.

If the SUPER USER password is set, pressing **P** the request of its insertion is shown, before being able to access Programming.

Pressing **E** will allow to jump the password request and to access as USER to the programming parameters.

If both passwords are set, the programming access will be granted depending on the inserted password.

Pressing the **P** key, the programming phase is started and on the display is shown the main menu. The selected menu item is indicated with uppercase characters and the presence, on the left, of an arrow. On the bottom right corner of the display a letter shows the current access level: U=User S=Super User

Password

E: login as user

Password

> HEATING	mode
wire	memory
gas	manual

S

- the main menu (HEATING, WIRE, GAS ...) contains submenus that allow access to the operating parameters; scroll through the menus using the **↑** (increase) and **↓** (decrease) keys, and access the submenus by pressing the **E** key.
- the parameters are activated with the **E** key.
- the parameter values are modified using the **↑** (increase) and **↓** (decrease) keys.
- modifications are confirmed with the **E** key.
- return to the main menu and exit from programming with the **P** key.

There follows an example of the programming operation with the messages that appear on the display and the keys pressed:

Keys pressed	Displayed message	Meaning
	> HEATING mode wire memory gas manual	Pressing the  key once gives access to the main menu; the submenu selected is the one written in capital letters and marked by the arrow (>)
 	heating mode wire memory > GAS manual	the menu required is selected by means of the  and  keys: each menu selected appears in capital letters and the arrow moves to it
	HEATING DELAY 1 > 3,0 s < HEATING DELAY 2 3,5 s	access to the chosen menu is made by pressing the  key the function selected is marked by two arrows (> xxxxxx <)
 	COOLING TIME 1 5,0 s COOLING TIME 2 > 5,0 s <	the command desired is chosen by means of the  and  keys
 +  	COOLING TIME 1 5,0 s COOLING TIME 2 > 7,0 s <	change the selected value: press the  key and change the selected value with the  and  keys
	COOLING TIME 1 5,0 s COOLING TIME 2 > 7,0 s <	confirm the new value selected by pressing the  key
	heating mode wire memory > GAS manual	return to the preceding menu by pressing the  key and exit from the programming section by pressing the key again

4.6.1 Menu content



Only the significative parameters for the selected OPERATING MODE and for the options activated, are shown.

Legenda

Symb.	Option	Symb.	Functioning mode
	Wire Feeder Option		Timer-controlled with temperature control
	Thermal Profile Option		Timer-controlled with one heating phase
	IXC Option (web server)		Timer-controlled with two heating phases
	Field Bus Interface		Synchronised functioning
	Tin Soldering Option		Fully automatic functioning
	DIAS Option (thermocameras management)		Functioning with two soldering wire feeders
	Stand-Alone option		Functioning with thermal profile
	Sensors-Only option		Functioning with thermal profile and two soldering wire feeders
	Analog Sensors option		
	Synchronized Sections option		

4.6.1.1 HEATING

Parameters	Options	Functioning modes								Notes
TEMPERATURE 1/2	-TS	A	B		S	AUT	A2W			
MAX.TEMPERATURE 1/2		A	B		S					Only with TCTY=WR,WM
MIN.TEMPERATURE 1/2		A	B		S					Only with TCTY=WR,WM
LOW TEMPERATURE 1/2		A	B		S					Only with TCTY=WR,WM
EMISSIONITY A 1		A	B			AUT	A2W	TP		
EMISSIONITY A 2		A	B			AUT		TP		
EMISSIONITY B 1/2						AUT	A2W	TP	T2W	
EMIS. ADJUST DELAY 1/2						AUT	A2W	TP	T2W	
PRE-HEATING TIME 1						AUT	A2W			
PRE-HEATING TIME 2						AUT				
HOLDING TIME A 1		A	B	C	S	AUT	A2W			
HOLDING TIME A 2		A	B	C	S	AUT				
HOLDING TIME B 1						AUT	A2W			
HOLDING TIME B 2						AUT				
POWER A 1		A	B	C	S	AUT	A2W			
POWER A 2	A	B	C	S	AUT					
POWER B 1			C		AUT	A2W				
POWER B 2			C		AUT					
POWER A TIME 1			C		AUT	A2W				
POWER A TIME 2			C		AUT					

Parameters	Options	Functioning modes								Notes
NUMBER OF STEPS 1	-TP							TP	T2W	
NUMBER OF STEPS 2								TP		
H1 STEPxx: TYPE								TP	T2W	
H1 STEPxx: TIME MAX								TP	T2W	
H1 STEPxx: TIME								TP	T2W	
H1 STEPxx: POWER								TP	T2W	Only parameters related to the steps of the heating cycle are visible according to the value of the corresponding NUMBER OF STEPS parameter
H1 STEPxx: POWER MAX								TP	T2W	
H1 STEPxx: TEMP.MAX								TP	T2W	
H1 STEPxx: TEMP.FIN								TP	T2W	
H1 STEPxx: TOLERAN								TP	T2W	
H2 STEPxx: TYPE								TP		Only parameters related to the steps of the heating cycle are visible according to the value of the corresponding NUMBER OF STEPS parameter
H2 STEPxx: TIME MAX								TP		
H2 STEPxx: TIME								TP		
H2 STEPxx: POWER								TP		
H2 STEPxx: POWER MAX								TP		
H2 STEPxx: TEMP.MAX								TP		
H2 STEPxx: TEMP.FIN								TP		
H2 STEPxx: TOLERAN								TP		
H1/2 STEPxx: WIRE								TP	T2W	
H1/2 T.OUT TOLER: SIGNAL								TP		
H1/2 T.OUT TOLER: TIME								TP		
TEMP SENSOR TYPE 1/2	-PYAN	A	B		S	AUT	A2W	TP	T2W	
AN.TEMP. MIN. 1		A	B					TP		Visible only if TEMP SENSOR TYPE = ANA
AN.TEMP. MAX. 1		A	B					TP		
AN.TEMP. MIN. 2		A	B					TP		Visible only if TEMP SENSOR TYPE = ANA
AN.TEMP. MAX. 2		A	B					TP		
PYROM. DIAGNOSIS 1		A	B		S	AUT	A2W	TP	T2W	Visible only if TEMP SENSOR TYPE = /SL
PYROM. DIAGNOSIS 2		A	B		S	AUT		TP		
CLOSE-UP LENS PYR.1		A	B		S	AUT	A2W	TP	T2W	Visible only if TEMP SENSOR TYPE = /SL
CLOSE-UP LENS PYR.2		A	B		S	AUT		TP		
UPP1/2 TEMP.RANGE LOW		A	B		S	AUT	A2W	TP	T2W	
UPP1/2 TEMP.RANGE HIGH		A	B		S	AUT	A2W	TP	T2W	Visible only if TEMP SENSOR TYPE = UPP and sensor is connected
UPP1/2 EXPOSIT.TIME		A	B		S	AUT	A2W	TP	T2W	
DIAS TIMEOUT	-DIAS	A	B		S	AUT	A2W	TP	T2W	Visible only if TEMP SENSOR TYPE = THC
DIAS SHUTTER T1/2		A	B		S	AUT	A2W	TP	T2W	
DIAS IP		A	B		S	AUT	A2W	TP	T2W	
AIMING LIGHT		A	B		S	AUT	A2W	TP	T2W	
SOLDERING COUNTER 1/2		A	B	C	S	AUT	A2W	TP	T2W	Visible only with a /900 generator connected
SOLDERING COUNTING		A	B	C	S	AUT	A2W	TP	T2W	
CONTROL TYPE		A	B	C	S	AUT	A2W	TP	T2W	
HEATING TIMEOUT		A	B	C	S	AUT	A2W	TP	T2W	
TEMPERATURE TIMEOUT 1/2		A	B		S	AUT	A2W	TP	T2W	
TEMPERATURE SETTING 1/2		A	B		S	AUT	A2W	TP	T2W	
POWER SETTING 1/2		A	B	C	S	AUT	A2W	TP	T2W	
VERIFY CALIBRATION		A	B	C	S	AUT	A2W	TP	T2W	
TEMPERATURE CONTROL 1/2		A	B		S	AUT	A2W	TP	T2W	
PID P 1/2		A	B		S	AUT	A2W	TP	T2W	Visible only with TEMPERATURE CONTROL = PID or TEMP SENSOR TYPE = S, K or C
PID I 1/2		A	B		S	AUT	A2W	TP	T2W	
PID D 1/2		A	B		S	AUT	A2W	TP	T2W	
PID T 1/2		A	B		S	AUT	A2W	TP	T2W	
POWER INCREASE 1/2		A	B		S					Only with TCTY=WR,WM
CONTROL TIME 1/2		A	B		S					Only with TCTY=WR,WM
TEMPERATURE DISPLAY		A	B		S					Only with TCTY=WR,WM
TEMPERATURE DELAY 1/2		A	B		S					Only with TCTY=WR,WM
TEMPER. SETTING		A	B	C	S	AUT	A2W	TP	T2W	

4.6.1.2 WIRE

Parameters	Options	Functioning modes								Notes
FEED A 1/2	WF TS					AUT	A2W	TP	T2W	
FEED RATE A 1/2						AUT	A2W	TP	T2W	
FEED B 1/2	TS					AUT	A2W	TP	T2W	
FEED RATE B 1/2						AUT	A2W	TP	T2W	
REWIND LENGTH 1/2	WF TS					AUT	A2W	TP	T2W	
REWIND RATE 1/2						AUT	A2W	TP	T2W	
CYLINDER DELAY 1/2						AUT	A2W	TP	T2W	
INJECTION CYCLE 1/2						AUT	A2W	TP	T2W	
INJECTION TIME 1/2						AUT	A2W	TP	T2W	
WIRE ADJUST FACTOR 1/2						AUT	A2W	TP	T2W	
MANUAL FEED RATE						AUT	A2W	TP	T2W	
MANUAL REWIND RATE						AUT	A2W	TP	T2W	
CYL STROKE TIME						AUT	A2W	TP	T2W	
MAX CURRENT 1/2						AUT	A2W	TP	T2W	
WIRE TIMEOUT						AUT	A2W	TP	T2W	
WIRE PRESENCE						AUT	A2W	TP	T2W	
FORWARD FEED						AUT	A2W	TP	T2W	
INJECTION						AUT	A2W	TP	T2W	
OUT J9-6 J14-6						AUT	A2W	TP	T2W	

4.6.1.3 GAS

Parameters	Options	Functioning modes								Notes
HEATING DELAY 1		A	B	C	S	AUT	A2W	TP	T2W	
HEATING DELAY 2		A	B	C	S	AUT		TP		
COOLING TIME 1		A	B	C	S	AUT	A2W	TP	T2W	
COOLING TIME 2		A	B	C	S	AUT		TP		
GAS TIMEOUT		A	B	C	S	AUT	A2W	TP	T2W	

4.6.1.4 MODE

Parameters	Options	Functioning modes								Notes
OPERATING MODE		A	B	C	S	AUT	A2W	TP	T2W	
HEATING SYSTEM		A	B	C	S	AUT	A2W	TP	T2W	
SECTIONS SYNCHRON.	-SYNS							TP		
SINGLE POWER CUBE		A	B	C	S	AUT	A2W	TP	T2W	Visible only if HEATING SYSTEM = SINGLE GENERATOR
HEAD 1 MODE		A	B	C	S	AUT	A2W	TP	T2W	
TEMPERATURE CONTROL						AUT	A2W	TP	T2W	
EMIS. ADJUSTMENT						AUT	A2W	TP	T2W	
LANGUAGE		A	B	C	S	AUT	A2W	TP	T2W	
+V SUPPLY		A	B	C	S	AUT	A2W	TP	T2W	
INPUT LOGIC		A	B	C	S	AUT	A2W	TP	T2W	
CHANGE PASSWORD		A	B	C	S	AUT	A2W	TP	T2W	
CHANGE QUICK ACCESS PASSWORD		A	B	C	S	AUT	A2W	TP	T2W	
USER PARAMETERS		A	B	C	S	AUT	A2W	TP	T2W	
RESET USER PASSWORD		A	B	C	S	AUT	A2W	TP	T2W	
SETUP QUICK ACCESS		A	B	C	S	AUT	A2W	TP	T2W	
ANALOG INPUT J19-x		A	B	C	S	AUT	A2W	TP	T2W	
ANALOG OUTPUT J20-x		A	B	C	S	AUT	A2W	TP	T2W	
AN.INP.NOISE REJ.		A	B	C	S	AUT	A2W	TP	T2W	
DEBOUNCE TIME		A	B	C	S	AUT	A2W	TP	T2W	
TEMP.DECIMAL DIGITS		A	B	C	S	AUT	A2W	TP	T2W	
READY SIGNAL MODE	-IXC	A	B	C	S	AUT	A2W	TP	T2W	
REVERSE DISPLAY		A	B	C	S	AUT	A2W	TP	T2W	
F.B. ID	-FB	A	B	C	S	AUT	A2W	TP	T2W	Fieldbus settings
F.B. DHCP		A	B	C	S	AUT	A2W	TP	T2W	
F.B. IP		A	B	C	S	AUT	A2W	TP	T2W	
F.B. NET.MASK		A	B	C	S	AUT	A2W	TP	T2W	
F.B. ENDIANNESS		A	B	C	S	AUT	A2W	TP	T2W	
F.B.ADDRESS		A	B	C	S	AUT	A2W	TP	T2W	
F.B. VERSION		A	B	C	S	AUT	A2W	TP	T2W	
M.C. HOSTNAME	-IXC	A	B	C	S	AUT	A2W	TP	T2W	Master Controller settings
M.C. DHCP		A	B	C	S	AUT	A2W	TP	T2W	
M.C. IP		A	B	C	S	AUT	A2W	TP	T2W	
M.C. NETMASK		A	B	C	S	AUT	A2W	TP	T2W	
M.C. GATEWAY		A	B	C	S	AUT	A2W	TP	T2W	

4.6.1.5 MEMORY

Parameters	Options	Functioning modes								Notes
STORES PARAMETERS IN USE IN CARD		A	B	C	S	AUT	A2W	TP	T2W	
STORES PARAMETERS IN USE IN THE CELL		A	B	C	S	AUT	A2W	TP	T2W	
LOADS NEW PARAMETERS FROM THE CELL		A	B	C	S	AUT	A2W	TP	T2W	
CLEAR PROGRAM CARD		A	B	C	S	AUT	A2W	TP	T2W	
BACKUP MEMORY TO PROGRAM CARD		A	B	C	S	AUT	A2W	TP	T2W	
RESTORE MEMORY FROM PROGRAM CARD		A	B	C	S	AUT	A2W	TP	T2W	
BINARY INP.CELL SEL		A	B	C	S	AUT	A2W	TP	T2W	
DOWNLOAD / UPLOAD CELLS	-IXC	A	B	C	S	AUT	A2W	TP	T2W	Commands available only with a remote connection
MOVE WORKING CELLS		A	B	C	S	AUT	A2W	TP	T2W	

4.6.1.6 MANUAL

Parameters	Options	Functioning modes				Notes
CYL 1	WF TS				AUT A2W TP T2W	
CYL 2	WF TS				AUT A2W TP T2W	
CYL 1 AND 2	WF TS				AUT A2W TP T2W	
WIRE 1	WF TS				AUT A2W TP T2W	
WIRE 2	WF TS				AUT A2W TP T2W	

4.6.2 Parameters list

In the following list, the presence of a symbol in the **Options** column, indicates that this parameter is available only for the option indicated by the symbol itself, and where none of the symbols is present, that parameter is available for all options.

Each parameter shall be send to the Controller, following the syntax:

[Command] <space> [value] <enter>

```
>
> OPM AUTO
```



All values set with the commands of the open serial interface, are not saved for the next power-up. To save them it's necessary to use the command ST x.
If saved with ST 0, they will be automatically available at the next power-up.

4.6.2.1 HEATING parameters

Parameter	Remote	Possible settings	Standard setting	Options
TEMPERATURE 1	TEM1	according to the kind of temperature sensor	600	
TEMPERATURE 2	TEM2			
This parameter selects the temperature that must be reached by the piece being worked upon. If HEATING > TEMPERATURE CONTROL 1/2 = WR or WM, this parameter is not visible.				

Parameter	Remote	Possible settings	Standard setting	Options
MAX.TEMPERATURE 1	TEM1	according to the kind of temperature sensor	600	
MAX.TEMPERATURE 2	TEM2			
This parameter selects the maximum temperature that must be reached by the piece being worked upon. This parameter is visible only with HEATING > TEMPERATURE CONTROL 1/2 = WR or WM.				

Parameter	Remote	Possible settings	Standard setting	Options
MIN.TEMPERATURE 1	TEM1L	according to the kind of temperature sensor	600	
MIN.TEMPERATURE 2	TEM2L			
This parameter selects the minimum temperature that must be reached by the piece being worked upon. This parameter is visible only with HEATING > TEMPERATURE CONTROL 1/2 = WR or WM .				

Parameter	Remote	Possible settings	Standard setting	Options
LOW TEMPERATURE 1	TEM1C	0 ÷ Max. according to the kind of temperature sensor	600	
LOW TEMPERATURE 2	TEM2C			
Threshold value used to synchronize the acquisition of a new max value. This parameter is visible only with HEATING > TEMPERATURE CONTROL 1/2 = WR or WM .				

Parameter	Remote	Possible settings	Standard setting	Options
EMISSIONITY A 1	EMA1	0.10 ÷ 1.00	0.70	
EMISSIONITY A 2	EMA2			
This parameter selects the value of emissivity of the piece being worked upon. It lets the optical pyrometer have a correct reading of temperature. See also the parameter EMIS. ADJUSTMENT .				

Parameter	Remote	Possible settings	Standard setting	Options
EMISSIONITY B 1	EMB1	0.10 ÷ 1.00	0.70	
EMISSIONITY B 2	EMB2			
This parameter selects the value of emissivity of the piece being worked upon, during the second phase of the heating. It lets the optical pyrometer have a correct reading of temperature, during the soldering wire melting phase.				
 This command is available only with EMIS. ADJUSTMENT = YES .				

Parameter	Remote	Possible settings	Standard setting	Options
EMIS. ADJUST DELAY 1	EAD1	0.1 – 10.0 s	2.0	
EMIS. ADJUST DELAY 2	EAD2			
Select the first phase length (in seconds) of the heating, performed using the coefficient EMISSIVITY A .				
 This command is available only with EMIS. ADJUSTMENT = YES				

Parameter	Remote	Possible settings	Standard setting	Options
PRE HEATING TIME 1	PH1	0.0 – 120.0 s	0.0	
PRE HEATING TIME 2	PH2			
Select the delay between the heating start and the beginning of the welding phase. Timing: ph . Steps of 1/10 s if below 10 seconds and steps of 1 s if over 10 seconds.				
 This parameter applies only in operating modes: AUT and A2W , with TEMPERATURE CONTROL = OFF .				

Parameter	Remote	Possible settings	Standard setting	Options
HOLDING TIME A 1	HTA1	0.00 s – 2250 hours	0.0	
HOLDING TIME A 2	HTA2			
See paragraph Timing for the explanation of the various operating modes. Timing: ta . From 0 to 10 s with steps of 1/100 s; from 10 to 100 s with steps of 1/10 s; from 101 to 600 seconds with steps of 1 s; from 11 to 2160 minutes with steps of 1 minute; from 36 to 2250 hours with steps of 1 hour.				

Parameter	Remote	Possible settings	Standard setting	Options
HOLDING TIME B 1	HTB1	0.00 s – 2250 hours	0.0	
HOLDING TIME B 2	HTB2			
See paragraph Timing for the explanation of the various operating modes. Timing: tb . From 0 to 10 s with steps of 1/100 s; from 10 to 100 s with steps of 1/10 s, from 101 to 600 seconds with steps of 1 s; from 11 to 2160 minutes with steps of 1 minute; from 36 to 2250 hours with steps of 1 hour.				
 This parameter applies only in operating modes: AUT and A2W .				

Parameter	Remote	Possible settings	Standard setting	Options
POWER A 1	PWA1	10 ÷ 99 %	10	
POWER A 2	PWA2			
In the operating modes with temperature control (TEMPERATURE CONTROL = ON) this parameter represents the degree of power (with reference to the maximum power that the apparatus can deliver) with which the piece is heated during a work cycle; naturally, the greater the power supplied to the piece the less the time needed for its heating. In the operating modes without temperature control (TEMPERATURE CONTROL = OFF) this parameter represents the power value (with reference to the maximum power output of the machine) with which the piece is heated during the course of the first heating phase (see previous paragraph). If the Controller is connected to a series 50 generator, and only upon request, the minimum possible value can be 2%.				

Parameter	Remote	Possible settings	Standard setting	Options
POWER B 1	PWB1	10 ÷ 99 %	10	
POWER B 2	PWB2			
This parameter represents the power value (with reference to the maximum power output of the machine) with which the piece is heated during the course of the second heating phase (see previous paragraph); the greater the power supplied to the piece the less the time needed for its heating. If the Controller is connected to a series 50 generator, and only upon request, the minimum possible value can be 2%.  This parameter applies only in operating modes: AUT , A2W and C with TEMPERATURE CONTROL = OFF.				

Parameter	Remote	Possible settings	Standard setting	Options
POWER A TIME 1	PAT1	0.0 – 600.0 s	1.0	
POWER A TIME 2	PAT2			
This parameter represents the duration of the "first heating phase" when the piece is being heated at the first power level, fixed by POWER A 1 [2] parameter. Timing: pl. Steps of 1/10 s if below 100 seconds and steps of 1 s if over 100 seconds.				
 This parameter applies only in operating modes: AUT, A2W and C with TEMPERATURE CONTROL = OFF.				

Parameter	Remote	Possible settings	Standard setting	Options
NUMBER OF STEPS 1	NST1	1 ÷ 20	2	
NUMBER OF STEPS 2	NST2			
Select the number of steps to be performed for the thermal profiling of each head.				
 This parameter is visible only in operating modes: TP and T2W .				

Parameter	Remote	Possible settings	Standard setting	Options
H1 STEPxx: TYPE	H1SxxTY	HEATING (HE) TEMP. CONTROL (TC) COOLING (CO)	HEATING (HE)	TP
Set the step type for step xx, section 1. This parameter is visible only in operating modes: TP and T2W.				

Parameter	Remote	Possible settings	Standard setting	Options
H1 STEPxx: TIME MAX	H1SxxTI	0.00 s – 2250 hours	1	TP
Set the timeout for step xx, on section 1. From 0 to 10 s with steps of 1/100 s; from 10 to 100 s with steps of 1/10 s; from 101 to 600 seconds with steps of 1 s; from 11 to 2160 minutes with steps of 1 minute; from 36 to 2250 hours with steps of 1 hour. This parameter is visible only in operating modes: TP and T2W.				

Parameter	Remote	Possible settings	Standard setting	Options
H1 STEPxx: TIME	H1SxxTI	0.00 s – 2250 hours	1	(TP)
Set the time needed to reach the set temperature on step xx, section 1. From 0 to 10 s with steps of 1/100 s; from 10 to 100 s with steps of 1/10 s; from 101 to 600 seconds with steps of 1 s; from 11 to 2160 minutes with steps of 1 minute; from 36 to 2250 hours with steps of 1 hour.				
 This parameter is visible only in operating modes: (TP) and (T2W) .				

Parameter	Remote	Possible settings	Standard setting	Options
H1 STEPxx: POWER	H1SxxPO	10 ÷ 99 %	10	(TP)
Set the generator power during step xx on section 1. If the Controller is connected to a series 50 generator, and only upon request, the minimum possible value can be 2%.				
 This parameter is visible only in operating modes: (TP) and (T2W) .				

Parameter	Remote	Possible settings	Standard setting	Options
H1 STEPxx: POWER MAX	H1SxxPO	10 ÷ 99 %	10	(TP)
Set the maximum generator power for step xx on section 1. If the Controller is connected to a series 50 generator, and only upon request, the minimum possible value can be 2%.				
 This parameter is visible only in operating modes: (TP) and (T2W) .				

Parameter	Remote	Possible settings	Standard setting	Options
H1 STEPxx: TEMP. MAX	H1SxxTE	according to the kind of temperature sensor	600	(TP)
Set the maximum temperature to be reached during step xx, on section 1.				
 This parameter is visible only in operating modes: (TP) and (T2W) .				

Parameter	Remote	Possible settings	Standard setting	Options
H1 STEPxx: TEMP.FIN	H1SxxTE	according to the kind of temperature sensor	600	(TP)
Set the final temperature to be reached on step xx, section 1.				
 This parameter is visible only in operating modes: (TP) and (T2W) .				

Parameter	Remote	Possible settings	Standard setting	Options
H1 STEPxx: TOLERAN	H1SxxTO	0 – 500°C	0	(TP)
Set the temperature allowed tolerance, in positive and in negative, during step xx, on section 1.				
 This parameter is visible only in operating modes: (TP) and (T2W) .				

Parameter	Remote	Possible settings	Standard setting	Options
H2 STEPxx: TYPE	H2SxxTY	HEATING (HE) TEMP.CONTROL. (TC)	HEATING (HE)	(TP)
Set the step type for step xx, section 2.				
 This parameter is visible only in operating mode: (TP) .				

Parameter	Remote	Possible settings	Standard setting	Options
H2 STEPxx: TIME MAX	H2SxxTI	0.0 s – 2250 hours	1	(-TP)
Set the timeout for step xx, on section 2. From 0 to 100 s with steps of 1/10 s; from 101 to 600 seconds with steps of 1 s; from 11 to 2160 minutes with steps of 1 minute; from 36 to 2250 hours with steps of 1 hour.				
This parameter is visible only in operating mode: TP .				

Parameter	Remote	Possible settings	Standard setting	Options
H2 STEPxx: TIME	H2SxxTI	0.0 s – 2250 hours	1	(-TP)
Set the time needed to reach the set temperature on step xx, section 2. From 0 to 100 s with steps of 1/10 s; from 101 to 600 seconds with steps of 1 s; from 11 to 2160 minutes with steps of 1 minute; from 36 to 2250 hours with steps of 1 hour.				
This parameter is visible only in operating mode: TP .				

Parameter	Remote	Possible settings	Standard setting	Options
H2 STEPxx: POWER	H2SxxPO	10 ÷ 99 %	10	(-TP)
Set the generator power during step xx on section 2. If the Controller is connected to a series 50 generator, and only upon request, the minimum possible value can be 2%.				
This parameter is visible only in operating mode: TP .				

Parameter	Remote	Possible settings	Standard setting	Options
H2 STEPxx: POWER MAX	H2SxxPO	10 ÷ 99 %	10	(-TP)
Set the maximum generator power for step xx on section 2. If the Controller is connected to a series 50 generator, and only upon request, the minimum possible value can be 2%.				
This parameter is visible only in operating mode: TP .				

Parameter	Remote	Possible settings	Standard setting	Options
H2 STEPxx: TEMP. MAX	H2SxxTE	according to the kind of temperature sensor	600	(-TP)
Set the maximum temperature to be reached during step xx, on section 2.				
This parameter is visible only in operating mode: TP .				

Parameter	Remote	Possible settings	Standard setting	Options
H2 STEPxx: TEMP.FIN	H2SxxTE	according to the kind of temperature sensor	600	(-TP)
Set the final temperature to be reached on step xx, section 2.				
This parameter is visible only in operating mode: TP .				

Parameter	Remote	Possible settings	Standard setting	Options
H2 STEPxx: TOLERAN	H2SxxTO	0 – 500°C	0	(-TP)
Set the temperature allowed tolerance, in positive and in negative, during step xx, on section 2.				
This parameter is visible only in operating mode: TP .				

Parameter	Remote	Possible settings	Standard setting	Options
H1 STEP: WIRE	FDS1	0 – 20	0	
H2 STEP: WIRE	FDS2			
Set in which step the soldering wire is fed.				
 This parameter is visible only in operating modes: TP and T2W .				

Parameter	Remote	Possible settings	Standard setting	Options
H1 T.OUT TOLER:SIGNAL	OTS1	END CYCLE (E) IMMEDIATE (I)	END CYCLE (E)	
H2 T.OUT TOLER:SIGNAL	OTS2			
Activation of the J15 output in case of a temperature out of range.				

Parameter	Remote	Possible settings	Standard setting	Options
H1 T.OUT TOLER.:TIME	OTT1	0.00 – 5.00 s	-	
H2 T.OUT TOLER.:TIME	OTT2			
In case the sensors read a temperature out of range, this shall last a minimum time as set in these parameters, in order to activate the “temperature out of range” event. Steps of 1/100 s.				

Parameter	Remote	Possible settings	Standard setting	Options
TEMP SENSOR TYPE 1	TSN1	CEIA /SL (/SL) THERMOCOUPLE K (TCK) THERMOCOUPLE S (TCS) THERMOCOUPLE C (TCC) UPP SENSOR (UPP) THERMAL CAMERA (THC) ANALOG (ANA)	CEIA /SL (/SL)	
TEMP SENSOR TYPE 2	TSN2			
It selects the temperature sensor installed.				
CEIA /SL	CEIA temperature sensors SH/SL series			
TERMOCOPPIA K	Type K insulated thermocouples from 50 to 1300°C			
TERMOCOPPIA S	Type S insulated thermocouples from 50 to 1700°C			
TERMOCOPPIA C	Type C insulated thermocouples from 50 to 2320°C			
SENSORE UPP	Pyrometers with UPP® protocol			
TERMOCAMERA	DIAS Thermal camera, connected via Ethernet		This is available only with  option	
ANALOG	External analog temperature sensors (0-10 V, 0-20 mA, 4-20 mA)		This is available only with  option	

Parameter	Remote	Possible settings	Standard setting	Options
AN.TEMP.MIN 1	ATE1N	1 – 4096	1	
AN.TEMP.MIN 2	ATE2N			
Select the temperature corresponding to the minimum value of the analog input.				
 This parameter is visible only if TEMP SENSOR TYPE 1/2 = ANALOG .				

Parameter	Remote	Possible settings	Standard setting	Options
AN.TEMP.MAX 1	ATE1X	1 – 4096	1	
AN.TEMP.MAX 2	ATE2X			
Select the temperature corresponding to the maximum value of the analog input.				
 This parameter is visible only if TEMP SENSOR TYPE 1/2 = ANALOG .				

Parameter	Remote	Possible settings	Standard setting	Options
PYROM. DIAGNOSIS 1	P1DIAG	ENABLED (Y) DISABLED (N)	DISABLED (N)	
PYROM. DIAGNOSIS 2	P2DIAG			
When this parameter is ENABLED, if the corresponding sensor is not connected a fault message is shown on the display and the cycle do not start.				

Parameter	Remote	Possible settings	Standard setting	Options
CLOSE-UP LENS PYR.1	P1CU	YES (Y) NO (N)	NO (N)	
CLOSE-UP LENS PYR.2	P2CU			
If the temperature sensor is of CEIA /SL type, fitted with close-up lens, this parameter shall be set in order to allows the system to apply the relative reading correction.				

Parameter	Remote	Possible settings	Standard setting	Options
UPP 1 TEMP.RANGE LOW	TRL1	according to the kind of UPP temperature sensor	-	
UPP 2 TEMP.RANGE LOW	TRL2			
Inside the UPP® sensor range limits, it is possible to define, through this parameters, a different range.				

Parameter	Remote	Possible settings	Standard setting	Options
UPP 1 TEMP.RANGE HIGH	TRH1	according to the kind of UPP temperature sensor	-	
UPP 2 TEMP.RANGE HIGH	TRH2			
Inside the UPP® sensor range limits, it is possible to define, through this parameters, a different range.				

Parameter	Remote	Possible settings	Standard setting	Options
UPP 1 EXPOSIT.TIME	ET1	0 ÷ 6	0	
UPP 2 EXPOSIT.TIME	ET2			
Acquisition range for the calculation of the average temperature. The value of 0 represent the direct reading, without the average calculation.				

Parameter	Remote	Possible settings	Standard setting	Options
DIAS TIMEOUT	DIATO	0.5 ÷ 3.0	1.5	
Set the data reading timeout on DIAS thermal camera.				

Parameter	Remote	Possible settings	Standard setting	Options
DIAS SHUTTER T1	DISH1	0 – 1000	0	
DIAS SHUTTER T2	DISH2			

Manual control of thermal camera shutter, so as to synchronize it with the moments the heating cycle is idle.
If set to 0, the control is performed with the fixed times set on the thermal camera.
If set to values higher than 0, the shutter is managed by the Controller, as following:

- In the first 15 minutes from the moment the thermal camera connection is active, a shutter each T1 seconds is performed, afterwards a shutter each T2 seconds (for example T1=60s, T2=900s).
- If a heating cycle using the thermal camera is running, when T1 or T2 expires, the shutter is postponed until the end of the cycle.
- If the start of a new cycle, using the thermal camera, is requested before DIAS TIMEOUT seconds from the last shutter performed, the generator start is postponed until the DIAS TIMEOUT is not completely expired.



With DISH1 and DISH2 values higher than 0, the thermal camera automatic shutter **shall be disabled by the operator**, following the thermal camera instructions. If not, the automatic shutter will be performed together with the manual one, managed by the Controller.

If the automatic shutter has not been disabled on the thermal camera, and DISH1 and DISH2 are set to 0, the thermal camera is never performing a shutter, resulting in a fault after a brief time.

Parameter	Remote	Possible settings	Standard setting	Options
DIAS IP	DIAIP	IPv4 address	-	
Setting of the IP address where a DIAS thermocamera should be.  WARNING! The indicated address shall be on the same subnet of the IXC board. See parameter MCIP.				

Parameter	Remote	Possible settings	Standard setting	Options
AIMING LIGHT	LAS	ON (Y) OFF (N) AUTO (AUTO)	AUTO (AUTO)	
This command activates and deactivates the light pointing device, while keeping the temperature measuring function permanently active. ON: Aiming light always ON; OFF: Aiming light always OFF; AUTO: Aiming light ON, but automatically OFF during the execution of the heating cycle;				

Parameter	Remote	Possible settings	Standard setting	Options
SOLDERING COUNTER 1	-	-	-	
SOLDERING COUNTER 2	-			
Displays the total number of solderings effected by each PowerCube connected. The two counters can be reset (contemporary) pressing the E key.				
 This parameter is visible only if the Controller is connected to a /900 series generator.				

Parameter	Remote	Possible settings	Standard setting	Options
SOLDERING COUNTING	SCNTING	Y / N	Y	
This command activates and deactivates the soldering counters shown with the instruction SOLDERING COUNTER.  This parameter is visible only if the Controller is connected to a /900 series generator.				

Parameter	Remote	Possible settings	Standard setting	Options
CONTROL TYPE	_C	VOLTAGE (V) CURRENT (C)	V	
Selects the type of generator control over the power stages: VOLTAGE: voltage control; CURRENT: current control. The standard value is VOLTAGE (voltage control) for generators of series 900 and 200, CURRENT (current control) for those of serie 50.				
 THIS PARAMETER SHOULD ONLY BE MODIFIED ON THE EXPRESS INSTRUCTIONS OF CEIA				

Parameter	Remote	Possible settings	Standard setting	Options
HEATING TIMEOUT	HTO	0.00 s – 2250 hours	0	
Selects the maximum generator activation time, after which the work cycle is halted.				
 Selecting 0 there is not a maximum activation time, allowing the continued operation.				
From 0 to 10 s with steps of 1/100 s; from 10 to 100 s with steps of 1/10 s; from 101 to 600 seconds with steps of 1 s; from 11 to 2160 minutes with steps of 1 minute; from 36 to 2250 hours with steps of 1 hour.				

Parameter	Remote	Possible settings	Standard setting	Options
TEMPERATURE TIMEOUT 1	TTO1	0.00 s – 2250 hours	0	
TEMPERATURE TIMEOUT 2	TTO2			

If the time needed to reach the work cycle (or Thermal Profile first step) set temperature is higher than the time set here, the Controller stop the cycle and show the message “A14 Heating timeout”.

This control is disabled if set to 0.

From 0 to 10 s with steps of 1/100 s; from 10 to 100 s with steps of 1/10 s; from 101 to 600 seconds with steps of 1 s; from 11 to 2160 minutes with steps of 1 minute; from 36 to 2250 hours with steps of 1 hour.

Parameter	Remote	Possible settings	Standard setting	Options
TEMPER.1 SETTING	TEM1SET	ANALOG, DIGITAL, REMOTE	DIGITAL	
TEMPER.2 SETTING	TEM2SET			

Sets the type of parameter control to: **TEMPERATURE 1** and **TEMPERATURE 2**.

If the **DIGITAL** control option is set, the system will control the temperature as a function of the values set during programming of the heating cycle.

If the **ANALOG** control option is set, the system will control the temperature as a function of the corresponding analog input. See “**Temperature regulation**” paragraph.

The **REMOTE** value is visible only if the FieldBus is enabled and F.B.VERSION=2. If the **REMOTE** control option is set, the system will control the temperature as a function of the corresponding value received on the fieldbus at the cycle start.

If the –TP option is active, the temperature value is updated even during the cycle.

Parameter	Remote	Possible settings	Standard setting	Options
POWER 1 SETTING	PW1SET	ANALOG, DIGITAL, REMOTE	DIGITAL	
POWER 2 SETTING	PW2SET			
Sets the type of parameter control to: POWER 1 and POWER 2 .				
If the DIGITAL control option is set, the system will control the power as a function of the values set during programming of the heating cycle.				
If the ANALOG control option is set, the system will control the power as a function of the corresponding analog input. See “ Power regulation ” paragraph.				
The REMOTE value is visible only if the FieldBus is enabled and F.B.VERSION=2. If the REMOTE control option is set, the system will control the generator power as a function of the corresponding value received on the fieldbus.				

Parameter	Remote	Possible settings	Standard setting	Options
VERIFY CALIBRATION	-	-	-	
This command activates the calibration verification procedure for the temperature sensors inputs.				

Parameter	Remote	Possible settings	Standard setting	Options
TEMPERATURE CONTROL 1	TCTY1	AUTO (AUTO) P.I.D. (PID)	AUTO	
TEMPERATURE CONTROL 2	TCTY2	WINDOW LIVE VAL. (WR) WINDOW MAX VAL. (WM)		

Select the kind of control for the temperature sensor.

If set to AUTO, when a pyrometer is set as temperature sensor, a continuous control will be executed, when a thermocouple is set, the temperature control is performed via P.I.D. algorithm.

If set to P.I.D., this kind of control will be executed also on a pyrometer.

WINDOW LIVE VAL. can be used when heating a piece in continuous movement.

WINDOW MAX VAL. can be used to heat consecutive pieces with or without an empty space in between or a single piece in a discontinuous movement.

 The WR and WM settings are visible only in operating modes: **A**, **B** and **S**.

Parameter	Remote	Possible settings	Standard setting	Options
PID P 1	PIDP1	0 ÷ 1023	9	
PID P 2	PIDP2			
The greater is the value of this coefficient, the lower is the influence of the proportional part of the control on the generator power correction.				
 Parameter visible only with TEMPERATURE CONTROL = PID or TEMP.SENSOR TYPE = K, S or C.				

Parameter	Remote	Possible settings	Standard setting	Options
PID I 1	PIDI1	0 ÷ 1023	40	
PID I 2	PIDI2			
The greater is the value of this coefficient, the higher is the influence of the integral part of the control on the generator power correction.				
 Parameter visible only with TEMPERATURE CONTROL = PID or TEMP.SENSOR TYPE = K, S or C.				

Parameter	Remote	Possible settings	Standard setting	Options
PID D 1	PIDD1	0 ÷ 1023	5	
PID D 2	PIDD2			
The greater is the value of this coefficient, the higher is the influence of the derivative part of the control on the generator power correction.				
 Parameter visible only with TEMPERATURE CONTROL = PID or TEMP.SENSOR TYPE = K, S or C.				

Parameter	Remote	Possible settings	Standard setting	Options
PID T 1	PIDT1	1 ÷ 256	1	
PID T 2	PIDT2			
Select how much time lasts between each cycle of temperature reading and generator power corrections. The time corresponds to the value set multiplied for 144ms for thermocouples, and 25ms for pyrometers.				
 Parameter visible only with TEMPERATURE CONTROL = PID or TEMP.SENSOR TYPE = K, S or C.				

Parameter	Remote	Possible settings	Standard setting	Options
POWER INCREASE 1	DPW1	0 – 20 %	0	
POWER INCREASE 2	DPW2			
This parameter indicates how much to increase or to decrease the generator power, for each control step. If set to 0, the power must be changed manually.				
 This parameter is available only if TEMPERATURE CONTROL is set to WR or WM .				

Parameter	Remote	Possible settings	Standard setting	Options
CONTROL TIME 1	TCNT1	0.0 – 20.0 s	0.0	
CONTROL TIME 2	TCNT2			
This value represents the time where the average temperature is evaluated, before deciding if the generator power needs to be changed.				
 This parameter is available only if TEMPERATURE CONTROL is set to WR or WM .				

Parameter	Remote	Possible settings	Standard setting	Options
TEMPERATURE DISPLAY	WMTD	AUTO (AUTO) AVG (AVG)	AUTO	
Select the temperature display mode: AUTO: If TEMPERATURE CONTROL 1/2 = WR the temperature shown is in real time, if TEMPERATURE CONTROL 1/2 = WM the temperature shown is the maximum of the actual piece. AVG: The temperature shown on display is the average on the whole control time.				
 This parameter is only visible with HEATING > TEMPERATURE CONTROL 1/2 = WR or WM .				

Parameter	Remote	Possible settings	Standard setting	Options
TEMPERATURE DELAY 1	DLT1	0.0 – 20.0 s	0.0	
TEMPERATURE DELAY 2	DLT2			
This value indicates for how much time to ignore the temperature measurement after a power change.				
 This parameter is available only if TEMPERATURE CONTROL is set to WR or WM .				

Parameter	Remote	Possible settings	Standard setting	Options
TEMPER. SETTING	TEMUN	CELSIUS (C) FAHRENHEIT (F)	CELSIUS (C)	
Select the measurement unit to use for the temperature display.				

4.6.2.2 WIRE parameters

Parameter	Remote	Possible settings	Standard setting	Options
FEED A 1	FDA1	0.0 ÷ 300.0 mm	0	 
FEED A 2	FDA2			
Select the soldering wire length for the first part of the workcycle. Steps of 0.1mm if below 10 mm and steps of 1mm if over 10mm.				

Parameter	Remote	Possible settings	Standard setting	Options
FEED RATE A 1	FRA1	1 ÷ 99 mm/s	1	 
FEED RATE A 2	FRA2			
Select the wire feeding rate for the first part of the workcycle.				

Parameter	Remote	Possible settings	Standard setting	Options
FEED B 1	FDB1	0.0 ÷ 300.0 mm	0	
FEED B 2	FDB2			
Select the soldering wire length for the second part of the workcycle. Steps of 0.1mm if below 10 mm and steps of 1mm if over 10mm.				

Parameter	Remote	Possible settings	Standard setting	Options
FEED RATE B 1	FRB1	1 ÷ 99 mm/s	1	
FEED RATE B 2	FRB2			
Select the wire feeding rate for the second part of the workcycle.				

Parameter	Remote	Possible settings	Standard setting	Options
REWIND LENGTH 1	RV1	0.0 ÷ 300.0 mm	0	 
REWIND LENGTH 2	RV2			
Select the soldering wire length to be rewinded at the end of its distribution, when the piece is in temperature. Steps of 0.1mm if below 10 mm and steps of 1mm if over 10mm.				

Parameter	Remote	Possible settings	Standard setting	Options
REWIND RATE 1	RVR1	1 ÷ 99 mm/s	1	 
REWIND RATE 2	RVR2			
Select the wire feeding rate during the rewind.				

Parameter	Remote	Possible settings	Standard setting	Options
CYLINDER DELAY 1	CD1	0,0 ÷ 99,0 s	0	 
CYLINDER DELAY 2	CD2			
Delay time for the advancing of wire feeder cylinder, respect to the heating activation. Timing: cd . After this time lapse, the cylinder moves toward the piece to heat, to be ready to the welding phase.				
 If time cd is set too much long, the soldering wire will be neared to the processed piece anyway, when the temperature set is reached, and this is valid for both the wire feeders, independently.				

Parameter	Remote	Possible settings	Standard setting	Options
INJECTION CYCLE 1	IJC1	0 ÷ 10	0	 
INJECTION CYCLE 2	IJC2			
This parameter manages the liquid injection frequency for the soldering wire, respect to the number of weldings. For example: with INJECTION CYCLE = 7 every 7 weldings an injection of liquid is performed.				
 If this parameter or the parameter INJECTION TIME are set to 0, the injection of liquid is never activated.				

Parameter	Remote	Possible settings	Standard setting	Options
INJECTION TIME 1	IJT1	0.00 ÷ 1.00 s	0	 
INJECTION TIME 2	IJT2			
This parameter indicates the duration of the injection of liquid.				
 If this parameter or the parameter INJECTION CYCLE are set to 0, the injection of liquid is never activated.				

Parameter	Remote	Possible settings	Standard setting	Options
WIRE ADJUST FACTOR 1	WADJF1	0.100 – 5.000	0.700	 
WIRE ADJUST FACTOR 2	WADJF2			
Select the conversion factor between the encoder impulses number (soldering wire feeder) and the wire advancing length.				
 THIS PARAMETER SHOULD ONLY BE MODIFIED ON THE EXPRESS INSTRUCTIONS OF CEIA.				

Parameter	Remote	Possible settings	Standard setting	Options
MANUAL FEED RATE	-	1 – 30 mm/s	15	
Select the advancing speed during the manual functioning.				

Parameter	Remote	Possible settings	Standard setting	Options
MANUAL REVERSE RATE	-	1 – 30 mm/s	2	
Select the rewinding speed during the manual functioning.				

Parameter	Remote	Possible settings	Standard setting	Options
CYL STROKE TIME	STKT	0.0 – 2.0 s	0	
Select the time lapse between the preset temperature reaching and the beginning of the soldering wire feeding. The parameter is relevant only if an excessive cylinder delay (cd) is set and allows to delay the wire feeding respect to the movement of the pneumatic cylinder.				
The HOLDING TIME (ta) and CYL STROKE TIME parameters are not added to each other, but is relevant only the bigger. If ta is set to 1 s and CYL STROKE TIME on 0,3 s, for example, the time lapse will be 1 s and not 1,3 s.				

Parameter	Remote	Possible settings	Standard setting	Options
MAX CURRENT 1	MAXCUR1	50 – 300 mA	100	 
MAX CURRENT 2	MAXCUR2			
Select the optimal current (optimum couple) for the wire advancing motor.				

Parameter	Remote	Possible settings	Standard setting	Options
WIRE TIMEOUT	WTO	0.0 s – 2250 hours	0	 

Select the maximum wire feeder activation time.

Generally speaking, a value greater than the normal time required for a work cycle should be selected.

From 0 to 100 s with steps of 1/10 s; from 101 to 600 seconds with steps of 1 s; from 11 to 2160 minutes with steps of 1 minute; from 36 to 2250 hours with steps of 1 hour.

Parameter	Remote	Possible settings	Standard setting	Options
WIRE PRESENCE	WPRES	ENABLED (Y) DISABLED (N)	DISABLED (N)	 

Enable /disable the control on the soldering wire presence.

Parameter	Remote	Possible settings	Standard setting	Options
FORWARD FEED	FFD	STANDARD (STD) PRE-HEAT (PRE)	STANDARD (STD)	 

Setting it as PRE-HEAT, distribution of the soldering wire is performed before turning on the generator, instead of after the piece reaches the temperature desired. At the cycle activation the stroke time is waited, after this the cylinder is lowered and the motor starts.

Parameter	Remote	Possible settings	Standard setting	Options
INJECTION	INJ	STANDARD (STD) PRE-HEAT (PRE)	STANDARD (STD)	 

Selection of the moment for the injection of anti-oxidizing gas: before (with PRE_HEAT) or after (with STANDARD) the distribution of soldering wire.

Parameter	Remote	Possible settings	Standard setting	Options
OUT J9-6 J14-6	OUT_6	INJECTION (INJ) WIRE MOV.SIGNAL (WIR)	INJECTION (INJ)	 

Selection of the meaning for Open Collector output 3 (pins 5 and 6 connector J9) for the soldering station 1 and 9 (pins 5 and 6 connector J14) for the soldering station 2. If set as INJECTION the outputs will drive the injection of anti-oxidizing gas, if set as WIRE MOV:SIGNAL, they will represent the movement of the soldering wire feeder motor.

4.6.2.3 GAS parameters

Parameter	Remote	Possible settings	Standard setting	Options
HEATING DELAY 1	HD1	0.0 ÷ 10.0 s	0	
HEATING DELAY 2	HD2			

Delay period in activation of heating: this represents the delay period in the heating of the piece being worked upon, with respect to the moment of activation of the cycle. The delay period is doubled in case of inert gas deactivation for at least 30 seconds. Timing: **hd**.

Parameter	Remote	Possible settings	Standard setting	Options
COOLING TIME 1	C01	0.0 ÷ 600 s	0	
COOLING TIME 2	C02			

Delay period in deactivation of cooling: this represents the duration of the cooling after the heating phase. In this time a relay for the control of an additional gas circuit is activated (see Installation chapter). Timing: **c**.
From 0 to 10 s with steps of 1/10 s; from 11 to 600 seconds with steps of 1 s.

Parameter	Remote	Possible settings	Standard setting	Options
GAS TIMEOUT	GTO	0.0 s – 2250 hours	0	

Selects the maximum gas distribution activation time, after which the work cycle is halted. Generally speaking, a value greater than the normal time required for a work cycle should be selected.
From 0 to 100 s with steps of 1/10 s; from 101 to 600 seconds with steps of 1 s; from 11 to 2160 minutes with steps of 1 minute; from 36 to 2250 hours with steps of 1 hour.

4.6.2.4 MODE parameters

Parameter	Remote	Possible settings	Standard setting	Options
OPERATING MODE	OPM	TIMER A (A) TIMER B (B) TIMER C (C) SYNCHRONISED (S) FULL AUTO (AUTO) FULL AUTO 2 WIRES (AUTO2W) THERMAL PROFILE (TPR) THERMAL PROFILE 2 WIRES (TP2W)	TIMER A (A)	
For a detailed explanation of all operating modes, see the chapter <i>Use</i>. The FULL AUTO 2 WIRES mode is available only with option  activated. The THERMAL PROFILE mode is available only with option  activated. The THERMAL PROFILE 2 WIRES mode is available only with options  and  activated.				

Parameter	Remote	Possible settings	Standard setting	Options
HEATING SYSTEM	HSYS	SINGLE GENERATOR (1) DUAL GENERATOR (2)	SINGLE GENERATOR (1)	
Select the type of functioning depending on the number of generators installed: - with one generator the Controller functions in alternating mode between the two heads; - with two generators installed the Controller simultaneously controls both heads.				

Parameter	Remote	Possible settings	Standard setting	Options
SECTIONS SYNCHRON.	SECSY	YES (Y) NO (N)	YES (Y)	
Allows the simultaneous activation of two CEIA generators on the same work piece.				

Parameter	Remote	Possible settings	Standard setting	Options
SINGLE POWERCUBE	SHEATC	ONE HEAD (1) TWO HEADS (2)	TWO HEADS (2)	
When only one generator is connected to the Controller, this parameter indicates how many heating heads are used. If set to ONE HEAD, the head used is the 1.  This command is available only with HEATING SYSTEM = SINGLE GENERATOR.				

Parameter	Remote	Possible settings	Standard setting	Options
HEAD 1 MODE	H1CONF	STANDARD (STD) 2-pyro (2PY)	STANDARD (STD)	
Setting this parameter to 2-pyro, the Controller uses two pyrometers on the same heating head. Refers to paragraph 3.5 for all information.  This command is available only with SINGLE POWERCUBE = ONE HEAD.				

Parameter	Remote	Possible settings	Standard setting	Options
TEMPERATURE CONTROL	TCON	ON (Y) OFF (N)	ON (Y)	
This parameter enable the temperature control. If the control is not enabled, the workcycle is controlled by parameters: PRE HEATING TIME, POWER A TIME, POWER A, POWER B (see paragraph "Functioning without temperature control").  This command is available only for operating modes AUT, A2W,  or T2W.				

Parameter	Remote	Possible settings	Standard setting	Options
EMISSIONITY ADJUSTMENT	EADJ	ON (Y) OFF (N)	ON (Y)	
OFF: Functioning with static emissivity, selected by parameter EMISSIONITY A. ON: Functioning with double emissivity: during the heating and the first soldering phase, (see parameter EMIS. ADJUST DELAY) the emissivity corresponds to the value of EMISSIONITY A; during the second soldering phase, the emissivity corresponds to the value of EMISSIONITY B. This command is available only for operating modes AUT, A2W, TP o T2W.				

Parameter	Remote	Possible settings	Standard setting	Options
LANGUAGE	LG	ENG, POL, SPA, DEU, FRA, ITA	ENG	
Selects the language of displayed messages.				

Parameter	Remote	Possible settings	Standard setting	Options
+V SUPPLY	OUTV	12V – 24V	12V	
Selects the supply voltage of digital inputs and outputs.				

Parameter	Remote	Possible settings	Standard setting	Options
INPUT LOGIC	ILOG	P, N, P+N	P+N	
Selects the logic for all digital inputs and outputs.				

Parameter	Remote	Possible settings	Standard setting	Options
CHANGE PASSWORD	-	6 alphanumeric characters	-	
Allows to change the password of current programming access level. By default the password is not set, the access to Programming is free.				

Parameter	Remote	Possible settings	Standard setting	Options
CHANGE QUICK ACCESS PASSWORD	-	6 alphanumeric characters	-	
Allows to change the password to access on the “Quick access” menu. By default that password is not set. When the password is programmed, pressing “Quick Access” the password is necessary to enter inside the menu.				

Parameter	Remote	Possible settings	Standard setting	Options
USER PARAMETER	UP	-	-	
Available only if logged in as SUPER USER. Allows to see and select which parameters are available to the USER access level.				

Parameter	Remote	Possible settings	Standard setting	Options
RESET USER PASSWORD	-	-	-	
Available only if logged in as SUPER USER, this permit to reset the USER password.				

Parameter	Remote	Possible settings	Standard setting	Options
SETUP QUICK ACCESS	QCKACC	-	-	
Allows to select the parameter list available on the “Quick Acces” menu. It is possible to put on the list only parameters that are available to the USER and that are significative for the current operating mode.				

Parameter	Remote	Possible settings	Standard setting	Options
ANALOG INPUT J19-3	AIT1	0 – 10 V 4 – 20 mA 0 – 20 mA	0 – 10 V	
ANALOG INPUT J19-5	AIT2			
ANALOG INPUT J19-7	AIT3			
ANALOG INPUT J19-9	AIT4			
Allows to set the analog input (1,2,3,4) mode.				

Parameter	Remote	Possible settings	Standard setting	Options
ANALOG OUTPUT J20-1	AOT1	0 – 10 V 4 – 20 mA 0 – 20 mA	0 – 10 V	
ANALOG OUTPUT J20-3	AOT2			
ANALOG OUTPUT J20-5	AOT3			
ANALOG OUTPUT J20-7	AOT4			
Allows to set the analog output (1,2,3,4) mode.				

Parameter	Remote	Possible settings	Standard setting	Options
AN.INP.NOISE REJ.	AINR	YES (Y) NO (N)	NO (N)	
Activation of a filter for a better noise reduction on the analogic inputs.				

Parameter	Remote	Possible settings	Standard setting	Options
DEBOUNCE TIME	DEBTIM	2 - 20 ms	2	
Select debounce time of the static inputs used for activation of the Cycle.				

Parameter	Remote	Possible settings	Standard setting	Options
REVERSE DISPLAY	DISREV	ENABLED (Y) DISABLED (N)	DISABLED (N)	
With REVERSE DISPLAY = ENABLED the display is shown with black letters on white background. If set to DISABLED, the letters are white on black background.				

Parameter	Remote	Possible settings	Standard setting	Options
TEMP.DECIMAL DIGITS	TEMDEC	0 – 1	1	
Select the number of decimal digits for the temperature indication.				

Parameter	Remote	Possible settings	Standard setting	Options
READY SIGNAL MODE	RDYSM	STANDARD (STD) STD+LOG (STD+LOG)	STANDARD (STD)	
Select the "Machine Ready" signal mode: STANDARD : the signal is given when the system is ready to start a new soldering cycle STD+LOG : the signal is given only when also the network card (with Data Logger) is ready				

Parameter	Remote	Possible settings	Standard setting	Options
F.B. ID	FBID	8 characters	-	
Controller identification name in order to identify it from other devices connected to the Field Bus module.				

Parameter	Remote	Possible settings	Standard setting	Options
F.B. DHCP	FBDHCP	Y / N	N	
This parameter sets the Controller in DHCP client mode.				

Parameter	Remote	Possible settings	Standard setting	Options
F.B. IP	FBIP	IPv4 address	-	(-FB)
With this parameter it is possible to manually assign an IP address to the Controller. It shall be a valid IP address, i.e. 192.168.100.1.				
This parameter is ignored if F.B. DHCP is set to Y .				

Parameter	Remote	Possible settings	Standard setting	Options
F.B. NET.MASK	FBMK	-	-	(-FB)
With this parameter it is possible to manually assign a network mask to the Controller. It shall be a valid IP address, i.e. 255.255.0.0				
This parameter is ignored if F.B. DHCP is set to Y .				

Parameter	Remote	Possible settings	Standard setting	Options
F.B.ENDIANNESS	FBENDI	LSB / MSB	LSB	(-FB)
This parameter defines the sending/receiving order of the bytes values (UINT16, UINT32) on the Field Bus. LSB : Least Significant Byte first MSB : Most Significant Byte first				

Parameter	Remote	Possible settings	Standard setting	Options
F.B.NODE ADDRESS	FBNADD	0 - 125	1	(-FB)
Selection of the device address on RS485 bus.				

Parameter	Remote	Possible settings	Standard setting	Options
F.B. VERSION	FBVER	1 - 2 - 3	1	(-FB)
Selection of the Cyclic Data table to be used. See related paragraph.				

Parameter	Remote	Possible settings	Standard setting	Options
M.C. HOSTNAME	MCHNAM	20 characters string	-	(-IXC)
Set the Controller name for the network.				

Parameter	Remote	Possible settings	Standard setting	Options
M.C. DHCP	MCDHCP	Y / N	N	(-IXC)
This parameter set the Controllore in client DHCP mode.				

Parameter	Remote	Possible settings	Standard setting	Options
M.C. IP	MCIP	IPv4 address	-	(-IXC)
With this parameter it is possible to manually assign an IP address to the Controller. It should be a valid IP address, i.e. 192.168.100.1				
This parameter is ignored if M.C. DHCP is set to Y .				

Parameter	Remote	Possible settings	Standard setting	Options
M.C. NET.MASK	MCMK	IPv4 address	-	(-IXC)
With this parameter it is possible to manually assign a network mask to the Controller. It should be a valid IP address, i.e. 255.255.0.0				
This parameter is ignored if M.C. DHCP is set to Y .				

Parameter	Remote	Possible settings	Standard setting	Options
M.C. GATEWAY	MCGW	IPv4 address	-	
With this parameter it is possible to manually assign the Gateway address of the IXC board.				

4.6.2.5 MEMORY parameters

Parameter	Remote	Possible settings	Standard setting	Options
STORES PARAMETERS IN USE IN CARD	-	from 1 to 100 6 characters string	-	
Stores the actual parameters set in one of the 100 memory cells of a single Program Card				
 The possibility to load a set of parameters from a Program Card is automatically suggested at the insertion of the Program Card into the reader.				

Parameter	Remote	Possible settings	Standard setting	Options
STORE PARAMETERS IN USE IN THE CELL	ST	from 1 to 100 6 characters string	-	
The command stores all the present functional parameters into memory. The memory cells allow various sets of parameters to be recorded, according to the various processing to be carried out. The memory used is not volatile and thus is not altered when the device is turned off. It is possible to assign a 6 characters name to the stored set of parameters.				
 By recording a parameter setting into a memory cell, all current data of the cell are cancelled. In the event that parameters are about to be copied onto a cell that is already programmed, the Power Controller warns the operator with this message beside. In this case the operator may choose changing cell or to go ahead by pressing the E key (the previous programming will be cancelled). CELL NOT EMPTY E=continue P=cancel				

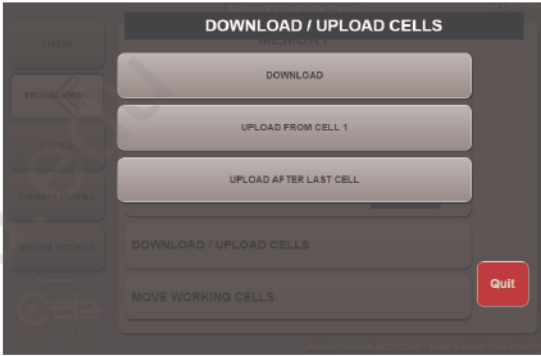
Parameter	Remote	Possible settings	Standard setting	Options
LOAD NEW PARAMETERS FROM THE CELL	LD	from 1 to 100 0 to undo	-	
This operation selects the various parameters, previously stored, activating them for the next work cycle. when confirming the loading, all the parameters included unto the Controller work memory are replaced. Such parameters are stored into memory cell n°0 and can be recalled with function "Load parameters in use in cell 0". The name of the loaded program will be shown in the bottom left corner of the normal use screen.				

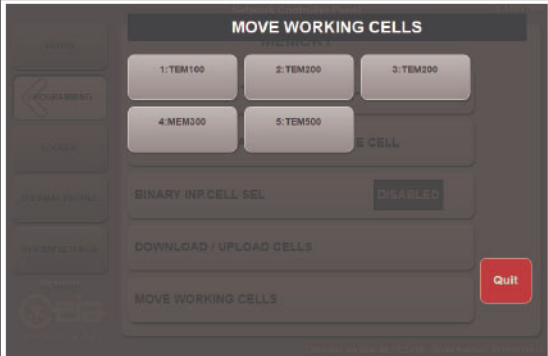
Parameter	Remote	Possible settings	Standard setting	Options
CLEAR PROGRAM CARD	-	-	-	
This command format the inserted Program Card, erasing the 100 memory cells.				

Parameter	Remote	Possible settings	Standard setting	Options
BACKUP MEMORY TO PROGRAM CARD	-	-	-	
This command stores all the 100 memory cells of the Controller into the inserted Program Card.				
 If the Program Card contains data, they will be overwritten.				

Parameter	Remote	Possible settings	Standard setting	Options
RESTORE MEMORY FROM PROGRAM CARD	-	-	-	
This command loads into the memory of the Controller, all the 100 memory cells of the inserted Program Card.				
If the internal memory cells contains data, they will be overwritten.				

Parameter	Remote	Possible settings	Standard setting	Options
BINARY INP.CELL SEL	-	ENABLED DISABLED	DISABLED	
With this function ENABLED, the Controller loads data from the memory cell indicated by the binary inputs in J13. See paragraph 3.3.4 for detailed information.				

Parameter	Remote	Possible settings	Standard setting	Options
DOWNLOAD/UPLOAD CELLS	-	-	-	
This command allows to download and upload memory cells through the web server.				
DOWNLOAD: Working cells are saved on a single ZIP file with, inside, all the 100 memory cells. UPLOAD FROM CELL 1: A previously saved ZIP file is sent to the Controller restoring all memory cells starting from cell number 1. UPLOAD AFTER LAST CELL: A previously saved ZIP file is sent to the Controller, restoring the memory cells it contains, starting from the first free cell on the Controller itself.				
 This function is only available by remote connection with the Controller and it is not available on the front panel.				

Parameter	Remote	Possible settings	Standard setting	Options
MOVE WORKING CELLS	-	-	-	
This command allows to move an existing memory cell in a new location.				
A list of all memory cells in the Generator, is given. After selecting one of them, the new position number is requested. If the inserted position is not empty, a confirmation will be requested. If the operation is confirmed, the destination cell content will be overwritten by the content of the starting cell.				
 This function is only available by remote connection with the Controller and it is not available on the front panel.				

4.6.2.6 MANUAL parameters

Parameter	Remote	Possible settings	Standard setting	Options
CYL 1 CYL 2	-	-	-	 
This command, when enabled, allows to move manually the soldering wire cylinder. To move it use  (backward) and  (forward)				

Parameter	Remote	Possible settings	Standard setting	Options
CYL 1 and 2	-	-	-	 
This command, when enabled, allows to move manually both soldering wire cylinders, together. To move them use  (backward) and  (forward)				

Parameter	Remote	Possible settings	Standard setting	Options
WIRE 1 WIRE 2	-	-	-	 
This command, when enabled, allows to feed manually the soldering wire. To feed the wire use  (backward) and  (forward)				

4.7 Use of the Web Server



ATTENTION! All functions described in this paragraph, are available only in presence of the **-IXC** option.

The **-IXC** option incorporates the Web Server function for the monitoring, remote programming, data logging and management of the thermal profile of CEIA induction generators. It thereby provides all the data required for the quality control process.

The Controller with **-IXC** option only requires an Ethernet connection for accessing the programming parameters and for archiving the data saved.

All the information is stored in the .csv and .txt formats. The board can save more than 100,000,000 samples.

The sampling interval is programmable; the resolution of the acquisition time is 10 milliseconds.

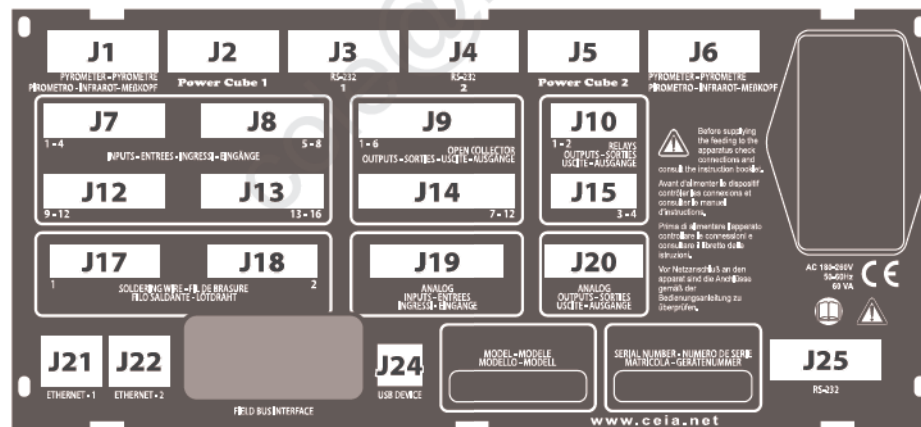
4.7.1 Requirements

The minimum requirements for using the Web Server are as follows:

- Personal computer
- Network cable (straight or cross over)
- CEIA Discovery software
- Internet browser (Google Chrome preferred. Alternatively, Mozilla Firefox, Apple® Safari or Microsoft® Internet Explorer).

4.7.2 Connection

Connect the V3+ Controller to the Ethernet network using J21 or J22 connectors.
In cascade connections the Controller acts as a bridge.



Connections on the back of the Controller

4.7.3 Settings

To display the setup information (e.g. IP address, sampling frequency, data logger status), press the “Status” button on the V3 Controller and then scroll through the various status displays until you reach the **LOGGER** display like the one shown here.

When you power up the Controller, the **LOGGER** display will appear after one minute, i.e. after the time needed to start up the operating system on the **IXC** card.



Figure 4-1 – **LOGGER** display on the Controller

Start the **CEIA Discovery** by double-clicking its icon.

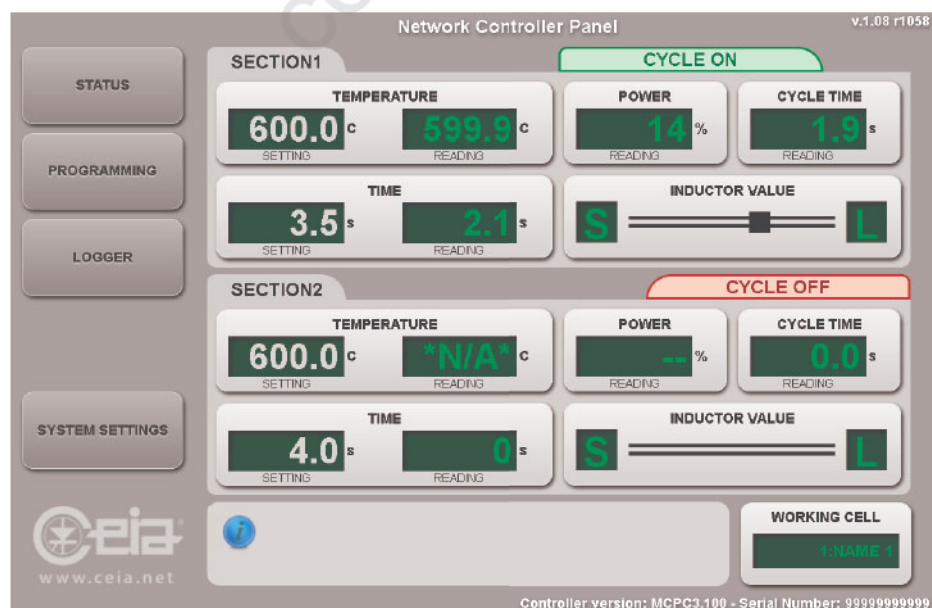


In order to download and install the **CEIA Discovery** software, contact our Sales Office.



The default browser will open up, showing a list of **CEIA** devices available in the network.

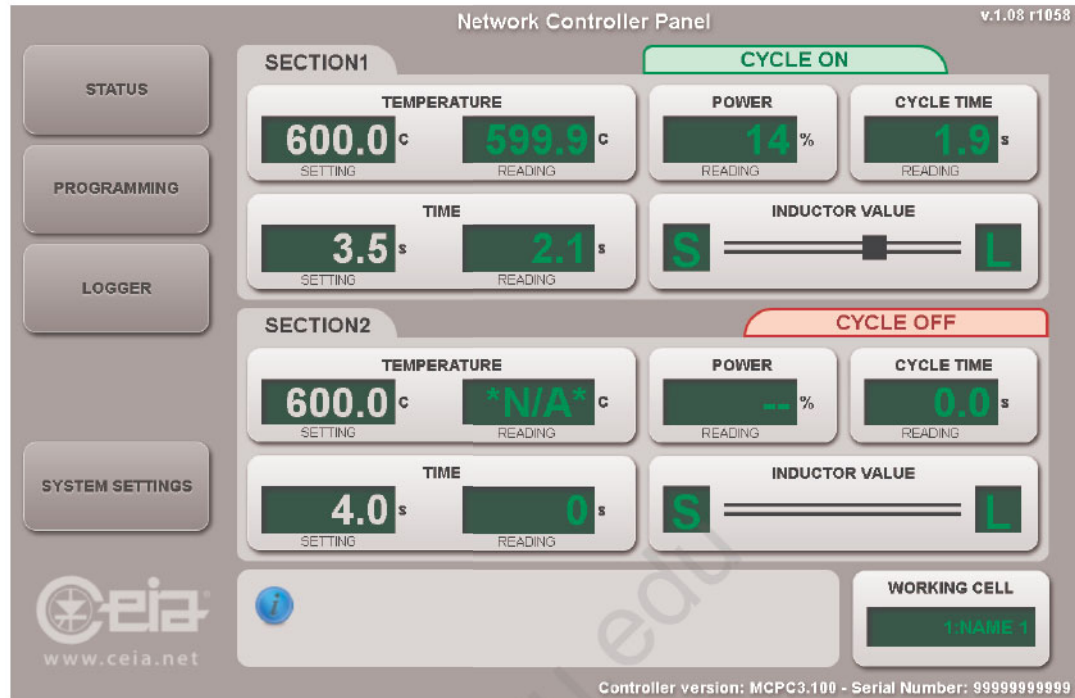
Click on the desired device name to select it and to access its **Remote panel**.



Main page on the Remote Panel

4.7.4 Use

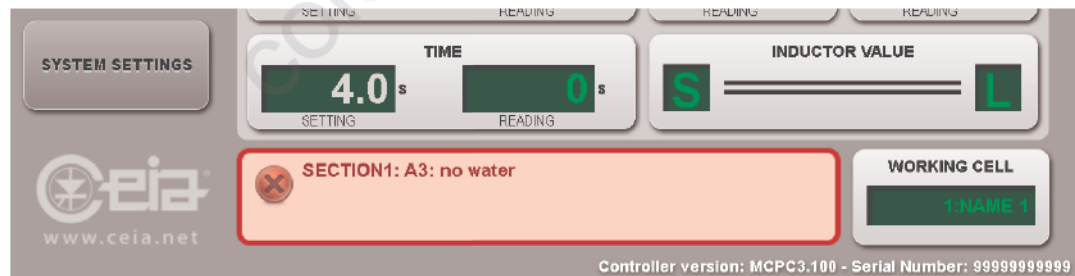
The main page of the Remote panel displays all the main operating data.



Main page for a two-section heating system where the cycle for section 1 is in progress.

For each section the page displays: the temperature (setpoint and reading); the power; the time; the inductor value. To modify a setting value, click on the Temperature (Setting) or Power (Setting) fields.

A dialog window like the one shown below will open at the bottom of the screen showing messages for the operator.



Dialog window showing an operator message.

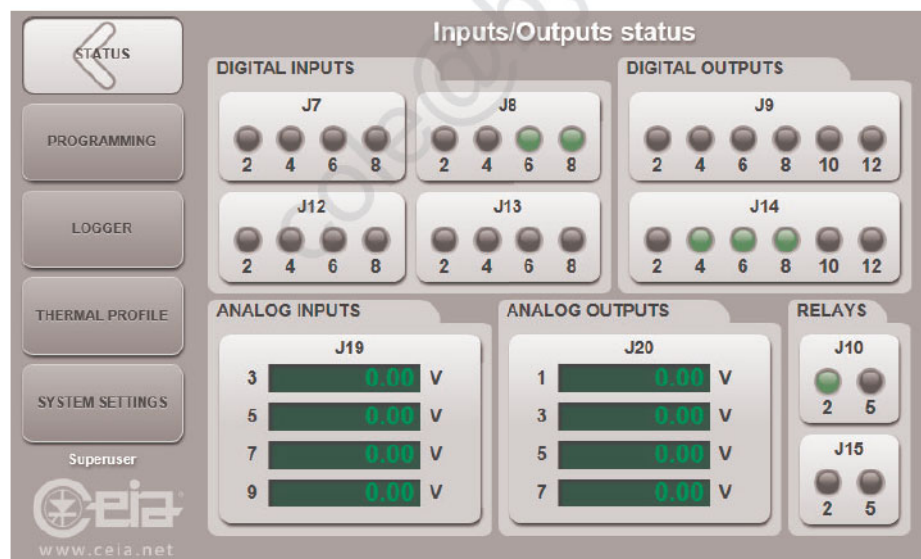
4.7.4.1 Status page

Pressing the STATUS button the page beside is shown, where it is possible to choose one of the various generator status pages.



Inputs/Outputs status

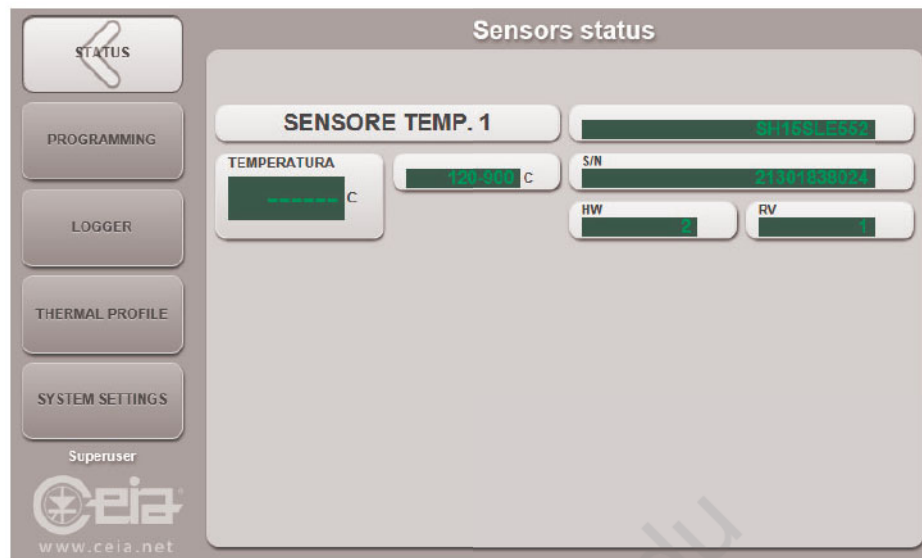
In this page it is possible to check all inputs and outputs connected to the internal Controller.



Pressing again the STATUS button, with the left arrow on it, the display will return back to the Status page.

Sensors status

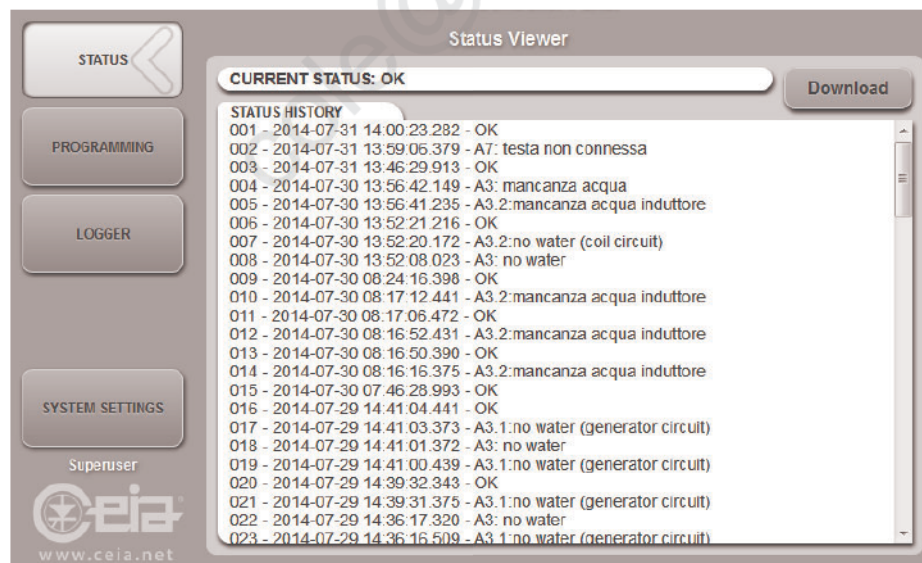
In this page it is possible to check the status of all CEIA temperature sensors of the system. In the example beside, a single SH15/SLE pyrometer is connected.



Pressing again the STATUS button to return back to the Status page.

Status viewer

In this page, the events and diagnosis log is shown. The log will show the last 100 events occurred.



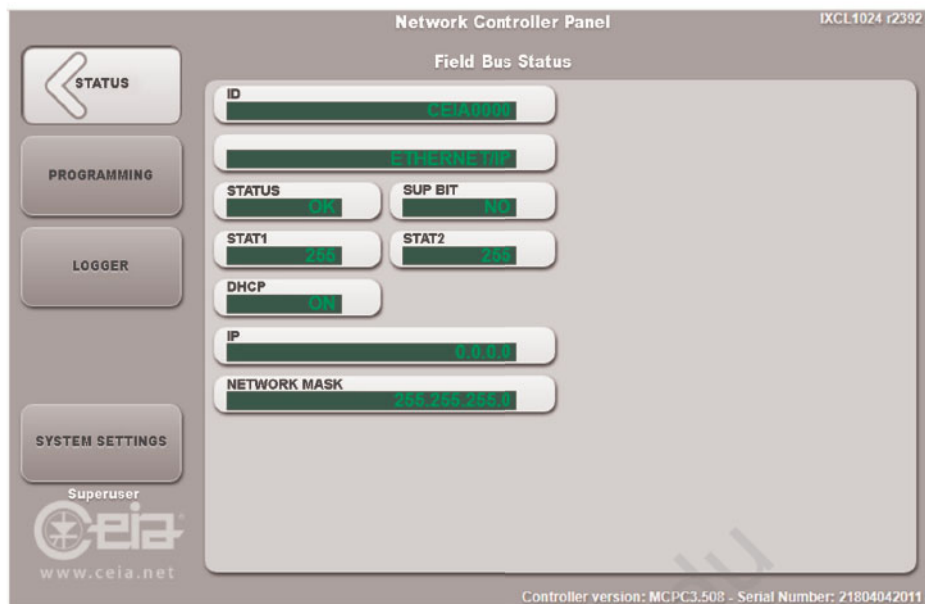
Pressing again the STATUS button, with the left arrow on it, the display will return back to the Status page.



The **Download** button, in the upper right, is available only using the remote connection and allows to save the status file.

Field Bus Status

In this page, the Field Bus Status is shown:



Network Controller Panel IXCL1024 r2392

Field Bus Status

ID CEIA0000

ETHERNET

STATUS OK SUP BIT NO

STAT1 255 STAT2 255

DHCP OK

IP 0.0.0.0

NETWORK MASK 255.255.255.0

Superuser

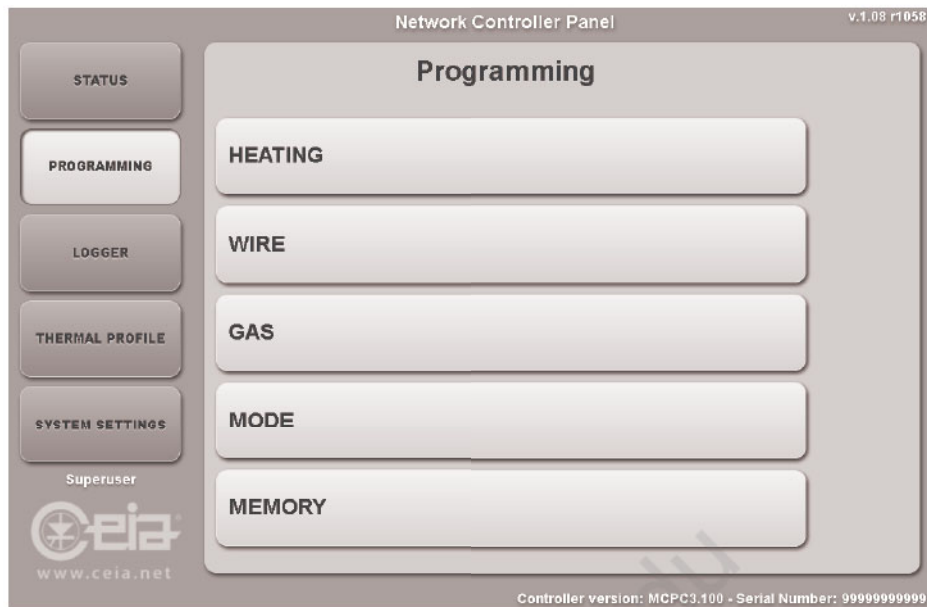
 www.ceia.net

Controller version: MCPC3.508 - Serial Number: 21804042011

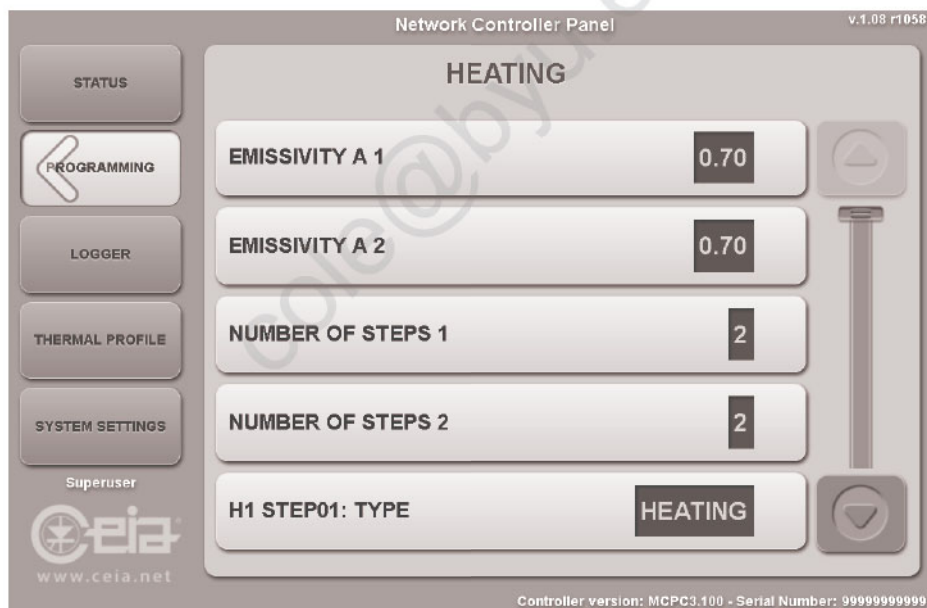
Pressing again the STATUS button, with the left arrow on it, the display will return back to the Status page.

4.7.4.2 PROGRAMMING page

The screenshots below shown the PROGRAMMING pages where you can program the Controller.



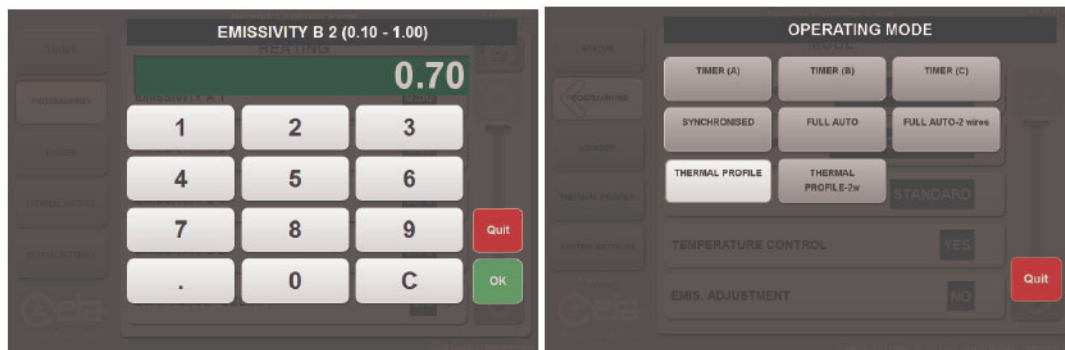
Programming page: Main menu.



Programming page: Heating menu.

The parameters are displayed as a tree list. Use the arrow buttons on the right to scroll backwards and forwards between the parameters. To return to the main menu, press the "HOME" icon at the top right-hand side of the window.

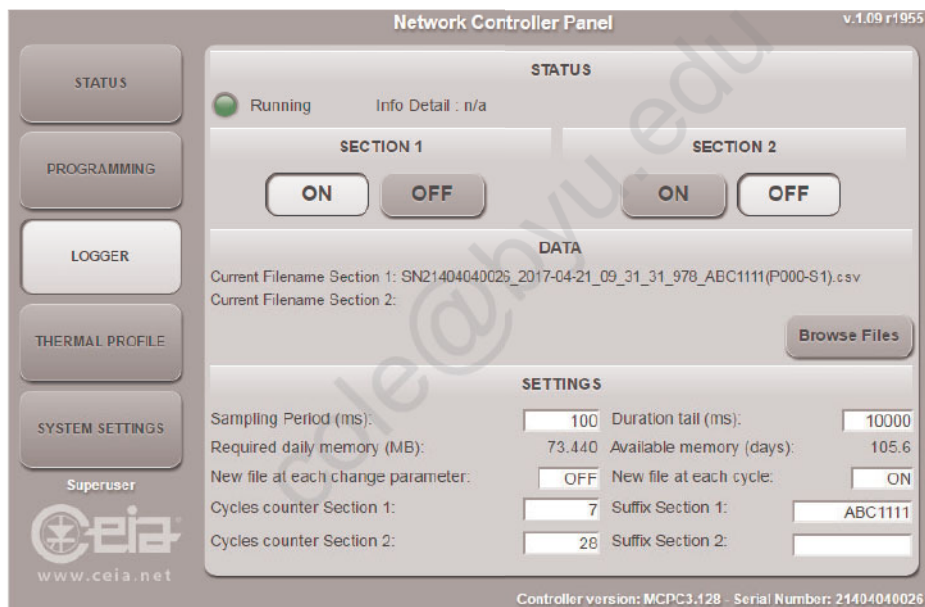
To modify the value of a parameter, first click on that parameter. Next, a numeric keypad or a list of possible values will appear over the parameter list. Select the value you require.



Modifying a parameter value

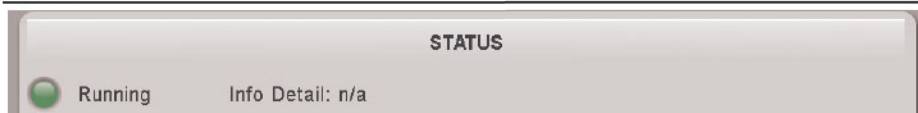
4.7.4.3 LOGGER page

Use this page to set all the parameters for the Data Logger function.



LOGGER page

STATUS area



LOGGER page – STATUS area

When the button icon in this area is green, this indicates that data logging for at least one of the sections is in progress.

If the button icon is grey, this indicates that data logging for both sections is OFF.

If the button icon is red, this indicates an error.

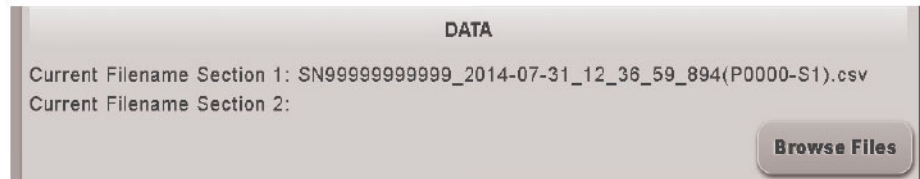
SECTION 1 and SECTION 2 areas



LOGGER page – SECTION 1 and SECTION 2 areas

Use this area to switch the data logging for both sections ON and OFF. In the example shown above, data logging for Section 1 is ON and data logging for Section 2 is OFF.

DATA area



LOGGER page – DATA area

This area displays the names of the current files for which data has been saved for both sections (where this function is ON). The files are saved in .csv format with the following name:

$\{SNXXXXXXXXXX\}_{Date}_{Time}_{Pxxxx} - \{Sx\}.csv$

where:

- $SNXXXXXXXXXX$: Controller serial number (e.g. SN21104002003).
- $Date$: Date of file creation in the format $YYYY-MM-DD$.
- $Time$: Time of file creation in the format $hh.mm.ss.ms$.
- $Pxxxx$: Consecutive number.
- Sx : Section 1 or Section 2.

When the file size reaches 65,000 lines a new file is created automatically and the consecutive number is increased accordingly.

A new file is created automatically each time a parameter is modified.

When you press the **Browse Files** button this will open a new window listing all the files saved to date. You can display and download these file from here.

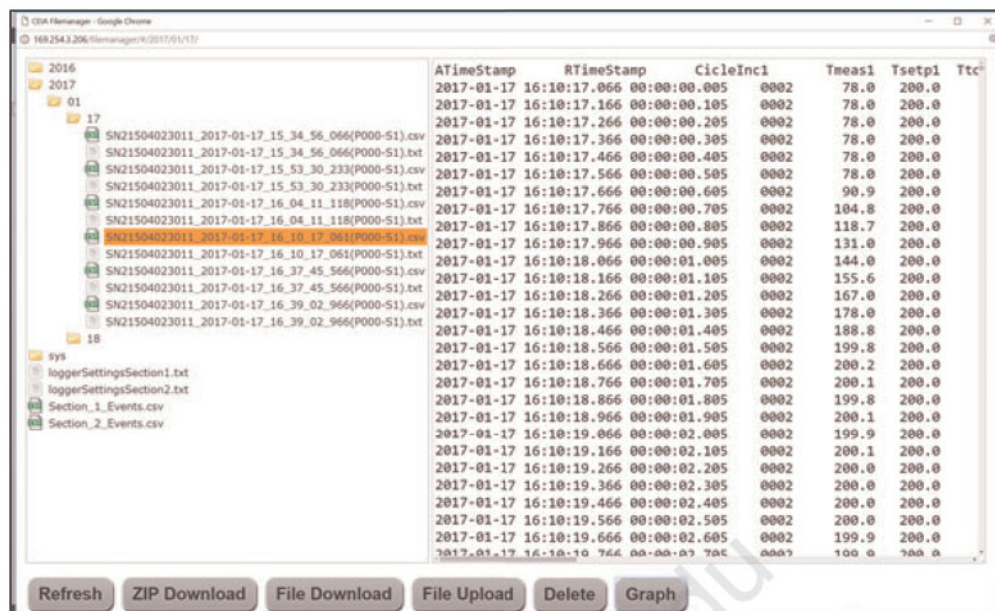


Figure 4-2 – Data logger file manager display

The left-hand side of page shows a list of the folders and files saved on the Ethernet Interface Board.

The files are saved in folders ordered by year, month and day.

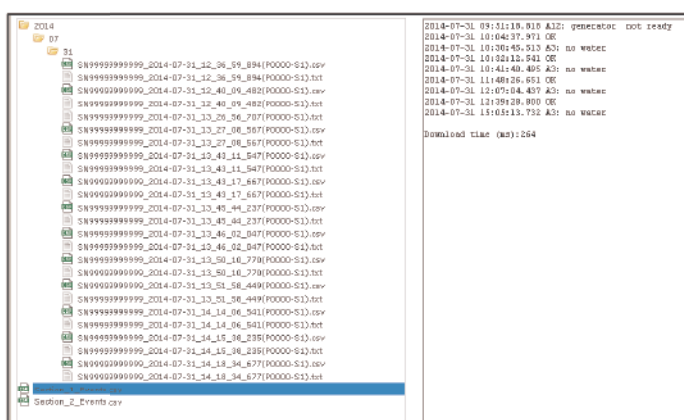
To display an automatic preview of file content, select and click the file that interests you. The preview will appear on the right-hand side of the page.

Once selected a .csv file, it is possible to download it on your computer, simply by clicking on the File Download button. It is possible to upload a .csv file inside the Ethernet Interface Board, using the File Upload button.

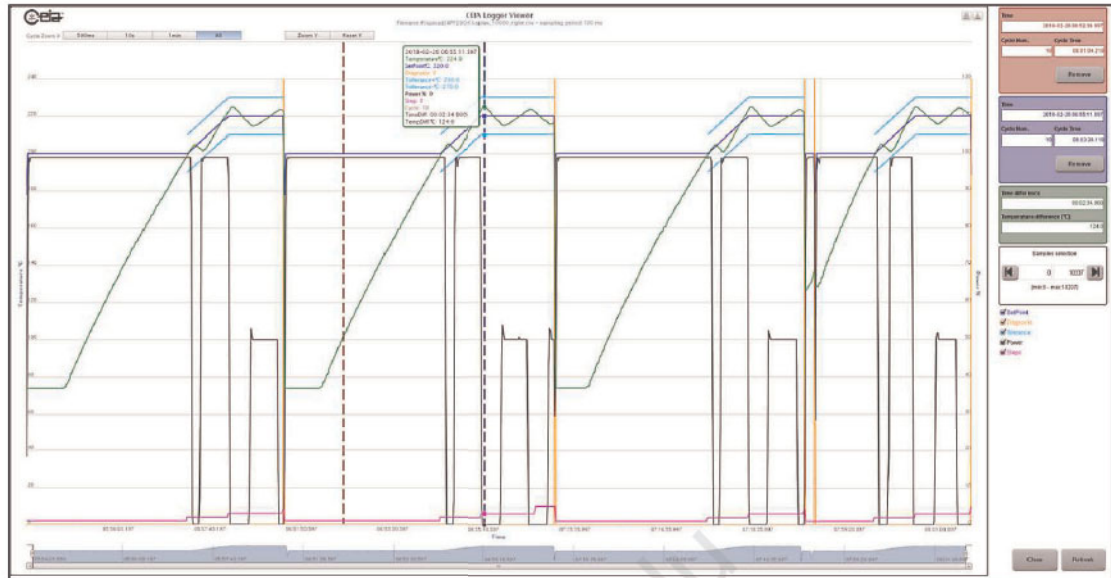
To select multiple files, press the SHIFT key or the CTRL key. To download the files selected, press the **Download ZIP** button. To delete the files selected, press the **Delete** button. You can download or delete the entire contents of a folder by selecting that folder and then pressing the button required.

From this window it is possible also to view and download the event logs related to the two sections.

Event log view.



To display a complete graph for the data contained in a .csv file, select the file and then press the **Graph** button. If you select multiple .csv files, a new window will open for each graph.



CEIA Logger Viewer

The graph shows the following information:

- Heating temperature read by pyrometer (green — inside tolerance limit, red — outside tolerance limit)
- Temperature setpoint (blue —)
- Temperature tolerances (sky blue —)
- Power delivered by generator (black —)
- Thermal profile steps (fuchsia —)
- Generator diagnostics (orange —)

Each graph line (with the exception of the temperature read) can be hidden or shown by clicking one of the coloured icons in the right sidebar of the page (SetPoint, Diagnosis, Tolerance, Power, Steps).

To display a detailed analysis of the thermal profile, use the zoom function on the Time (X) axis and on the Temperature (Y) axis.

The vertical cursor can be moved along the graph. It will display a summary box containing all the data for the specific point intersected by the grey line and highlighted by the coloured symbols on each of the graph lines.

Moving the mouse on screen, all data inside the summary box are updated in real time.

2014-07-31 13:50:17:170
Temperature°C : 400.4
SetPoint°C : 400.0
Diagnosis : 0
Tolerance+°C : 415.0
Tolerance-°C : 385.0
Power % : 32
Step : 3
Cycle: 18
TimeDiff: 00:02:34.900
TempDiff °C: 124.0

Clicking in any point of the graph line, a cursor is placed (**red dotted vertical line**) and data related with that point are indicated in the **red info box** on the right sidebar.

Clicking again on the graph line, a second cursor is placed (**blue dotted vertical line**), whose data are indicated in the **blue info box**, on the right sidebar.

The green info box, when both cursors are defined, shows the time and temperature differences between the two cursors.

Each cursor can be moved along the graph line by drag&drop or removed using the button on the related info box.

Time

2019-02-26 06:52:36.907

Cycle Num.

18

Cycle Time

00:01:04.219

Remove

Time

2019-02-26 06:55:11.897

Cycle Num.

18

Cycle Time

00:03:39.119

Remove

Time difference

00:02:34.900

Temperature difference (°C)

124.0

Use the Print and Save buttons on the top right of the page to print the graph or to save an image of the graph.
The graph will be saved in .png format.



Example file

ATimeStamp	RTimeStamp	CicleInc1	Tmeas1	Tsetp1	Ttol1	Tmax1	Pw1	C1	Di1	Acquisition start time:	2014-07-31 13:51:58.449
2014-07-31 13:52:02.504	00:00:00.000	20	78	0	0	0	0	1	0	Sampling period (ms):	200
2014-07-31 13:52:02.549	00:00:00.145	20	114.4	113.5	15	116.1	24	2	0	SN:	9999999999
2014-07-31 13:52:02.849	00:00:00.345	20	142.5	143.5	15	143.5	25	2	0	Temperature scale:	°C
2014-07-31 13:52:03.049	00:00:00.545	20	172.9	173.5	15	172.9	21	2	0		
2014-07-31 13:52:03.249	00:00:00.745	20	203	203.5	15	203	33	2	0		
2014-07-31 13:52:03.449	00:00:00.945	20	232.4	233.5	15	232.8	36	2	0		
2014-07-31 13:52:03.649	00:00:01.145	20	262.9	263.5	15	262.9	37	2	0		
2014-07-31 13:52:03.849	00:00:01.345	20	293.5	293.5	15	293.5	39	2	0		
2014-07-31 13:52:04.049	00:00:01.545	20	322.4	323.5	15	322.4	40	2	0		
2014-07-31 13:52:04.249	00:00:01.745	20	352.9	353.5	15	352.9	44	2	0		
2014-07-31 13:52:04.449	00:00:01.945	20	382.5	383.5	15	382.5	44	2	0		
2014-07-31 13:52:04.649	00:00:02.145	20	400.3	400	15	401.3	29	3	0		
2014-07-31 13:52:04.849	00:00:02.345	20	400.1	400	15	401.3	30	3	0		
2014-07-31 13:52:05.049	00:00:02.545	20	400	400	15	401.3	24	3	0		
2014-07-31 13:52:05.249	00:00:02.745	20	399.4	400	15	401.3	24	3	0		
2014-07-31 13:52:05.449	00:00:02.945	20	399.9	400	15	401.3	22	3	0		
2014-07-31 13:52:05.649	00:00:03.145	20	399.9	400	15	401.3	18	2	0		

ATimeStamp	Date and time of data acquisition
CicleInc1	Consecutive number of the heating cycle completed
Tmeas1 (Tmeas2)	Temperature read by the pyrometer 1(2)
Tsetp1 (Tsetp2)	Temperature setpoint for the pyrometer 1(2)
Ttol1 (Ttol2)	Temperature tolerance for the pyrometer 1(2)
Tmax1 (Tmax2)	Maximum temperature recorded during the cycle
Pw1 (Pw2)	Power delivered by generator 1(2)
C1 (C2)	Step number
Di1 (Di2)	Generator diagnostics 1(2) – see table below

The corresponding .txt lists all the operating parameters and their values.

Di1 or Di2	Description of diagnostic messages
0	Generator ready, no alarm present
100	Controller buffer is full.
110	Parameter modification has been completed
111	Temperature outside tolerance limit in section 1
112	Temperature outside tolerance limit in section 2
113	Temperature returned to inside tolerance limit in section 1
114	Temperature returned to inside tolerance limit in section 2
120	Cycle aborted due to a signal on "Reset" input

List of diagnostic messages in .csv file.



Read paragraph 5.2.1 **Selfdiagnosis messages** for the complete list of fault messages.

SETTINGS area

SETTINGS

Sampling Period (ms):

100

Duration tail (ms):

10000

Required daily memory (MB):

73.440

Available memory (days):

105.6

New file at each change parameter:

OFF

New file at each cycle:

ON

Cycles counter Section 1:

7

Suffix Section 1:

ABC1111

Cycles counter Section 2:

28

Suffix Section 2:

Use this area to modify the sampling interval and the acquisition time after the end of the cycle.

To do it, simply click on the input field and enter a new value.

Sampling Period (10 - 60000)

200

1

2

3

4

5

6

7

8

9

Quit

.

0

C

OK

Sampling period

This is the time which elapses between one sampling and another.

Enter a value between 10 ms and 600,000 ms (10 minutes). Any modification you make to this parameter will only be implemented after the **Set** button has been pressed.

The Controller automatically calculates the amount of memory required to save the data for a day of continuous sampling.

When the memory available finishes, the system will automatically start to overwrite the oldest files.

Duration tail

This is the time (in milliseconds) after the end of cycle signal during which the system will continue to acquire information.

If you set this parameter to 0, data logging will stop at the same time as the end of cycle signal.

New file at each change parameter

When this parameter is set to ON, a new Log file is created each time a parameter is modified.

New file at each cycle

When this parameter is set to ON, a new Log file is created at the end of each work cycle.

Cycles counter

This parameter shows the number of performed work cycles.

This value is editable. To reset it, insert 0.

When its value is over 9999, it is automatically set back to 1.

Suffix

This field defines a string that will be used as a suffix for the log filename.

4.7.5 THERMAL PROFILE page

This menu is available only when the **OPERATING MODE** parameter is set to **THERMAL PROFILE**.

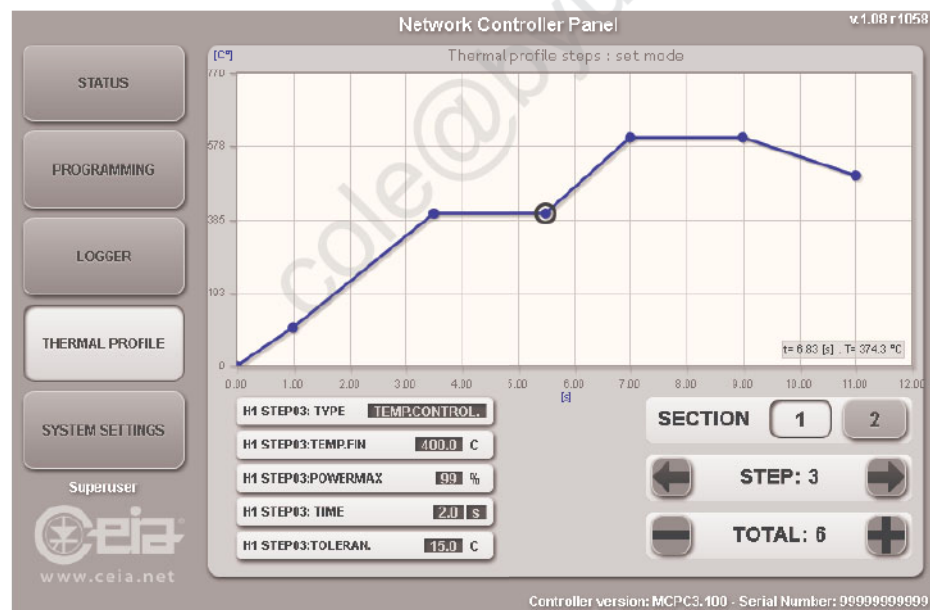
4.7.5.1 Set mode

If no cycle is in progress, the **Thermal profile** of each section can be set using the Web programming panel.

The parameters such as Time and Temperature can be modified directly on the page.

Once these parameters have been set by hand, the step type, the heating temperature, the power, the time and the permitted tolerance will be listed on the same pages and can be fine tuned by inserting the numeric value directly.

This makes it possible for the operator to modify the thermal profile with a few simple operations.

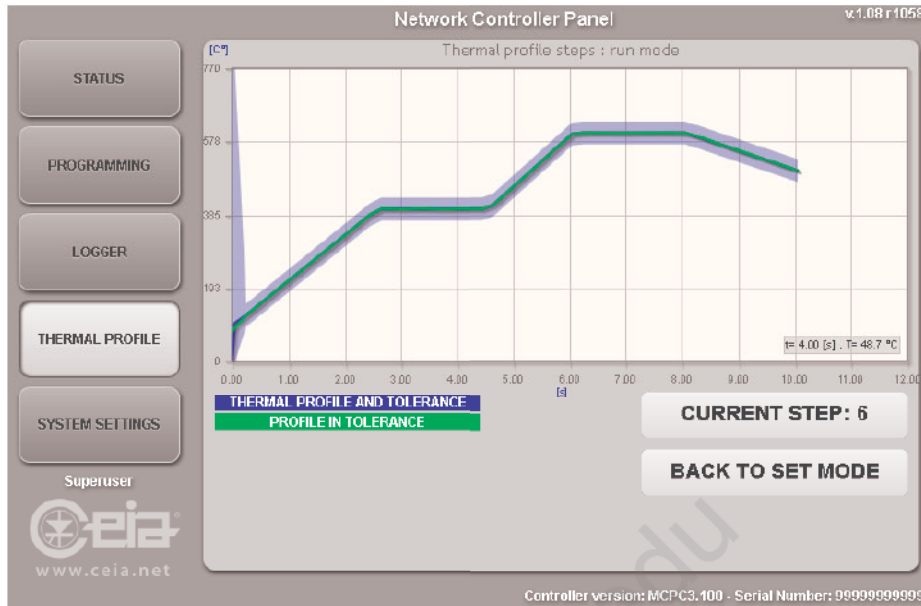


Example setting of a thermal profile

Steps can only be added to or removed from the end of the profile.

4.7.5.2 Run mode

During the heating cycle the thermal profile for each section as read by the pyrometer is shown in real time.

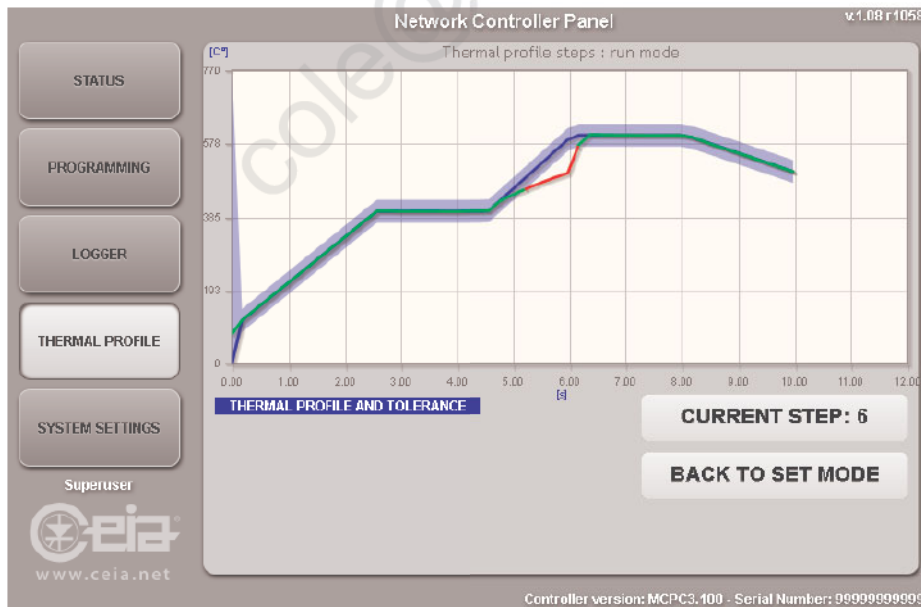


View of the thermal profile in real time

The blue line shows the thermal profile set by the operator.

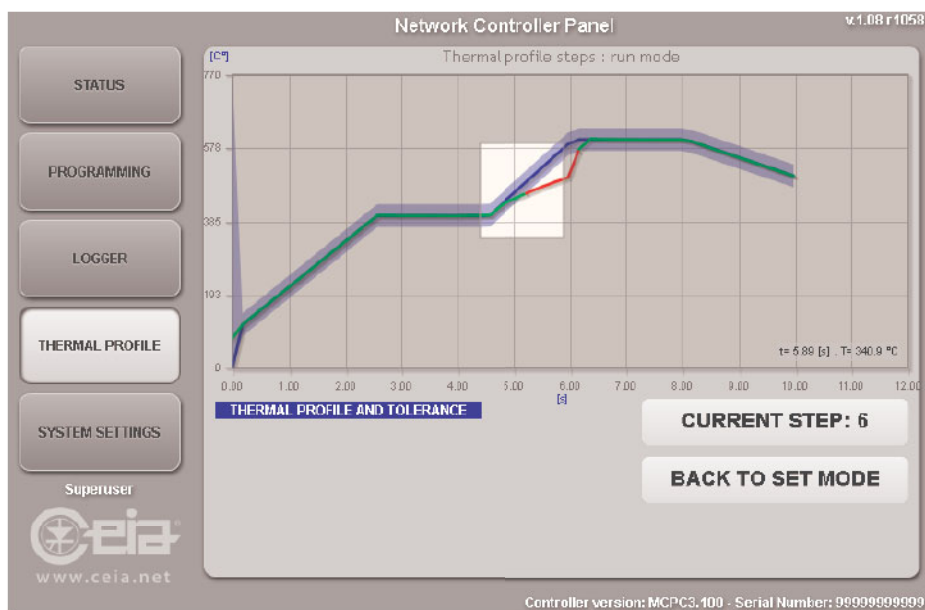
The transparent blue line shows the tolerance required.

A green line indicates that the temperature read by the pyrometer during the work cycle is inside the tolerance limit.



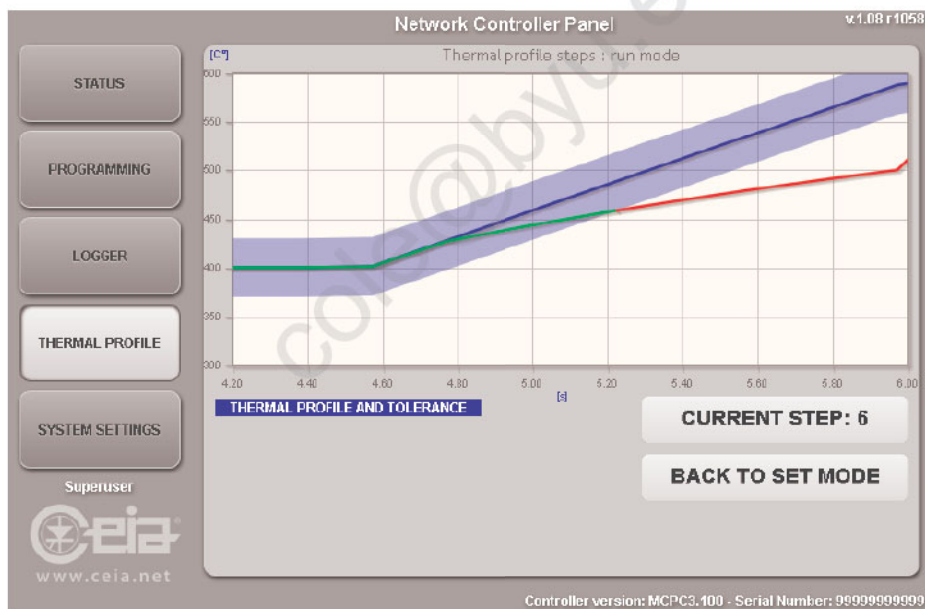
View of the thermal profile in real time

A red line (see the example above) indicates that the temperature is outside the tolerance limit.



View of the thermal profile - Selecting a detail view

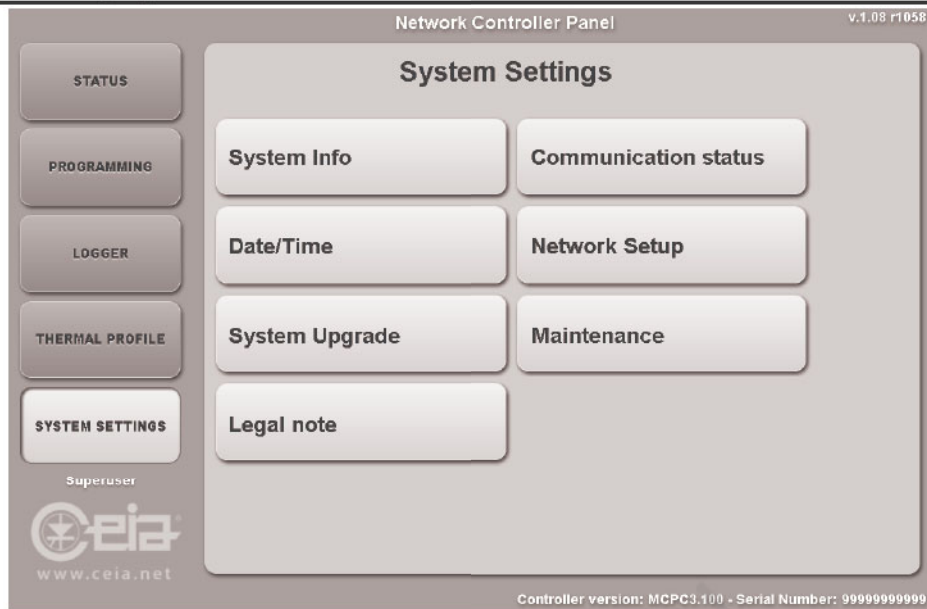
Use the mouse to select a specific area of the thermal profile to display a detailed, close-up view of that area.



View of the thermal profile - Detailed view

Double-click on the page to return to a display of the entire profile.

4.7.6 SETTINGS page

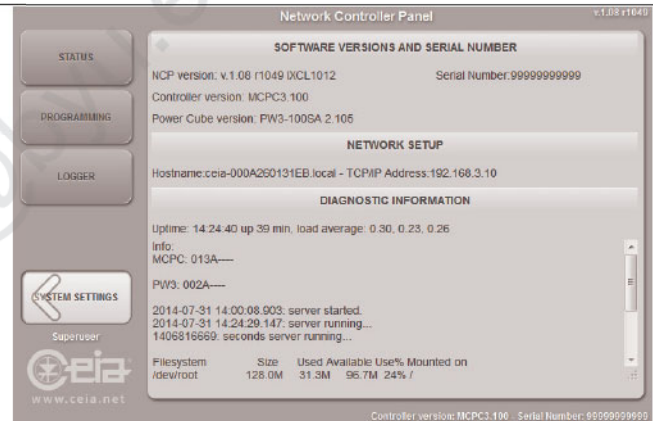


SETTINGS page

4.7.6.1 System Info

Pressing this button, all data about the system, as the installed software versions, the serial number, network configuration data and other diagnostic information, are shown.

Press the **System Settings** button to return back to the previous screen.



4.7.6.2 Communication status

Pressing this button the communication status, between the Controller and the Control Panel, is shown.

4.7.6.3 Date/Time

This area shows the date and time on the PowerCube and on the computer connected up.

If the dates are different, they can be synchronised by pressing the Synchronise Date/Time button.

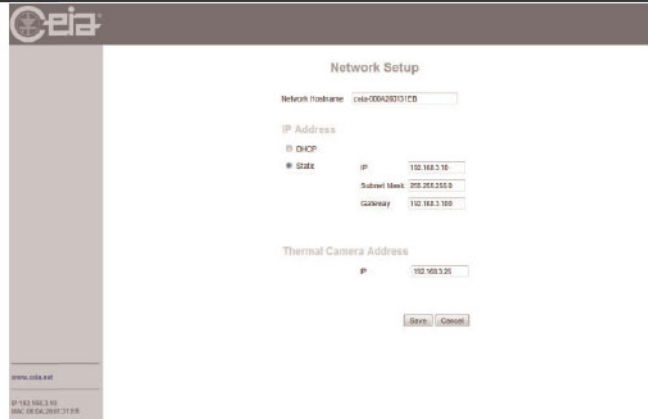


4.7.6.4 Network setup

This page allows to setup the IP address for the Controller V3. It can be set as dynamic (DHCP) or Static (default), according to the network type and to the User preferences.

An useful software for the connection with the Controller, whatever the settings, is the CEIA Discovery software, which can be downloaded free from the site.

When installed, simply write the hostname on the URL field of a browser, to automatically access the Controller internal Web Server.




IP addresses from 169.254.0.0 to 169.254.0.16 are reserved and can't be assigned.

4.7.6.5 System upgrade

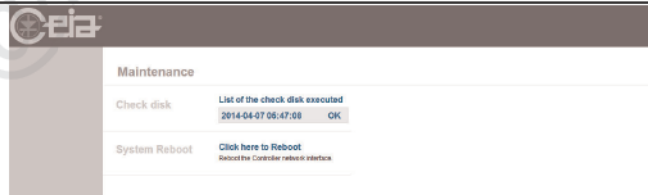
Click **Select file** and then **Send** to upgrade the Ethernet Interface Board firmware.



4.7.6.6 Maintenance

Check disk shows the result of the last disk check.

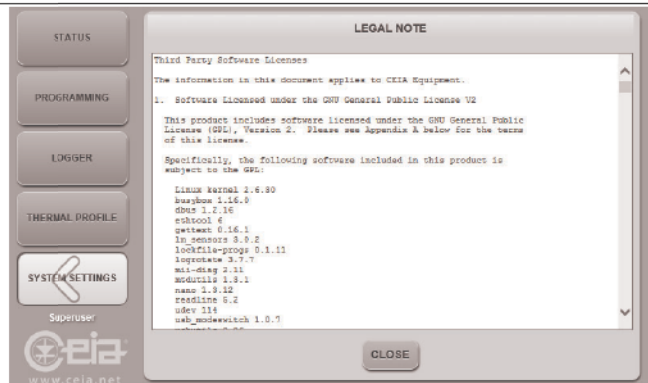
System reboot allows to reboot the Web Server..



4.7.6.7 Legal note

Press this button to read the legal notes about the software use.

Press the **Close** button to return back to the previous screen.



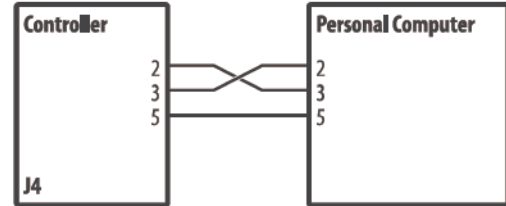
4.8 Remote Programming

The Power Cube series controllers can be managed and programmed via serial line using a terminal or a personal computer.

4.8.1 Terminal J4 configuration

Connect the RS232 port on the Controller to a terminal configured to the following line parameters:

*baud rate: 9600 or 57600; no parity
8 data bit; 1 stop bit
No flow control*



A serial communication cable can be used for this connection:

4.8.2 Description of the commands

Command	Syntax Description	Allowed Values
AOXDT	AOXDT <i>x</i> <CR> With AOXDT = 1, if more than 30 seconds are passing between the gas stopping and the start of a new working cycle, the time set on "Heating delay" is waited twice, before powering on the generator. With AOXDT = 0, the double time is disabled.	0 / 1
BR	BR <i>x</i> <CR> Set the new baud rate. Possible values: 9600 and 57600. The new settings is effective only after a PE command.	9600 / 57600
FBTOUT	FBTOUT <i><x></i> <CR> Set the FieldBus timeout. After this time, without a connection with the PLC, the A121: PLC COMM diagnosis is activated. Default value: 2 s	0 – 10 s
FLx	FL <i><x></i> <CR> This command reads the minimum flow threshold allowed on generator <i><x></i> before sending an A3 alarm. In case that a value is not available, the command answer with the string <N/A>.	1 / 2
Hn	H <i><n></i> <CR> Starts a work cycle in section <i><n></i>	1 / 2
HnSxx	H <i><n></i> S <i><xx></i> <CR> Reads the value of all parameters related to the section <i><n></i> and step <i><x></i> . The reply is provided in the format: <i><step_type></i> <space> <i><temperature></i> <space> <i><power></i> <space> <i><time></i> <space> <i><tolerance></i> <CR><LF>#	<i><n></i> : 1 / 2 <i><xx></i> : 01 ÷ 20
LD	LD <i>x</i> <CR> Loads the operating parameters contained in program cell X (1 ≤ X ≤ 100). The "_" character which follows LD is the "underscore" character (ASCII code: 95). If LD answer with the prompt, this means that the loading has been executed properly. If the indicated cell is empty, the LD will returns the message "NOT FOUND". If the indicated cell is corrupted, the answer will be "CORRUPTED". Using the underscore character between the command LD and the cell number, will load the parameters, but it will not save them and they will not be available at the next power-up. Using a space, instead of the underscore, will save the parameters, that will be ready at the next power-up.	1 / 100
PE	PE<CR> Programming end.	-
PYIBR	PYIBR <i><x></i> <CR> Set the transmission speed for all UPP sensors. Default value: 57600	57600 / 38400 19200 / 9600
PT	PT<CR> This command prints a table with all work cycle parameters together with their actual value. The format and the list of parameters depends on the operating mode selected.	-

Command	Syntax	Allowed Values
	Description	
PV	PV<CR>	-
	This command returns the Controller program version.	
RM	RM<CR>	-
	Reads the value of some of the Controller's current operating parameters. The reply is provided in the format: 	

Command	Syntax	Allowed Values										
	Description											
RS	RS<CR>	-										
	Reads the status of the Controller. The reply is provided following the format: <space><status_string_1><tab>< status_string_2><CR><LF>#											
	where: <status_string_1> refers to work station 1 and < status_string_2> refers to work station 2. Each of these may take on the following values:											
	<table><tr><th>Value</th><th>Meaning</th></tr><tr><td>P</td><td>Cycle underway</td></tr><tr><td>R</td><td>Machine ready to start a cycle</td></tr><tr><td>R<diagnosis_code></td><td>Machine ready to start a cycle, with the preceding cycle improperly concluded. The diagnosis_code can take on the values specified in the Controller manual: A4/A5/A6/A7/A8/A13/A14/A15</td></tr><tr><td>N<alarm_code></td><td>Machine in alarm (a new work cycle can not be started). The alarm_code can take on the values specified in the Controller manual: A1/A2/A3/A10/A11/A12/A16</td></tr></table>		Value	Meaning	P	Cycle underway	R	Machine ready to start a cycle	R<diagnosis_code>	Machine ready to start a cycle, with the preceding cycle improperly concluded. The diagnosis_code can take on the values specified in the Controller manual: A4/A5/A6/A7/A8/A13/A14/A15	N<alarm_code>	Machine in alarm (a new work cycle can not be started). The alarm_code can take on the values specified in the Controller manual: A1/A2/A3/A10/A11/A12/A16
	Value		Meaning									
P	Cycle underway											
R	Machine ready to start a cycle											
R<diagnosis_code>	Machine ready to start a cycle, with the preceding cycle improperly concluded. The diagnosis_code can take on the values specified in the Controller manual: A4/A5/A6/A7/A8/A13/A14/A15											
N<alarm_code>	Machine in alarm (a new work cycle can not be started). The alarm_code can take on the values specified in the Controller manual: A1/A2/A3/A10/A11/A12/A16											
RTC ^a	RTC<sensor><CR>	1, 2, 12										
	Starts to send, continuously, the temperature reading of the selected sensor. Each new measure is send each time interval set with RTT (see below). If the cycle is stopped, the string "*****" is send. If the cycle is running but the temperature read is below the minimum readable temperature, the string "-----" Is send. The data is always sent using four characters, with zeroes before the number, if needed. With <sensor> = 1 the data sent refers to the sensor 1, with <sensor> = 2 the data refers to sensor 2, with <sensor> = 12 the data refers to both sensor, with the measures separated by a space.											
RTT ^a	RTT <time><CR>	1 ÷ 40										
	Sets the time interval, in milliseconds, in which the command RTC sends a new data to the serial interface. The measurement unit is in tenth of milliseconds, so RTT=1 sends data each 10 ms. Setting RTT=0 will send data at the maximum available speed (usually the serial connection is 6-7ms for only one sensor). <time> range is 0 – 200.											
RTS ^a	RTS <samples><CR>	0 ÷ 200										
	Sets number of samples on which a mobile average is performed before sending data with RTC. A new sample is acquired each 1 ms. <sample> range is 1- 40.											
Sx	S<x><CR>	1 / 2										
	Terminates the cycle in section <x>. In the case of an automatic cycle it functions as an interruption (abort); in a manual cycle it operates as a stop command											
SN	SN<CR>	-										
	This command returns the Controller serial number.											
ST	ST<space>x<CR>	1 ÷ 100										
	Saves the operating parameters from memory to the programme cell X (1 ≤ X ≤ 100).											
_Tx	T<x><CR>	1 / 2										
	This command reads the cooling water temperatures on the indicated generator <x>. The answer will be composed by a number of values as per the number of modules installed on the indicated generator <x>. In case that a value is not available, the command answer with the string <N/A>.											
WCMD	WCMD <x><CR>	0 ÷ 250										
	This parameter allows to set a delay between the signal of a new piece and the beginning of the temperature tracking for that piece. This is used to avoid to start the temperature tracking meanwhile the piece is still moving. Default value: 0.											

Command	Syntax	Allowed Values
	Description	
WCOC	WCOC <x><CR>	0 / 1
	If set to 0, the outputs for a piece reject (too cold or too hot) are driven on the basis of the temperature read in real time. If set to 1, the outputs for a piece reject are driven on the basis of the average temperature of a full control time. Default value: 0.	
_Wx	W<x><CR>	1 / 2
	This command reads the values from the flux meters. The answer will be composed by a number of values as per the number of modules installed on the indicated generator <x>. In case that a value is not available, the command answer with the string <N/A>.	

^a These parameters will be available only if the -TP option is activated.



In case of wrong syntax, out of range parameter or unrecognised command, the controller will show the message: ???

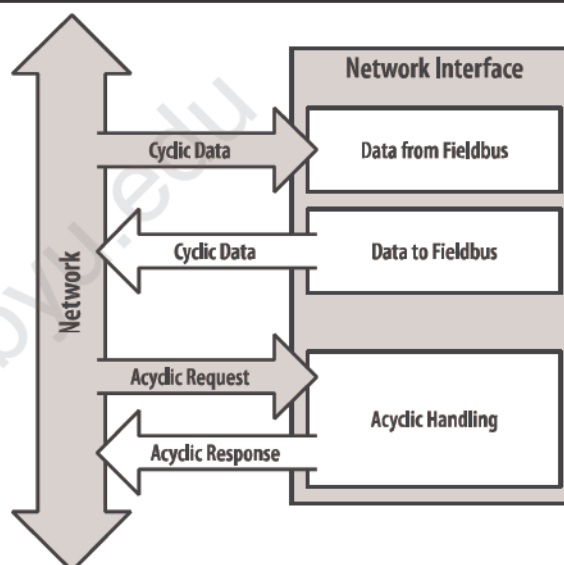
4.9 Field Bus Interface

The CEIA Field Bus Interface primary function is to exchange information with other devices on the network. This is performed by exchanging cyclic or acyclic I/O and making it available to the host device via two memory buffers.

The Field Bus connection is performed through the Anybus-CompactCom modules (produced by HMS), that converts data sent through the serial interface in corresponding data on the chosen network protocol.

Actually the Field Bus Interface is supplied with the modules for the following protocols:

- **Ethercat**. No additional settings required and automatically identified by the PLC
- **EthernetIP** and **Profinet IO**. Requires IP address and network mask. A valid address can be obtained from the network if set as DHCP



The network interface can start, stop and monitor heating cycles through the cyclic data exchanged on bus. It also allows to read and set all working parameters through acyclic messages.

4.9.1 Cyclic data

This is the I/O data transmitted in real-time on the field bus.

- In **Ethernet/IP** they are called “*implicit messages*”, they are assigned to instances 100 and 150 of the “*Assembly Object*” (04h).
- In **EtherCAT** they are called “*Process Data Objects*”, they are automatically assigned to the TxPDO and RxPDO available on the Anybus-CC module.
- In **Profinet** they are “*Profinet input/output data*”.
- In **Profibus** they are “*Profibus input/output data*”.
- In **ModBus** there are no Cyclic data. See paragraph 4.10.4.

Cyclic data is updated every 6 ms.



In case of **Profinet** or **Profibus**, the tables order shall be: Outputs before and inputs after.

4.9.1.1 From Controller to PLC (PLC inputs)

Name	Data Type	Description
Temperature Measure 1	UINT16	temperature (in tenths of degrees, i.e. 1234 = 123.4 °C) measured from sensor 1
Temperature Setpoint 1	UINT16	temperature (in tenths of degrees) requested ¹ , as programmed by parameters of section 1
Temperature Measure 2	UINT16	temperature measure from sensor 2
Temperature Setpoint 2	UINT16	temperature requested ¹ , as programmed by parameters of section 2
Power output 1	UINT16	power output (in tenths of %) of head 1, generator 1
Power output 2	UINT16	power output of head 2, generator 1 (or head 1, generator 2, if HEATING SYSTEM = DUAL GENERATOR)
Status section 1	UINT8	0 cycle stopped on section 1
		1 ÷ 20 cycle in progress on section 1, the number indicates which step of the thermal profile has been reached
		251 cycle stopped AND inhibited start, because Start Command is not 0
		the Abort Command is not 0
		the RESET1 input is activated
Status section 2	UINT8	252 cycle stopped AND inhibited start, because the PLC is not checking the connection with the Controller (enable SM watchdog on EtherCAT)
		255 cycle stopped on section 1, unable to start a new cycle because of diagnosis
		0 cycle stopped on section 2
		1 ÷ 20 cycle in progress on section 2, the number indicates which step of the thermal profile has been reached
		251 cycle stopped AND inhibited start, because Start Command is not 0
Diagnosis section 1	UINT8	the Abort Command is not 0
		the RESET2 input is activated
Diagnosis section 2	UINT8	252 cycle stopped AND inhibited start, because the PLC is not checking the connection with the Controller (enable SM watchdog on EtherCAT)
		255 cycle stopped on section 2, unable to start a new cycle because of diagnosis
Working cell	UINT8	0 no diagnosis on section 1 (status OK)
Spare	UINT8	>=1 numeric diagnosis code for section 1
		0 no diagnosis on section 2 (status OK)
		>=1 numeric diagnosis code for section 2
		Working cell in use, if value is between 1 and 100 or “no working cell loaded” if value is 255.
		Reserved

¹ When the heating cycle is OFF, the value is 0.

4.9.1.2 From PLC to Controller (PLC outputs)

With F.B. VERSION=1

Name	Data Type	Description
Start Command	UINT8	bit 0x01 operates on section 1; bit 0x10 operates on section 2 For each bit, commuting from 0 to 1 indicates to start the cycle.
		1 hold the cycle activated (applicable in manual mode, for example TIMER A with HOLDING TIME = 0)
		0 stop the cycle (applicable in manual mode, for esempio TIMER A with HOLDING TIME = 0)
		bit 0x02 operates on section 1; bit 0x20 operates on section 2 For each bit, commuting from 0 to 1 indicates to start of the maximum temperature tracking of a new piece (available when the Temperature in-window with maximum values mode is selected).
Abort Command ¹	UINT8	bit 0x01 operates on section 1; bit 0x10 operates on section 2 Effect of each bit:
		bit 0 Inhibition of section 1: if = 1 inhibition activated
		bit 1 Forward motor of section 1: if = 1 (and bit 2 = 0) the motor runs at "forward manual speed"
		bit 2 Motor forward of section 1: if = 1 (and bit 1 = 0) the motor rewinds at "reverse manual speed"
		bit 3 Cylinder of section 1: 0: retracted cylinder, 1: extended cylinder
		bit 4 Inhibition of section 2: if = 1 inhibition activated
		bit 5 Forward motor of section 2: if = 1 (and bit 6 = 0) the motor runs at "forward manual speed".
		bit 6 Motor forward of section 2: if = 1 (and bit 5 = 0) the motor rewinds at "reverse manual speed"
		bit 7 Cylinder of section 2: 0: retracted cylinder, 1: extended cylinder

¹ To enable the movements it is required that the relative abort bit is set to 1. Manual movement via fieldbus only has effect if the heating cycles are stopped in both sections.

Example: to move forward motor 1 with extended cylinder and simultaneously move back motor 2 with retracted cylinder, the value to be sent is 0x5B (0101 1011).

If the display is positioned on the programming menu for manual movement, the commands from PLC have no effect.

With F.B. VERSION=2

Name	Data Type	Description
Start Command	UINT8	bit 0x01 operates on section 1; bit 0x10 operates on section 2 For each bit, commuting from 0 to 1 indicates to start the cycle.
		1 hold the cycle activated (applicable in manual mode, for example TIMER A with HOLDING TIME = 0)
		0 stop the cycle (applicable in manual mode, for esempio TIMER A with HOLDING TIME = 0)
		bit 0x02 operates on section 1; bit 0x20 operates on section 2 For each bit, commuting from 0 to 1 indicates to start of the maximum temperature tracking of a new piece (available when the Temperature in-window with maximum values mode is selected).
Abort Command	UINT8	bit 0x01 operates on section 1; bit 0x10 operates on section 2 For each bit:
		0 no effect
		1 stop the current cycle and avoid the start of a new cycle
Remote Temperature setpoint 1	UINT16	Temperature to reach on section 1 in tenth of °C.
Remote Temperature setpoint 2	UINT16	Temperature to reach on section 2 in tenth of °C.
Remote Power Setpoint 1	UINT16	Power set for section 1 in tenth of %.
Remote Power Setpoint 2	UINT16	Power set for section 2 in tenth of %.

With F.B. VERSION=3

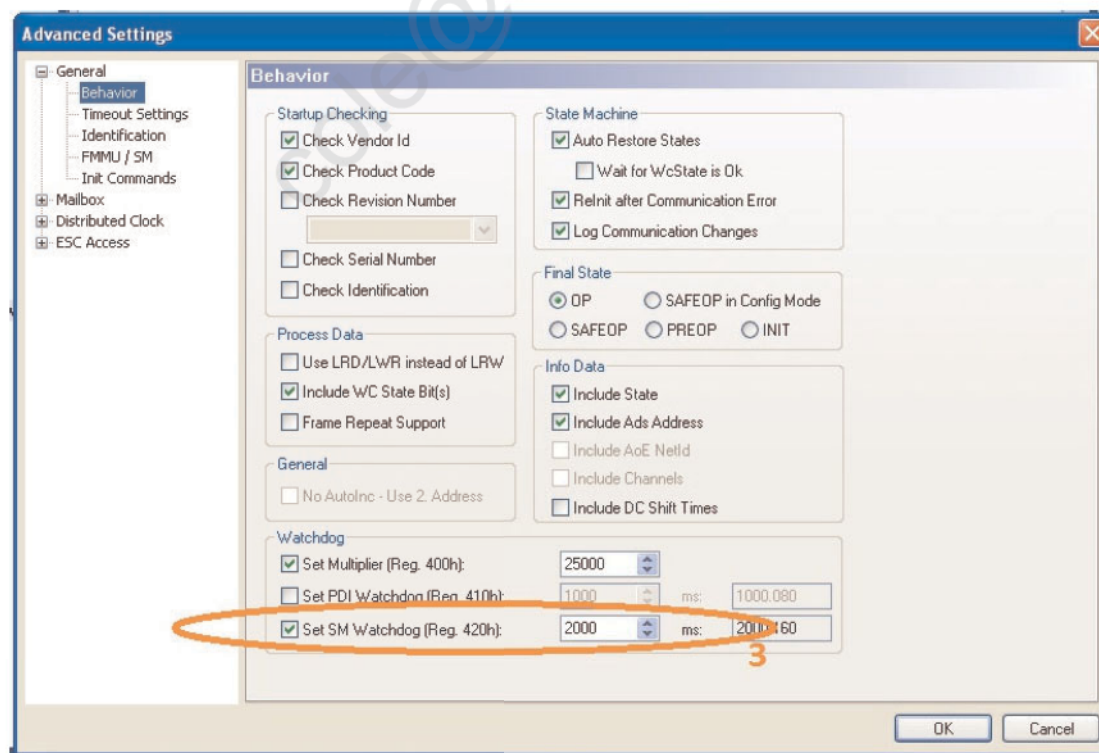
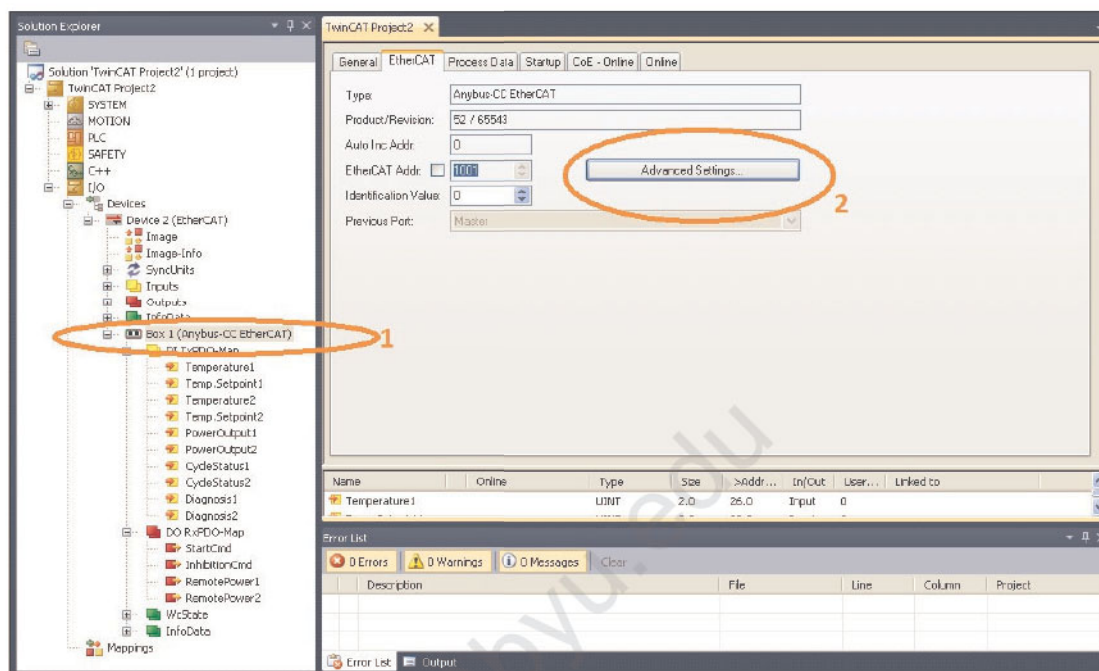
Name	Data Type	Description
...	...	As in F.B. VERSION=2
Cell Selection	UINT8	Selection of the cell to load, range 1-100
Cell Command	UINT8	Bit 0x01 executes the loading of the selected cell when it changes from 0 to 1

4.9.2 Field Bus connection monitoring

When a cycle is started from the Field Bus, using the cyclic data, the connection will be monitored. If the connection falls, the heating cycle will abort.

When using EtherCAT, it is necessary to enable the *SM watchdog* (suggested value: 2s) or the heating cycle will not start because the connection monitor is not enabled.

The following is an example on how to enable the Watchdog on Beckhoff TwinCAT3 software:



Using **ModBus**, it is important to send commands (read or write) with a period time of less than 2 s, in order to maintain the connection active.

4.9.3 Acyclic data (messages)

“Explicit messages” are used to get and set the working parameters of the Controller. The working parameters are associated to “Applicative Data Instances” (ADI), according to the table in the next page.

Each instance has the following attributes:

1. *name (first byte contains the length of the string)*
2. *data type*
3. *number of elements*
4. *access descriptor (read only, read/write)*
5. *value (or values if number of elements is more than 1)*
6. *maximum value*
7. *minimum value*
8. *default value*

When using **Ethernet/IP**, data are accessible through the instances of the object A2h.

When using **EtherCAT**, the parameters list (with name, data type and value) is available through the SDO protocol.

When using **Profinet IO**, the parameter list is accessible by means of record data read/write services, using API=0, SLOT=0, SUBSLOT=1 and INDEX>=1. INDEX selects the corresponding line of the Parameters list.

When using **Profibus** the access to the table is performed through “DP-V1 Read/Write services”. To select the ADI use Profibus SLOT and INDEX, according to the following formula:

$$ADI = SLOT \cdot 255 + INDEX + 1$$

that is

$$SLOT = (ADI - 1) / 255$$

$$INDEX = (ADI - 1) \text{ MOD } 255$$

For **ModBus TCP**, see the relative paragraph.

When accessing the working parameters, we recommend to use the “name” attribute instead of using the instance number, to ensure better compatibility with future software revisions which may alter the order of the instances.



The parameters settings performed through the Field Bus are lost when the Controller is powered off. It is possible to save them, so as to retrieve them and the next power on, using the MEMST command (instance 116) with the cell 0.

ADI	Type	Name	Range	Number of elements	Measurement unit	Access	Description
1	CHAR	CEIA_SN	str 11 elements	11	-	R	Serial number
2	UINT8	CEIA_REV	-	1	-	R	Data structure revision
3	CHAR	F.B.ID	Str 8 elements	8	-	R	Controller ID
4	UINT16	Temperature1	-	1	1/10 of degree	R	Measured Temperature 1
5	UINT16	Temp.Setpoint1	-	1	1/10 of degree	R	Target Temperature 1
6	UINT16	Temperature2	-	1	1/10 of degree	R	Measured Temperature 2
7	UINT16	Temp.Setpoint2	-	1	1/10 of degree	R	Target Temperature 2
8	UINT16	PowerOutput1	-	1	1/10 of %	R	Power Output 1
9	UINT16	PowerOutput2	-	1	1/10 of %	R	Power Output 2
10	UINT8	CycleStatus1	-	1	-	R	Status 1
11	UINT8	CycleStatus2	-	1	-	R	Status 2
12	UINT8	Diagnosis1	-	1	-	R	Diagnosis 1
13	UINT8	Diagnosis2	-	1	-	R	Diagnosis 2
14	UINT8	Work Cell	-	1	-	R	Loaded memory cell (or 255 if no cell)
15	UINT8	Reserved	-	1	-	R	Reserved
16	UINT8	StartCmd	-	1	-	R	Reading of "Start Command" status
17	UINT8	InhibitionCmd	-	1	-	R	Reading of "Inhibition Command" status
18	UINT16	RANGE TEMP1	-	-	1/10 of degree	R	Range of temperature sensor 1
19	UINT16	RANGE TEMP2	-	-	1/10 of degree	R	Range of temperature sensor 2
20	UINT16	TEM1	T _{1min} ÷ T _{1max}	1	1/10 of degree	R/W	Temperature 1
21	UINT16	TEM2	T _{2min} ÷ T _{2max}	1	1/10 of degree	R/W	Temperature 2
22	UINT16	TEM1L	T _{1min} ÷ T _{1max}	1	1/10 of degree	R/W	Minimum temperature 1
23	UINT16	TEM2L	T _{2min} ÷ T _{2max}	1	1/10 of degree	R/W	Minimum temperature 2
24	UINT16	TEM1C	0 ÷ T _{1max}	1	1/10 of degree	R/W	Low temperature 1
25	UINT16	TEM2C	0 ÷ T _{2max}	1	1/10 of degree	R/W	Low temperature 2
26	UINT16	EMAx	10-100	2	1/100 of unit	R/W	Emissivity A 1&2
27	UINT16	EMBx	10-100	2	1/100 of unit	R/W	Emissivity B 1&2
28	UINT16	EADx	0-1000	2	1/100 of s	R/W	Emissivity adjustment delay 1&2
29	UINT16	PHx	0-12000	2	1/100 of s	R/W	Pre-heating time 1&2
30	UINT16	HTAx cs	0-60000	2	1/100 of s	R/W	Holding time A 1&2
31	UINT16	HTAx m	10-2160	2	Minutes	R/W	
32	UINT16	HTAx h	36-2250	2	Hours	R/W	
33	UINT16	HTBx cs	0-60000	2	1/100 of s	R/W	Holding time B 1&2
34	UINT16	HTBx m	10-2160	2	Minutes	R/W	
35	UINT16	HTBx h	36-2250	2	Hours	R/W	
36	UINT16	PWAx	100-990	2	1/10 of %	R/W	Power A 1&2
37	UINT16	PWBx	100-990	2	1/10 of %	R/W	Power B 1&2
38	UINT16	PATx	0-60000	2	1/100 of s	R/W	Power A time 1&2
39	UINT16	NSTx	1-20	2	-	R/W	Number of Steps 1&2
40	UINT16	H1SxxTE	T _{1min} ÷ T _{1max}	20	1/10 of degree	R/W	Thermal profile temperature on step xx sections 1&2
41	UINT16	H2SxxTE	T _{2min} ÷ T _{2max}	20	1/10 of degree	R/W	
42	UINT16	H1SxxTO	0-5000	20	1/10 of degree	R/W	Thermal profile temperature tolerance on step xx, sections 1&2
43	UINT16	H2SxxTO	0-5000	20	1/10 of degree	R/W	
44	UINT16	H1SxxPO	100-990	20	1/10 of %	R/W	Thermal profile power set on step xx, section 1&2
45	UINT16	H2SxxPO	100-990	20	1/10 of %	R/W	
46	UINT16	H1SxxTI cs	0-60000	20	1/100 of s	R/W	Thermal profile times of step xx, Section 1
47	UINT16	H1SxxTI m	10-2160	20	Minutes	R/W	
48	UINT16	H1SxxTI h	36-2250	20	Hours	R/W	
49	UINT16	H2SxxTI cs	0-60000	20	1/100 of s	R/W	Thermal profile times of step xx, Section 2
50	UINT16	H2SxxTI m	10-2160	20	Minutes	R/W	
51	UINT16	H2SxxTI h	36-2250	20	Hours	R/W	
52	ENUM	H1SxxTY	HE, TC, CO	20	-	R/W	Step xx Type sect.1
53	ENUM	H2SxxTY	HE, TC, CO	20	-	R/W	Step xx Type sect.2
54	UINT16	FDSx	1-20	2	-	R/W	Wire feeding steps 1&2

ADI	Type	Name	Range	Number of elements	Measurement unit	Access	Description
55	ENUM	TSNx	/SL, TCK, TCS, TCC,UPP,THC	2	-	R/W	Temperature sensor type 1&2
56	ENUM	PxDIAG	N, Y	2	-	R/W	Pyrometer 1&2 Diagnosis Enable
57	UINT16	TRL1	T1 _{min} -(TRH1-50)	1	degrees	R/W	Impac minimum Temperature 1
58	UINT16	TRL2	T2 _{min} -(TRH2-50)	1	degrees	R/W	Impac minimum Temperature 2
59	UINT16	TRH1	(TRL1+50)-T1 _{max}	1	degrees	R/W	Impac max Temperature 1
60	UINT16	TRH2	(TRL2+50)-T2 _{max}	1	degrees	R/W	Impac max Temperature 2
61	UINT16	ETx	0-6	2	-	R/W	Impac exposition time 1&2
62	UINT16	DIATO	50-300	1	1/100 of s	R/W	Dias Timeout
63	ENUM	LAS	N, AUTO, Y	1	-	R/W	Aiming Light
64	UINT32	SCNTERx	0	2	-	R/W	Soldering counter reading.
			The special value 4294967295 (0xFFFFFFFF) means that the connected generator does not support counters. During writing it is possible to send "0" to reset the counter.				
65	ENUM	SCNTING	N, Y	1		R/W	Pieces counter
66	ENUM	_C	V, C	1	-	R/W	Control Type
67	ENUM	Reserved	-	1	-	R/W	Reserved
68	ENUM	Reserved	-	1	-	R/W	Reserved
69	ENUM	Reserved	-	1	-	R/W	Reserved
70	UINT16	HTO_cs	0-60000	1	1/100 of s	R/W	Heating timeout
71	UINT16	HTO_m	10-2160	1	minutes	R/W	
72	UINT16	HTO_h	36-2250	1	hours	R/W	
73	UINT16	TTOx_cs	0-60000	2	1/100 of s	R/W	Temperature timeout on section 1&2
74	UINT16	TTOx_m	10-2160	2	minutes	R/W	
75	UINT16	TTOx_h	36-2250	2	hours	R/W	
76	ENUM	TEMxSET	DIG, ANA	2	-	R/W	Temperature setting 1&2
77	ENUM	PWxSET	DIG, ANA	2	-	R/W	Power setting 1&2
78	ENUM	TCTYx	AUTO, PID, WR, WM	2	-	R/W	Temperature control type 1&2
79	UINT16	PIDPx	1-1023	2	-	R/W	PID P 1&2
80	UINT16	PIDIx	0-1023	2	-	R/W	PID I 1&2
81	UINT16	PID Dx	0-1023	2	-	R/W	PID D 1&2
82	UINT16	PIDTx	1-256	2	-	R/W	PID T 1&2
83	UINT16	DPWx	0-200	2	1/10 of %	R/W	Power increasing 1&2
84	UINT16	TCNTx	0-2000	2	1/100 of s	R/W	Control time 1&2
85	UINT16	DLTx	0-2000	2	1/100 of s	R/W	Control delay 1&2
86	ENUM	TEMUN	C, F	1	-	R/W	Temperature measurement unit
87	UINT16	FDAx	0-3000	2	1/10 of mm	R/W	Wire quantity A 1&2
88	UINT16	FRAx	1-99	2	mm/s	R/W	Wire speed A 1&2
89	UINT16	FDBx	0-3000	2	1/10 of mm	R/W	Wire quantity B 1&2
90	UINT16	FRBx	1-99	2	mm/s	R/W	Wire speed B 1&2
91	UINT16	RVx	0-3000	2	1/10 of mm	R/W	REWIND LENGTH 1&2
92	UINT16	RVRx	1-99	2	mm/s	R/W	REWIND RATE 1&2
93	UINT16	CDx	0-9900	2	1/100 of s	R/W	Cylinder delay 1&2
94	UINT16	IJCx	0-10	2	-	R/W	Injection cycle 1&2
95	UINT16	IJTx	0-100	2	1/100 of s	R/W	Injection time 1&2
96	UINT16	STKT	0-200	1	1/100 of s	R/W	Cylinder movement time 1&2
97	UINT16	MAXCURx	50-300	2	mA	R/W	Max Current 1&2
98	UINT16	WTO_cs	0-60000	1	1/100 of s	R/W	Wire timeout
99	UINT16	WTO_m	10-2160	1	minutes	R/W	
100	UINT16	WTO_h	36-2250	1	hours	R/W	
101	ENUM	WPRES	N, Y	1	-	R/W	Wire presence
102	ENUM	FFD	STD, PRE	1	-	R/W	Wire sequence
103	ENUM	INJ	STD, PRE	1	-	R/W	Deoxidizing injection
104	UINT16	HDx	0-1000	2	1/100 of s	R/W	Heating delay 1&2
105	UINT16	COx	0-60000	2	1/100 of s	R/W	Cooling time 1&2

ADI	Type	Name	Range	Number of elements	Measurement unit	Access	Description
106	UINT16	GTO_cs	0-60000	1	1/100 of s	R/W	Gas Timeout
107	UINT16	GTO_m	10-2160	1	minutes	R/W	
108	UINT16	GTO_h	36-2250	1	hours	R/W	
109	ENUM	OPM	A, B, C, S, AUTO, AUTO2W, TPR, TP2W	1	-	R/W	Functioning mode
selectable values based on Controller enabled options							
110	ENUM	HSYS	1, 2	1	-	R/W	Heating system
111	ENUM	SHEATC	2, 1	1	-	R/W	Single generator adjustment
112	ENUM	H1CONF	STD, 2PY	1	-	R/W	Head 1 functioning mode
113	ENUM	TCON	N, Y	1	-	R/W	Temperature control
114	ENUM	EADJ	N, Y	1	-	R/W	Emissivity adjustment
115	ENUM	Reserved	-	1	-	R/W	Reserved
116	UINT16	MEMST	0-100	1	-	R/W	Writing an x value between 1 and 100, all parameters currently in use are saved on non-volatile memory cell x. Writing the value 0, the parameters currently in use are transferred to the Controller's internal work cell; thus, the parameters will be available after the Controller has been switched off and on again. The controller is ready for a new operation when the read value is 65535
117	UINT16	MEMLD	1-100	1	-	R/W	Writing an x value between 1 and 100 all parameters are loaded from memory cell x. Note: the parameters are loaded into the Controller's internal volatile work cell. In order to ensure that the loaded working parameters are operative even after switching the Controller off and on again, it is necessary to execute, after the MEMLD x command, the command MEMST 0. The controller is ready for a new operation when the read value is 65535
118	UINT16	_Wx	-	8	-	R	Reading of flowmeters
			first 4 values are from generator 1 (Master unit and eventually until 3 Slaves), the other 4 values are for generator 2				
119	UINT16	FLx	-	2	-	R	Minimum thresholds for water flows in order to give A3 fault (generator 1 & 2)
120	UINT16	_Tx	-	8	-	R	Reading of water temperatures
			first 4 values are from generator 1 (Master unit and eventually until 3 Slaves), the other 4 values are for generator 2				
121	UINT16	_Vx	-	2	-	R	Reading of Vcoil for Generator 1 & 2
122	UINT16	_Ix	-	2	-	R	Reading of I for Generator 1 & 2
123	UINT16	DISH1	0-1000	1	seconds	R/W	Dias Shutter Time 1
124	UINT16	DISH2	0-1000	1	seconds	R/W	Dias Shutter Time 2
125	UINT16	Reserved	-	1	-	R/W	Reserved
126	UINT16	Reserved	-	1	-	R/W	Reserved
127	ENUM	PxCU	N, Y	2	-	R/W	Close-up lens 1 and 2
128	UINT8	Start Cmd	-	1	-	R	Reading of "Start Command"
129	UINT8	Inhib. Cmd	-	1	-	R	Reading of "Abort Command"
130	UINT16	Remote Temper Setp1	-	1	-	R	Reading of "Remote Temperature Setpoint 1"
131	UINT16	Remote Temper Setp2	-	1	-	R	Reading of "Remote Temperature Setpoint 2"
132	UINT16	Remote Power Setp1	-	1	-	R	Reading of "Remote Power Setpoint 1"
133	UINT16	Remote Power Setp2	-	1	-	R	Reading of "Remote Power Setpoint 2"
134	UINT8	Reserved	-	1	-	R	Reserved
135	UINT8	Reserved	-	1	-	R	Reserved
136	UINT8	Reserved	-	1	-	R	Reserved
137	UINT8	Reserved	-	1	-	R	Reserved
138	ENUM	WMTD	AUTO, AVG	1	-	R/W	Window Mode Temper. To Display

ADI	Type	Name	Range	Number of elements	Measurement unit	Access	Description
139	UINT16	AN1PK	-	1		R/W	Reserved
140	UINT16	ATEXN		2		R/W	Minimum analog input value
141	UINT16	ATEXX		2		R/W	Maximum analog input value
142	UINT32	EFBD	-	8	1-6: Date and time from the last partial reset 7: Partial kW counter 8: Global kW counter	R/W	SA generator power consumption data. <i>Write 0 on any of the elements to reset the partial kW counter</i>
143	UINT16	LFBDA	0-9999	20	1: Reading and setting of the cycle counter section 1 2: Reading and setting of the cycle counter section 2 3-11: Reading and setting of the filename suffix section 1 12-20: Reading and setting of the filename suffix section 2	R/W	Cycle counters and Filenames Suffix of Data Logger. Accepted values: Cycle Counter: from 1 to 9999. The 0 (zero) value is ignored. Suffix: Capital Letters (ASCII codes from 65 to 90) Numeric Strings (ASCII code from 48 to 57) 0 (zero), as end of string if the Suffix is shorter than 9 characters
144	UINT32	GFREQ		2		R	Generator operative frequency
145	UNIT16	OTTx	0.0 – 5.0	2	1/10 second	R/W	Minimum time for "Temperature out of tolerance"
146	ENUM	OTSx	I, E	2		R/W	Selection of the moment to warn about a "Temperature out of tolerance" I: Immediate; E: End cycle
147	UNIT16	UPPTCK	T _{min} – T _{max}	8	1/10 degree	R	Temperature sensors reading: Byte 0: UPP SENSOR #1 Byte 1: UPP SENSOR #2 Byte 2: TCK #1 Byte 3: TCK #2



Instances from 4 to 17, and from 128 to 133, are a copy of the cyclic data



All parameters representable as hundredth of second, minutes or hours (for example HTAx), only one measurement unit at a time can be used. If a value is set in minutes, for example, the other two are automatically set to zero



For all temperature parameters, the indicated "degree" measurement unit stand for Celsius or Fahrenheit, depending on TEMUN parameter setting.

For parameters of type ENUM, the following tables show the correspondence between the strings and the values of the enumeration. Some types of fieldbuses will use strings, other numerical values. For example, **EthernetIP** uses numerical values, and **EtherCAT** uses strings.

Instance name	String	Numeric value
HnSxxTY	HE	0
	TC	1
	CO	2
OPM	A	0
	B	1
	C	2
	S	3
	AUTO	4
	AUTO2W	5
	TPR	6
	TP2W	7
PWxSET, TEMxSET	DIG	0
	ANA	1
_C	V	0
	C	1
TEMUN	C	0
	F	1
FFD, INJ	STD	0
	PRE	1
HSYS	1	0
	2	1

Instance name	String	Numeric value
SHEATC	2	0
	1	1
H1CONF	STD	0
	2PY	1
LAS	N	0
	AUTO	1
	Y	2
PxDIAG, SCNTING, WPRES, TCON, EADJ, PxCU	N	0
	Y	1
TSNx	/SL	0
	TCK	1
	TCS	2
	UPP	3
	TCC	4
	THC	5
WMTD	AUTO	0
	AVG	1

4.9.4 ModBus TCP

Use the Ethernet 502 port.

To access cyclic and acyclic data, the “Read Holding Register (function code 3)”, “Write Single Register (function code 6)” and “Write Multiple Registers (function code 16)” are used.

The cyclic data concept is not present on ModBus, however the Controller cyclic data are accessible on the ModBus register listed below. The PLC software will have to periodically read these registers.

From Controller to PLC

Name	Modbus register
Measured temperature 1	256
Target temperature 1	257
Measured temperature 2	258
Target temperature 2	259
Power output 1	260
Power output 2	261

Name	Modbus register
Section 1 status	262
Section 2 status	263
Section 1 diagnosis	264
Section 2 diagnosis	265
Work cell	266
Spare	267

From PLC to Controller

Name	Modbus register
Start command	0
Inhibition command	1

For the acyclic data, each row of the table (ADI) is associated to a block of 64 ModBus registers. The base register of each block is calculated according to the following formula:

$$blk_{addr} = (ADI - 1) \cdot 64 + 528$$

Inside the block the single registers has different meaning according to ADI type.

Type	Element number	Description
ENUM, UINT8, UINT16	1	The block base register contains the parameter value
ENUM, UINT8	>1	Each ModBus register (16bit) contains 2 elements of the ADI (8bit). ADI_0 ADI_1 ADI_2 ADI_3 ADI_N vector of 8 bit elements of the ADI, becomes ADI_1 ADI_0 ADI_3 ADI_2 ADI_N ADI_N-1 vector of 16bit registers
UINT16	>1	Each ModBus register (16bit) contains 1 ADI element (16bit).
UINT32	any	2 ModBus registers (16bit) contain 1 ADI element (32bit). The ADI less significant part is in the first of the two registers.

4.10 User parameters setting

A SUPER USER can decide which parameters the USER can see and modify when accessing to the programming phase.

Access to the programming as SUPER USER and select the command **MODE > USER PARAMETERS**. A first list of the programming menus will be displayed.

USER PARAMETERS

```

heating
> gas
mode
memory

```

Each menu contains the full list of parameters available to the SUPER USER, together with a Y/N value. If set to Y, the parameter will be available to the USER, if set to N it will be invisible.

```

> HEATING DELAY 1.....Y<
HEATING DELAY 2.....Y
COOLING TIME 1.....Y
COOLING TIME 2.....Y
GAS TIMEOUT.....N

```

4.11 Quick Access setting

A list of frequently changed parameters can be set to be quickly accessed through the Quick Access key.

Access to the programming as SUPER USER and select the command:

MODE > SETUP QUICK ACCESS.

The list of the actual Quick Access parameters will be displayed.

Scroll the list to reach the desired position and press E to change the related parameters. The line will start to blink and through the arrow keys it will be possible to scroll through the parameters list and to choose one by pressing E again.

The positions on the list, showing four dashes "----", are free.

SETUP QUICK ACCESS

```

> 01.POWER A1
02.TEMPERATURE 2
03. ----
04.POWER A 2
05. ----
06. ----
07.COOLING TIME 2
08. ----
09. ----
10. ----
11. ----
12. ----

```

During the normal operations, press the Quick Access key to show the pre-set list of parameters, together with their value.

Select the parameter you want to change and press E. Through the arrow keys, it is possible now to change its value.

QUICK ACCESS

```

> POWER A1          98%
<
TEMPERATURE 2      520.0C

POWER A 2          RE

COOLING TIME 2     0.0s

```

5 MAINTENANCE



Before performing any maintenance operation on the unit, read carefully the *Safety instructions - Warnings* section of this manual.

5.1 Periodical maintenance

All periodical maintenance operations are listed below, together with the preferred frequency:

Intervention	Frequency	Operator	Procedure
Cleaning	Periodically	Operator	See paragraph 5.2.1



The operator shall evaluate the system status and its working capability, according eventually with the maintenance personnel.

5.2 Periodical maintenance procedures

5.2.1 Cleaning

Do not wash the device with liquid detergents or chemical substances. Any cleaning should be done using a slightly-damp, non-abrasive cloth.

5.3 Troubleshooting

See paragraph 5.5 for the self-diagnosis messages list.

5.4 Extraordinary maintenance procedures

5.4.1 Fuse replacement

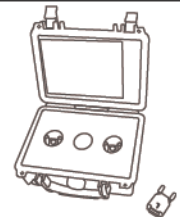
The following table shows position and values of the device fuses.

Ref.	Type	Description	Position
P1	1A, 250V; time-leg 5x20mm	Power supply protection fuse	Under the Main Switch

5.4.2 Calibration verification for the temperature sensors inputs

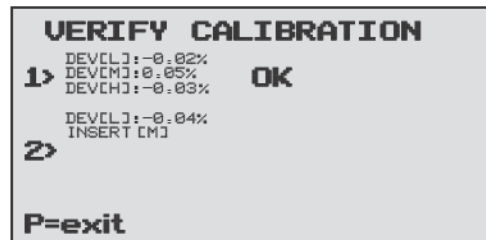
In case the Quality Control procedures requires the calibration of the temperature sensors, an automatic function and, upon request, a calibration verification kit for the temperature sensors inputs, are available.

The kit includes three serial adapters, marked by **L**, **M** and **H** letters, to be inserted in the place of temperature sensors, when requested by the automatic procedure.



The screenshot beside shows a moment of the automatic procedure, started through the command **HEATING > VERIFY CALIBRATION**.

In case the procedure is performed positively for both inputs, it will be necessary to calibrate only the temperature sensors, otherwise also the Controller needs to be re-calibrated.



5.5 Self-diagnosis messages

The microcomputer controller notifies the operator in case of malfunctioning of one of the devices connected to the Master Controller by means of a **self-diagnosis** display message.

The Master Controller display is divided, in this case, into two sections: the first line at the top displays the status of station 1, the second line displays the status of station 2 (see illustration below).

Also in cases where the self-diagnosis system gives a signal, the signal will appear in one of the two zones depending on whether it occurred in the first or second station.

The illustration beside shows an example of self-diagnosis that relates to station 1 (generator off).

```
1> A12: generator
not ready
```

```
2> T= 700C S=====L
TIME= 10s POW.=99%
```

The remote operator shall send the command RS to be able to see the self-diagnosis message related to the fault.

The illustration beside shows an example of self-diagnosis message that tells that section 1 is off (N[A2]) and section 2 is ready to operate (R).

```
#
#
# RS N[A2] R
```

Message	Remote message	Probable cause / Action	Signal on Fieldbus
–	R	No diagnosis, ready to operate	0
No actions required			
–	P	No diagnosis, cycle running	0
No actions required			
A1: water temperature too high	N[A1]	Water temperature too high	1
Check the temperature of cooling water that have to be less than 50° (intervention threshold =55°C)			
A2: internal fault	N[A2]	Tuning failure	2
Contact the technical department of CEIA or the nearest distributor			
A3: no water	N[A3]	The flow of cooling water has stopped	3
Check the presence of water in the cooling unit.			
Check the fluxometer input in the J4-J5 connectors.			
Check the presence of blocking-up in the generator inlets and outlets			
A4: increase coil dimension A4: add a loop	R[A4]	Too small dimensions of the inductor.	4
Increase coil dimensions			
A4: increase coil dimension A4: add a loop	R[A4]	Serie 50 generators: wrong coupling between the inductor and the piece to heat (the message A4 is shown after 20s from the start of the heating workcycle)	4
Verify the correct dimensioning of the inductor through the free software PowerCube Assistant (see paragraph 5.5.1)			
A4: check short circuit in adductors	R[A4]	Short circuit in the inductor	4
Check the absence of short circuit of lead in wires			

Message	Remote message	Probable cause / Action	Signal on Fieldbus
A5: decrease coil dimensions	R[A5]	Coil too big	5
Decrease coil dimensions. Eventually verify and optimize through the free software PowerCube Assistant (see paragraph 5.5.1)			
A5: decrease coil dimensions	R[A5]	Rims too far	5
Check that the distance between the rims of the head is not greater than the minimum required distance (see the Generator Manual). Eventually verify and optimize through the free software PowerCube Assistant (see paragraph 5.5.1)			
A5: decrease length of adductors	R[A5]	Rims too long	5
Diminish the length of the head spouts (centimetre by centimetre). The reduction of the length must be carried out only in the event that the shape of the coil cannot be changed for the type of processing in progress and when the distance between the lead-in wires has been checked . Eventually verify and optimize through the free software PowerCube Assistant (see paragraph 5.5.1)			
A5: decrease length of adductors	R[A5]	Serie 50 generators: wrong coupling between the inductor and the piece to heat (the message A5 is shown after 30s from the start of the heating workcycle)	5
Verify the correct dimensioning of the inductor through the free software PowerCube Assistant (see paragraph 5.5.1)			
A6: internal fault	R[A6]	Possible breakdown of an internal component	6
Contact the technical department of CEIA or the nearest distributor			
A7: head not connected	R[A7]	Disconnection of a head	7
Check the connection between the heads and the electronic unit			
A8: generator timeout	R[A8]	Generator timeout	8
Message coming only from Spectacle Frames Line generators: The generator activation time is too long: Check the programming and reduce the activation time.			
A9: no wire	R[A9]	Soldering wire not present	8
There is no more soldering wire in the feeder. Load the feeder with wire. Verify the correct positioning of the soldering wire in respect to the sensor.			
If a sensor is not installed, verify that the "Wire presence" parameter is set to OFF			
A10: low line voltage	N[A10]	Generator power supply under the range	10
Insert (between the mains and the heater) a transformer with an input voltage equal to that available on the line and an output voltage correct for the connected Generator			
A11: high line voltage	N[A11]	Generator power supply over the range	11
Insert (between the mains and the heater) a transformer with an input voltage equal to that available on the line and an output voltage correct for the connected Generator			
A12: generator not ready	N[A12]	PowerCube not connected	12
Check the connection of Power Cubes and their functioning			
A13: gas timeout	R[A13]	Gas timeout (internal timer)	13
Check the proper operating of external devices.			
Check the functioning parameters.			
A14: heating timeout	N[A14]	Heating timeout (internal timer)	14
Check the proper operating of external devices			
Check the functioning parameters.			

Message	Remote message	Probable cause / Action	Signal on Fieldbus
A15: wire timeout	R[A15]	Wire feeding timeout	15
Check the proper operating of the soldering wire feeder. Check the parameters programming			
A16: phase missing	N[A16]	Phase missing	16
Message coming from tri-phase generators only. Check the presence of all of the three phases in the main.			
A101: temp. sensor 1 not found	N[A101]	Temperature sensor 1 not connected	101
Check the connections. Check the temperature sensor status. This diagnosis has to be activated with P1DIAG ON			
A102: temp. sensor 2 not found	N[A102]	Temperature sensor 2 not connected	102
Check the connections. Check the temperature sensor status. This diagnosis has to be activated with P2DIAG ON			
A121: PLC COMM	R[A121]	Missing PLC communication	121
Check the PLC connections and programming. With connections re-established and programming verified, start a new cycle to reset the diagnosis			
UPP sen. not found	-	UPP® sensor not found	-
Check the sensor power supply, its serial connection and in the end, the parameters. On the Controller, verify that the hidden parameter MODE > output supply is set on 24V. On UPP sensor check the following parameters: baud rate = 57600; address = 00; wait time = 0; switch for serial interface = RS232			
UPP sensor not ready	-	UPP® sensor not ready	-
Wait until the sensor ends the self-balancing procedure.			
UPP sensor internal fault	-	UPP® sensor faulty	-
Contact the Assistance service for the sensor.			
-	-	Temperature 1 out of tolerance	111
Check the programming parameters.			
-	-	Temperature 2 out of tolerance	112
Check the programming parameters.			

As soon as the malfunctioning occurs, the apparatus notifies the operator by means of a short alarm signal.

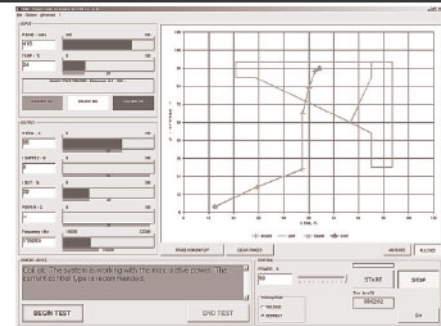
5.5.1 PowerCube Assistant

The PowerCube Assistant has been designed to optimize the heating coils, usable on serie 50, 200 and 900 – 1800 generators.

The software is free and downloadable, upon registration, at the address:

<http://www.ceia-power.com/monitoringsoftware.aspx>

The file is located in the Reserved Area of the site, accessible only after the registration.



5.5.2 Error messages during use of the memory cells

Message	Cause	Possible intervention
NO PARAMETERS SET FOUND Press any key	Attempt to load a set of parameters from an empty memory cell	Press the and key to give confirmation and repeat the loading operation using a correctly programmed memory
NOT VALID OR DEFECTIVE CARD Extract the card	Attempt to load a set of parameters, or to save it on a defective or not valid Program Card	Extract the card and repeat the action with a valid Program Card.
STORING ERROR Press any key	An error has been found during the copying of a set of parameters onto a memory cell; it may have been caused by a malfunctioning during the copying or by the use of a defective memory	Let the copying go ahead until the end and then repeat it. The set of parameters is recorded anyway and when loaded into the Power Controller there will be a (possible error) message such as the one described in the following point.
PARAMETERS SET CORRUPTED E = continue P = abort	Attempt to load a set of parameters from a memory cell, that was not correctly recorded	It is possible to interrupt the loading by pressing the P key or to continue by pressing E. In cases where the loading is carried out, it is advisable to check the parameters before using the apparatus.

5.6 Deinstallation

5.6.1 Disassembly

Power off the device and disconnect all cables from the back panel.

5.6.2 Packing back

Insert the Controller in the box, laying it down on the package base support	Lay down the intermediate support on the Controller and place on it the supply cable, the serial cable and all other contents.	Cover it with the upper support and place on it the Instruction Manual as shown in the picture.

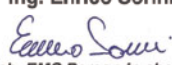
5.7 Apparatus disposal

Disposal of parts with environmental impact: follow the regulations in force in the country where the device is being used (refer to the Maintenance section).

6 APPENDICES

6.1 Declaration of Conformity CE

6.1.1 Master Controller V3+

EU DECLARATION OF CONFORMITY <i>(in accordance with EN ISO/IEC 17050-1:2010)</i> DECLARATION DE CONFORMITE UE (conformément à la norme EN ISO/IEC 17050-1:2010) EU-KONFORMITÄTSEKTLÄRUNG (nach EN ISO/IEC 17050-1:2010) DECLARACION DE CONFORMIDAD UE (según EN ISO/IEC 17050-1:2010) DICHIARAZIONE DI CONFORMITÀ UE (in conformità con EN ISO/IEC 17050-1:2010) <i>This declaration of conformity is issued under the sole responsibility of the manufacturer / La présente déclaration de conformité est établie sous la seule responsabilité du fabricant / Die alleinige Verantwortung für die Ausstellung dieser Konformitätserklärung trägt der Hersteller / La presente declaración de conformidad se expide bajo la exclusiva responsabilidad del fabricante / La presente dichiarazione di conformità è rilasciata sotto la responsabilità esclusiva del fabbricante</i>							
Manufacturer: Fabricant / Hersteller / Fabricante / Costruttore:	CEIA S.p.A. Zona industriale Viciomaggio 54/G-56 52041 Viciomaggio Arezzo – ITALY						
Declares that the product déclare que ce produit / erklärt, daß das Produkt / declara que el producto / dichiara che il prodotto:							
<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 40%;"> Product name: Nom du produit: / Produktname: Nombre del producto: / Nome: </td> <td style="width: 60%;"> Computerized soldering work-station Station de brasage contrôlée par microcomputer / Zentrale Elektronische Steuereinheit Estación computarizada de soldadura / Stazione computerizzata di saldatura </td> </tr> <tr> <td> Model: Modèle / Modell / Modelo / Modello: </td> <td> Master Controller MASTER-C-V3+ </td> </tr> <tr> <td> Product Options: Options / options: Opciones / opzioni: </td> <td> This declaration covers all options Cette déclaration est valide pour toutes les options / Diese Erklärung ist gültig für alle Optionen Este declaración es válida para todas las opciones / Questa dichiarazione è valida per tutte le opzioni </td> </tr> </table>		Product name: Nom du produit: / Produktname: Nombre del producto: / Nome:	Computerized soldering work-station Station de brasage contrôlée par microcomputer / Zentrale Elektronische Steuereinheit Estación computarizada de soldadura / Stazione computerizzata di saldatura	Model: Modèle / Modell / Modelo / Modello:	Master Controller MASTER-C-V3+	Product Options: Options / options: Opciones / opzioni:	This declaration covers all options Cette déclaration est valide pour toutes les options / Diese Erklärung ist gültig für alle Optionen Este declaración es válida para todas las opciones / Questa dichiarazione è valida per tutte le opzioni
Product name: Nom du produit: / Produktname: Nombre del producto: / Nome:	Computerized soldering work-station Station de brasage contrôlée par microcomputer / Zentrale Elektronische Steuereinheit Estación computarizada de soldadura / Stazione computerizzata di saldatura						
Model: Modèle / Modell / Modelo / Modello:	Master Controller MASTER-C-V3+						
Product Options: Options / options: Opciones / opzioni:	This declaration covers all options Cette déclaration est valide pour toutes les options / Diese Erklärung ist gültig für alle Optionen Este declaración es válida para todas las opciones / Questa dichiarazione è valida per tutte le opzioni						
conforms to the following Product Specifications est conforme aux spécifications suivantes / folgenden Produktspezifikationen entspricht es conforme a las siguientes especificaciones / è conforme alle seguenti specifiche di prodotto:							
Low Voltage (LVD) Directive 2014/35/EU EN 60204-1:2006 + EN60204-1:2006/A1:2009 + EN60204-1:2006/AC:2010 Safety of machinery - Electrical equipment of machines – Part 1: General requirements This product complies with the requirements of the Low Voltage Directive 2014/35/EU. Le produit ci-dessus répond aux exigences de la Directive 2014/35/UE concernant la basse tension. Dieses Produkt entspricht den Anforderungen an Niederspannungsgeräte gemäß der Norm 2014/35/EU El producto indicado cumple los requisitos de la Low Voltage Directive 2014/35/UE. Il prodotto è conforme alle norme della direttiva 2014/35/UE sulla bassa tensione							
Electromagnetic Compatibility (EMC) Directive 2014/30/EU EN 55011:2009 + EN 55011:2009/A1:2010 Group 2 - Class A Industrial, scientific and medical (ISM) radio-frequency equipment - Electromagnetic disturbance characteristics - Limits and methods of measurement EN 61000-6-2:2005 + EN 61000-6-2:2005/AC:2005 Electromagnetic compatibility (EMC) -- Part 6-2: Generic standards - Immunity for industrial environments This product complies with the requirements of the EMC Directive 2014/30/EU. Le produit ci-dessus répond aux exigences de la Directive 2014/30/UE concernant les interférences électromagnétiques. Dieses Produkt entspricht den Anforderungen der EMC-Norm 2014/30/UE. El producto indicado cumple los requisitos de la Directiva EMC 2014/30/UE. Il prodotto è conforme alle norme della direttiva EMC 2014/30/UE.							
The product can work only if connected to a proper CEIA device Le produit ne peut fonctionner que s'il est connecté à un dispositif CEIA approprié Das Produkt kann nur funktionieren, wenn es mit einem geeigneten CEIA-Gerät verbunden ist El producto sólo puede funcionar si está conectado a un dispositivo apropiado CEIA Il prodotto può funzionare solo se collegato a un appropriato dispositivo CEIA							
Signed for and on behalf of: Signé par et au nom de: / Unterzeichnet für und im Namen von: / Firmado por y en nombre de: / Firmato in vece e per conto di: CEIA S.p.A. Zona industriale Viciomaggio 54/G-56 52041 Viciomaggio Arezzo – ITALY							
Arezzo, 2016-11-25	 Ing. Enrico Sorini  Lab. EMC Person in charge Resp. Laboratoire EMC / Verantwortlicher für EMC-Labor Resp. Laboratorio EMC / Resp. Lab. EMC						

Doc. N° MASTER-C-V3+ rev 1

Mod. P0401-0025 rev 0

6.1.2 Optical Pyrometer SH15/SLE-550-D1

EU DECLARATION OF CONFORMITY

(in accordance with EN ISO/IEC 17050-1:2010)

DECLARATION DE CONFORMITE UE (conformément à la norme EN ISO/IEC 17050-1:2010) EU-KONFORMITÄTSERKLÄRUNG (nach EN ISO/IEC 17050-1:2010)

DECLARACION DE CONFORMIDAD UE (según EN ISO/IEC 17050-1:2010) DICHIARAZIONE DI CONFORMITÀ UE (in conformità con EN ISO/IEC 17050-1:2010)

This declaration of conformity is issued under the sole responsibility of the manufacturer / La présente déclaration de conformité est établie sous la seule responsabilité du fabricant / Die alleinige Verantwortung für die Ausstellung dieser Konformitätserklärung trägt der Hersteller / La presente declaración de conformidad se expide bajo la exclusiva responsabilidad del fabricante / La presente dichiarazione di conformità è rilasciata sotto la responsabilità esclusiva del fabbricante

Manufacturer: CEIA S.p.A.

Fabricant / Hersteller / Fabricante / Costruttore:

Zona industriale Viciomaggio 54/G-56
52041 Viciomaggio Arezzo – ITALY

Declares that the product

déclare que ce produit / erklärt, daß das Produkt / declara que el producto / dichiara che il prodotto:

Product name: Optical Pyrometer

Norm du produit: / Produktname: Pyromètre Optique / Optischen Pyrometer
Nombre del producto: / Nome: Sensor Optic: / Pirometro Ottico

Model: Modelé / Modell / Modelo / Modello: SH15/SLE-550-D1

Product Options: This declaration covers all options

Options / options: Cette déclaration est valide pour toutes les options / Diese Erklärung ist gültig für alle options
Opciones / opzioni: Este declaración es válida para todas las opciones / Questa dichiarazione è valida per tutte le opzioni

conforms to the following Product Specifications

est conforme aux spécifications suivantes / folgenden Produktspezifikationen entspricht
es conforme a las siguientes especificaciones / è conforme alle seguenti specifiche di prodotto:

Low Voltage (LVD) Directive 2014/35/EU

EN 61010-1:2010 Safety requirements for electrical equipment for measurement, control, and laboratory use -- Part 1: General requirements

This product complies with the requirements of the Low Voltage Directive 2014/35/EU.

Le produit ci-dessus répond aux exigences de la Directive 2014/35/UE concernant la basse tension.
Dieses Produkt entspricht den Anforderungen an Niederspannungsgeräte gemäß der Norm 2014/35/EU
El producto indicado cumple los requisitos de la Low Voltage Directive 2014/35/UE.
Il prodotto è conforme alle norme della direttiva 2014/35/UE sulla bassa tensione.

Electromagnetic Compatibility (EMC) Directive 2014/30/EU

EN 61000-6-2:2005 + EN 61000-6-2:2005/AC:2005 Electromagnetic compatibility (EMC) -- Part 6-2: Generic standards - Immunity for industrial environments
EN 61000-6-4:2007+ EN 61000-6-4:2007/A1:2011 Electromagnetic compatibility (EMC) -- Part 6-4: Generic standards - Emission standard for industrial environments

This product complies with the requirements of the EMC Directive 2014/30/EU.

Le produit ci-dessus répond aux exigences de la Directive 2014/30/UE concernant les interférences électromagnétiques.
Dieses Produkt entspricht den Anforderungen der EMC-Norm 2014/30/EU.
El producto indicado cumple los requisitos de la Directiva EMC 2014/30/UE.
Il prodotto è conforme alle norme della direttiva EMC 2014/30/UE.

The product can work only if connected to a proper CEIA device

Le produit ne peut fonctionner que s'il est connecté à un dispositif CEIA approprié
Das Produkt kann nur funktionieren, wenn es mit einem geeigneten CEIA-Gerät verbunden ist
El producto sólo puede funcionar si está conectado a un dispositivo apropiado CEIA
Il prodotto può funzionare solo se collegato a un appropriato dispositivo CEIA

Signed for and on behalf of: / Signé par et au nom de: / Unterzeichnet für und im Namen von: / Firmado por y en nombre de: / Firmato in vece e per conto di:
CEIA S.p.A. Zona industriale Viciomaggio 54/G-56 52041 Viciomaggio Arezzo – ITALY

Arezzo, 2016-11-25

 SpA

Ing. Enrico Sorini



Lab. EMC Person in charge

Resp. Laboratoire EMC / Verantwortlicher für EMC-Labor
Resp. Laboratorio EMC / Resp. Lab. EMC

Doc. N° SH15/SLE-550-D1 rev 1

Mod. P0401-0025 rev 0

6.1.3 Optical Pyrometer SH15/SLE-550-D2

EU DECLARATION OF CONFORMITY

(In accordance with EN ISO/IEC 17050-1:2010)

DECLARATION DE CONFORMITE UE (conformément à la norme EN ISO/IEC 17050-1:2010) EU-KONFORMITÄTSEKTLÄRUNG (nach EN ISO/IEC 17050-1:2010)
 DECLARACION DE CONFORMIDAD UE (según EN ISO/IEC 17050-1:2010) DICHIARAZIONE DI CONFORMITÀ UE (in conformità con EN ISO/IEC 17050-1:2010)

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Manufacturer: CEIA S.p.A.

Fabricant / Hersteller / Fabricante / Costruttore.

Zona industriale Viciomaggio 54/G-56
 52041 Viciomaggio Arezzo – ITALY

Declares that the product

déclare que ce produit / erklärt, daß das Produkt / declara que el producto / dichiara che il prodotto:

Product name: Optical Pyrometer

Nom du produit: / Produktname: Pyromètre Optique / Optischen Pyrometer
 Nombre del producto: / Nome: Sensor Optic: / Pirometro Ottico

Model: Modelé / Modell / Modelo / Modello: SH15/SLE-550-D2**Product Options: This declaration covers all options**

Options / options: Cette déclaration est valide pour toutes les options / Diese Erklärung ist
 Opciones / opzioni: gültig für alle options
 Este declaración es valida para todas las opciones / Questa dichiarazione
 è valida per tutte le opzioni

conforms to the following Product Specifications

est conforme aux spécifications suivantes / folgenden Produktspezifikationen entspricht
 es conforme a las siguientes especificaciones / è conforme alle seguenti specifiche di prodotto:

Low Voltage (LVD) Directive 2014/35/EU

EN 61010-1:2010 Safety requirements for electrical equipment for measurement, control, and laboratory use -- Part 1: General requirements

This product complies with the requirements of the Low Voltage Directive 2014/35/EU.

Le produit ci-dessus répond aux exigences de la Directive 2014/35/UE concernant la basse tension.

Dieses Produkt entspricht den Anforderungen an Niederspannungsgeräte gemäß der Norm 2014/35/EU.

El producto indicado cumple los requisitos de la Low Voltage Directive 2014/35/UE.

Il prodotto è conforme alle norme della direttiva 2014/35/UE sulla bassa tensione.

Electromagnetic Compatibility (EMC) Directive 2014/30/EU

EN 61000-6-2:2005 + EN 61000-6-2:2005/AC:2005 Electromagnetic compatibility (EMC) -- Part 6-2: Generic standards - Immunity for industrial environments
 EN 61000-6-4:2007+ EN 61000-6-4:2007/A1:2011 Electromagnetic compatibility (EMC) -- Part 6-4: Generic standards - Emission standard for industrial environments

This product complies with the requirements of the EMC Directive 2014/30/EU.

Le produit ci-dessus répond aux exigences de la Directive 2014/30/UE concernant les interférences électromagnétiques.

Dieses Produkt entspricht den Anforderungen der EMC-Norm 2014/30/EU.

El producto indicado cumple los requisitos de la Directiva EMC 2014/30/UE.

Il prodotto è conforme alle norme della direttiva EMC 2014/30/UE.

The product can work only if connected to a proper CEIA device

Le produit ne peut fonctionner que s'il est connecté à un dispositif CEIA approprié

Das Produkt kann nur funktionieren, wenn es mit einem geeigneten CEIA-Gerät verbunden ist

El producto sólo puede funcionar si está conectado a un dispositivo apropiado CEIA

Il prodotto può funzionare solo se collegato a un appropriato dispositivo CEIA

Signed for and on behalf of: / Signé par et au nom de: / Unterzeichnet für und im Namen von: / Firmado por y en nombre de: / Firmato in vece e per conto di:

CEIA S.p.A. Zona Industriale Viciomaggio 54/G-56 52041 Viciomaggio Arezzo – ITALY

Arezzo, 2016-11-25



Ing. Enrico Sorini



Lab. EMC Person in charge

Resp. Laboratoire EMC / Verantwortlicher für EMC-Labor
 Resp. Laboratorio EMC / Resp. Lab. EMC

6.1.4 Optical pyrometer SH15/SLE-550-D3

EU DECLARATION OF CONFORMITY

(in accordance with EN ISO/IEC 17050-1:2010)

DECLARATION DE CONFORMITE UE (conformément à la norme EN ISO/IEC 17050-1:2010) EU-KONFORMITÄTSERKLÄRUNG (nach EN ISO/IEC 17050-1:2010)

DECLARACION DE CONFORMIDAD UE (según EN ISO/IEC 17050-1:2010) DICHIARAZIONE DI CONFORMITÀ UE (in conformità con EN ISO/IEC 17050-1:2010)

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Manufacturer: CEIA S.p.A.

Fabricant / Hersteller / Fabricante / Costruttore:

Zona industriale Viciomaggio 54/G-56
52041 Viciomaggio Arezzo – ITALY

Declares that the product

déclare que ce produit / erklärt, daß das Produkt / declara que el producto / dichiara che il prodotto:

Product name: Optical Pyrometer

Nom du produit: / Produktname:

Pyromètre Optique / Optischen Pyrometer

Nombre del producto: / Nome:

Sensor Optic: / Pirometro Ottico

Model: Modelé / Modell / Modelo / Modello: SH15/SLE-550-D3

Product Options: This declaration covers all options

Options / options:

Cette déclaration est valide pour toutes les options / Diese Erklärung ist

Opciones / opzioni:

gültig für alle options

Esta declaración es válida para todas las opciones / Questa dichiarazione

è valida per tutte le opzioni

conforms to the following Product Specifications

est conforme aux spécifications suivantes / folgenden Produktspezifikationen entspricht
es conforme a las siguientes especificaciones / è conforme alle seguenti specifiche di prodotto:

Low Voltage (LVD) Directive 2014/35/EU

EN 61010-1:2010 Safety requirements for electrical equipment for measurement, control, and laboratory use -- Part 1: General requirements

This product complies with the requirements of the Low Voltage Directive 2014/35/EU.

Le produit ci-dessus répond aux exigences de la Directive 2014/35/UE concernant la basse tension.

Dieses Produkt entspricht den Anforderungen an Niederspannungsgeräte gemäß der Norm 2014/35/EU

El producto indicado cumple los requisitos de la Low Voltage Directive 2014/35/UE.

Il prodotto è conforme alle norme della direttiva 2014/35/UE sulla bassa tensione.

Electromagnetic Compatibility (EMC) Directive 2014/30/EU

EN 61000-6-2:2005 + EN 61000-6-2:2005/AC:2005 Electromagnetic compatibility (EMC) -- Part 6-2: Generic standards - Immunity for industrial environments
EN 61000-6-4:2007+ EN 61000-6-4:2007/A1:2011 Electromagnetic compatibility (EMC) -- Part 6-4: Generic standards - Emission standard for industrial environments

This product complies with the requirements of the EMC Directive 2014/30/EU.

Le produit ci-dessus répond aux exigences de la Directive 2014/30/UE concernant les interférences électromagnétiques.

Dieses Produkt entspricht den Anforderungen der EMC-Norm 2014/30/EU.

El producto indicado cumple los requisitos de la Directiva EMC 2014/30/UE.

Il prodotto è conforme alle norme della direttiva EMC 2014/30/UE.

The product can work only if connected to a proper CEIA device

Le produit ne peut fonctionner que s'il est connecté à un dispositif CEIA approprié

Das Produkt kann nur funktionieren, wenn es mit einem geeigneten CEIA-Gerät verbunden ist

El producto sólo puede funcionar si está conectado a un dispositivo apropiado CEIA

Il prodotto può funzionare solo se collegato a un appropriato dispositivo CEIA

Signed for and on behalf of: / Signé par et au nom de: / Unterzeichnet für und im Namen von: / Firmado por y en nombre de: / Firmato in vece e per conto di:
CEIA S.p.A. Zona industriale Viciomaggio 54/G-56 52041 Viciomaggio Arezzo – ITALY

Arezzo, 2016-11-25



Ing. Enrico Sorini

Enrico Sorini

Lab. EMC Person in charge

Resp. Laboratoire EMC / Verantwortlicher für EMC-Labor

Resp. Laboratorio EMC / Resp. Lab. EMC

6.1.5 Optical pyrometer SH15/SLE-550-D4

EU DECLARATION OF CONFORMITY

(In accordance with EN ISO/IEC 17050-1:2010)

DECLARATION DE CONFORMITE UE (conformément à la norme EN ISO/IEC 17050-1:2010) EU-KONFORMITÄTSEKTLÄRUNG (nach EN ISO/IEC 17050-1:2010)
 DECLARACION DE CONFORMIDAD UE (según EN ISO/IEC 17050-1:2010) DICHIARAZIONE DI CONFORMITÀ UE (in conformità con EN ISO/IEC 17050-1:2010)

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Manufacturer: CEIA S.p.A.

Fabricant / Hersteller / Fabricante / Costruttore.

Zona industriale Viciomaggio 54/G-56
 52041 Viciomaggio Arezzo – ITALY

Declares that the product

déclare que ce produit / erklärt, daß das Produkt / declara que el producto / dichiara che il prodotto:

Product name: Optical Pyrometer

Nom du produit: / Produktname: Pyromètre Optique / Optischen Pyrometer
 Nombre del producto: / Nome: Sensor Optic: / Pirometro Ottico

Model: Modelé / Modell / Modelo / Modello: SH15/SLE-550-D4**Product Options: This declaration covers all options**

Options / options: Cette déclaration est valide pour toutes les options / Diese Erklärung ist
 Opciones / opzioni: gültig für alle options
 Este declaración es valida para todas las opciones / Questa dichiarazione
 è valida per tutte le opzioni

conforms to the following Product Specifications

est conforme aux spécifications suivantes / folgenden Produktspezifikationen entspricht
 es conforme a las siguientes especificaciones / è conforme alle seguenti specifiche di prodotto:

Low Voltage (LVD) Directive 2014/35/EU

EN 61010-1:2010 Safety requirements for electrical equipment for measurement, control, and laboratory use -- Part 1: General requirements

This product complies with the requirements of the Low Voltage Directive 2014/35/EU.

Le produit ci-dessus répond aux exigences de la Directive 2014/35/UE concernant la basse tension.

Dieses Produkt entspricht den Anforderungen an Niederspannungsgeräte gemäß der Norm 2014/35/EU.

El producto indicado cumple los requisitos de la Low Voltage Directive 2014/35/UE.

Il prodotto è conforme alle norme della direttiva 2014/35/UE sulla bassa tensione.

Electromagnetic Compatibility (EMC) Directive 2014/30/EU

EN 61000-6-2:2005 Electromagnetic compatibility (EMC) -- Part 6-2: Generic standards - Immunity for industrial environments

EN 61000-6-4:2007+ EN 61000-6-4:2007/A1:2011 Electromagnetic compatibility (EMC) -- Part 6-4: Generic standards - Emission standard for industrial environments

This product complies with the requirements of the EMC Directive 2014/30/EU.

Le produit ci-dessus répond aux exigences de la Directive 2014/30/UE concernant les interférences électromagnétiques.

Dieses Produkt entspricht den Anforderungen der EMC-Norm 2014/30/EU.

El producto indicado cumple los requisitos de la Directiva EMC 2014/30/UE.

Il prodotto è conforme alle norme della direttiva EMC 2014/30/UE.

The product can work only if connected to a proper CEIA device

Le produit ne peut fonctionner que s'il est connecté à un dispositif CEIA approprié

Das Produkt kann nur funktionieren, wenn es mit einem geeigneten CEIA-Gerät verbunden ist

El producto sólo puede funcionar si está conectado a un dispositivo apropiado CEIA

Il prodotto può funzionare solo se collegato a un appropriato dispositivo CEIA

Signed for and on behalf of: / Signé par et au nom de: / Unterzeichnet für und im Namen von: / Firmado por y en nombre de: / Firmato in vece e per conto di:

CEIA S.p.A. Zona Industriale Viciomaggio 54/G-56 52041 Viciomaggio Arezzo – ITALY

Arezzo, 2016-11-25



Ing. Enrico Sorini



Lab. EMC Person in charge

Resp. Laboratoire EMC / Verantwortlicher für EMC-Labor
 Resp. Laboratorio EMC / Resp. Lab. EMC

6.1.6 Optical pyrometer SH2C/SLE

EU DECLARATION OF CONFORMITY

(in accordance with EN ISO/IEC 17050-1:2010)

DECLARATION DE CONFORMITE UE (conformément à la norme EN ISO/IEC 17050-1:2010) EU-KONFORMITÄTSEKTLÄRUNG (nach EN ISO/IEC 17050-1:2010)

DECLARACION DE CONFORMIDAD UE (según EN ISO/IEC 17050-1:2010) DICHIARAZIONE DI CONFORMITÀ UE (in conformità con EN ISO/IEC 17050-1:2010)

This declaration of conformity is issued under the sole responsibility of the manufacturer / La présente déclaration de conformité est établie sous la seule responsabilité du fabricant / Die alleinige Verantwortung für die Ausstellung dieser Konformitätserklärung trägt der Hersteller / La presente declaración de conformidad se expide bajo la exclusiva responsabilidad del fabricante / La presente dichiarazione di conformità è rilasciata sotto la responsabilità esclusiva del fabbricante

Manufacturer: CEIA S.p.A.

Fabricant / Hersteller / Fabricante / Costruttore:

Zona industriale Viciomaggio 54/G-56
52041 Viciomaggio Arezzo – ITALY

Declares that the product

déclare que ce produit / erklärt, daß das Produkt / declara que el producto / dichiara che il prodotto:

Product name: Optical Pyrometer

Nom du produit: / Produktname:

Pyromètre Optique / Optischen Pyrometer

Nombre del producto: / Nome:

Sensor Optic: / Pirometro Ottico

Model: Modelé / Modell / Modelo / Modello: SH2C/SLE

Product Options: This declaration covers all options

Options / options:

Cette déclaration est valide pour toutes les options / Diese Erklärung ist

Opciones / opzioni:

gültig für alle options

Esta declaración es válida para todas las opciones / Questa dichiarazione

è valida per tutte le opzioni

conforms to the following Product Specifications

est conforme aux spécifications suivantes / folgenden Produktspezifikationen entspricht
es conforme a las siguientes especificaciones / è conforme alle seguenti specifiche di prodotto:

Low Voltage (LVD) Directive 2014/35/EU

EN 61010-1:2010 Safety requirements for electrical equipment for measurement, control, and laboratory use -- Part 1: General requirements

This product complies with the requirements of the Low Voltage Directive 2014/35/EU.

Le produit ci-dessus répond aux exigences de la Directive 2014/35/UE concernant la basse tension.

Dieses Produkt entspricht den Anforderungen an Niederspannungsgeräte gemäß der Norm 2014/35/EU

El producto indicado cumple los requisitos de la Low Voltage Directive 2014/35/UE.

Il prodotto è conforme alle norme della direttiva 2014/35/UE sulla bassa tensione.

Electromagnetic Compatibility (EMC) Directive 2014/30/EU

EN 61000-6-2:2005 + EN 61000-6-2:2005/AC:2005 Electromagnetic compatibility (EMC) -- Part 6-2: Generic standards - Immunity for industrial environments
EN 61000-6-4:2007+ EN 61000-6-4:2007/A1:2011 Electromagnetic compatibility (EMC) -- Part 6-4: Generic standards - Emission standard for industrial environments

This product complies with the requirements of the EMC Directive 2014/30/EU.

Le produit ci-dessus répond aux exigences de la Directive 2014/30/UE concernant les interférences électromagnétiques.

Dieses Produkt entspricht den Anforderungen der EMC-Norm 2014/30/EU.

El producto indicado cumple los requisitos de la Directiva EMC 2014/30/UE.

Il prodotto è conforme alle norme della direttiva EMC 2014/30/UE.

The product can work only if connected to a proper CEIA device

Le produit ne peut fonctionner que s'il est connecté à un dispositif CEIA approprié

Das Produkt kann nur funktionieren, wenn es mit einem geeigneten CEIA-Gerät verbunden ist

El producto sólo puede funcionar si está conectado a un dispositivo apropiado CEIA

Il prodotto può funzionare solo se collegato a un appropriato dispositivo CEIA

Signed for and on behalf of: / Signé par et au nom de: / Unterzeichnet für und im Namen von: / Firmado por y en nombre de: / Firmato in vece e per conto di:
CEIA S.p.A. Zona industriale Viciomaggio 54/G-56 52041 Viciomaggio Arezzo – ITALY

Arezzo, 2016-11-25

 SpA

Ing. Enrico Sorini



Lab. EMC Person in charge

Resp. Laboratoire EMC / Verantwortlicher für EMC-Labor

Resp. Laboratorio EMC / Resp. Lab. EMC

Doc. N° SH2C/SLE rev 1

Mod. P0401-0025 rev 0

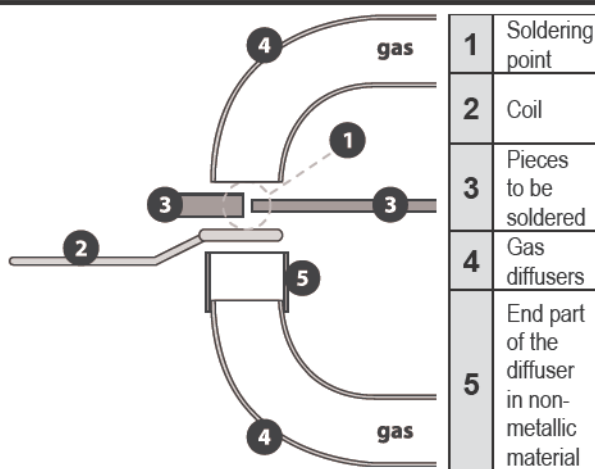
6.2 Use of anti-oxidizing gas

With the aim of improving the quality of the soldering, it is useful to use distribution valves for the diffusion of anti-oxidizing gas in the soldering area.

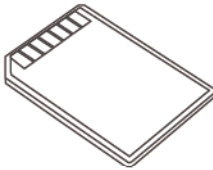
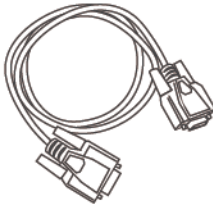
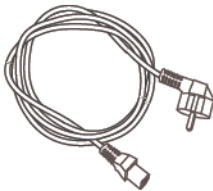
These diffusers should be as close as possible to the heating coil and to the piece being processed, so as to create, during the soldering, an oxygen-free "chamber".

The end part of the diffusers should be constructed in non-metallic material (for example, teflon), in the event that such parts come very close to the coil.

Beside an example of an installation with gas diffusers.



6.3 Spare parts

Spare part		Order code
	Program Card (100 memory cells)	55265
	RS232 Serial connection cable (1 m length)	19623
	RS232 Serial connection cable (3 m length)	23243
-	Pyrometer connection cable (4 m length)	49439
	Power supply cable with EU plug	1559

6.4 Credits

UPP® is a registered trademark of IMPAC® Infrared GmbH.

6.5 Manufacturer

CEIA S.p.A.
Costruzioni Elettroniche Industriali Automatismi
 Zona Industriale 54/56, 52041 Vicinaggio – Arezzo (ITALY)